



Goodbye to Herbicides

Development of an Autonomous Weeding
Robot - June 18th 2020

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Chapter 1

L^AT_EX -Styleguide

Nachfolgend wird die Codierung einiger oft verwendeter Elemente kurz beschrieben. Versucht euch an die entsprechenden Formate zu halten.

1.1 Gliederungen

Ein Text kann mit den Befehlen `\chapter{.}`, `\section{.}`, `\subsection{.}` und `\subsubsection{.}` gegliedert werden (gemäss definierter Kapitelstruktur). Subsubsections werden nicht im Inhaltsverzeichnis aufgeführt und können von euch frei als Überschriften verwendet werden. Wenn ihr euren Text weiter strukturieren wollt, könnt ihr eine weitere Überschrift unter den Subsubsections machen. Dazu verwendet ihr den `\emph{.}` Befehl. Das Resultat ist abgebildet (siehe fig. 1.1). Haltet euch bitte an die Listenebenen wie sie in der Kapitelstruktur definiert sind.

Chapter Title

1.1 Section Title

1.1.1 Subsection Title

Subsubsection

Heading

Followup Text.

Figure 1.1: Kapitelstruktur

1.2 Referenzen und Verweise

Literaturreferenzen werden mit dem Befehl `\citep{.}` und `\citet{.}` erzeugt. Beispiele: ein Buch [1], ein Buch und ein Journal Paper [1, 2], ein Konferenz Paper mit Erwähnung des Autors: Pratt and Williamson [3].

Zur Erzeugung von Fussnoten wird der Befehl `\footnote{.}` verwendet. Auch hier ein Beispiel¹.

¹Bla bla.

Querverweise im Text werden mit `\label{.}` verankert und mit `\cref{.}` erzeugt. Beispiel einer Referenz auf das zweite Kapitel: chapter 2.

1.3 Aufzählungen

Für Aufzählungen verwenden wir folgende Vorlagen für Aufzählungen mit und ohne Nummerierung. Zusätzlicher Text kann nach einem Zeilenumbruch hinzugefügt werden, allerdings mit einem kleinen Zeilenabstand. Aufzählung ohne Nummerierung:

- Punkt 1
Zusätzliche Informationen zu Punkt 1.
- Punkt 2

wurde erzeugt mit:

```
\begin{itemize}
  \item Punkt 1

  Zusätzliche Informationen zu Punkt 1.
  \item Punkt 2
\end{itemize}
```

Aufzählung mit Nummerierung:

1. Punkt 1
Zusätzliche Informationen zu Punkt 1.
2. Punkt 2

wurde erzeugt mit:

```
\begin{enumerate}
  \item Punkt 1

  Zusätzliche Informationen zu Punkt 1.
  \item Punkt 2
\end{enumerate}
```

1.4 Erstellen einer Tabelle

Ein Beispiel einer Tabelle:

Kennzahl	Einheit	ECE	EUDC	NEFZ
Dauer	s	780	400	1180
Distanz	km	4.052	6.955	11.007
Durchschnittsgeschwindigkeit	km/h	18.7	62.6	33.6
Leerlaufanteil	%	36	10	27

Table 1.1: Daten der Fahrzyklen ECE, EUDC, NEFZ.

Die Tabelle wurde erzeugt mit:

```

\begin{table}[h]
\begin{center}
\caption{Daten der Fahrzyklen ECE, EUDC, NEFZ.}\vspace{1ex}
\label{tab:tabnefz}
\begin{tabular}{ll|ccc}
\hline
Kennzahl & Einheit & ECE & EUDC & NEFZ \\ \hline
Dauer & s & 780 & 400 & 1180 \\
Distanz & km & 4.052 & 6.955 & 11.007 \\
Durchschnittsgeschwindigkeit & km/h & 18.7 & 62.6 & 33.6 \\
Leerlaufanteil & \% & 36 & 10 & 27 \\
\hline
\end{tabular}
\end{center}
\end{table}

```

1.5 Einbinden einer Grafik

Das Einbinden von Graphiken kann wie folgt bewerkstelligt werden:

```

\begin{figure}[h!]
\centering
\includegraphics[width=0.75\textwidth]{images/k_surf.pdf}
\caption{Ein Bild.}
\label{fig:k_surf}
\end{figure}

```

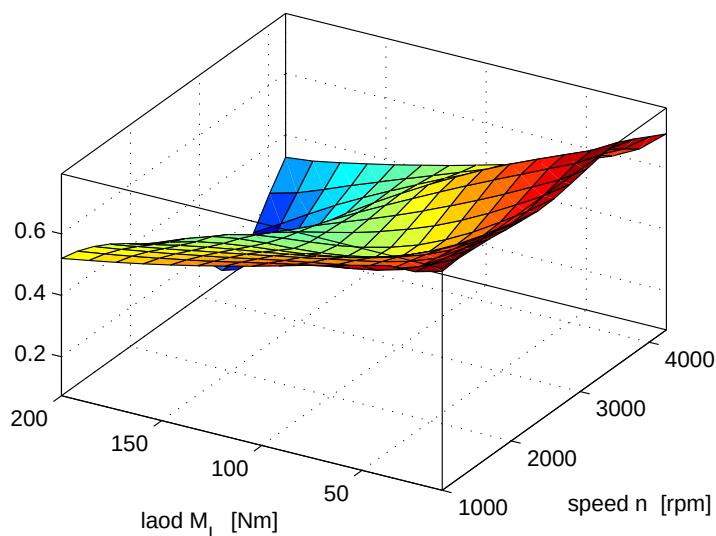


Figure 1.2: Ein Bild

oder bei zwei Bildern nebeneinander mit:

```

\begin{figure}[h!]
\begin{minipage}[t]{0.48\textwidth}
\includegraphics[width = \textwidth]{images/cycle_we.pdf}
\end{minipage}
\end{figure}

```

```

\hfill
\begin{minipage}[t]{0.48\textwidth}
  \includegraphics[width = \textwidth]{images/cycle_ml.pdf}
\end{minipage}
\caption{Zwei Bilder nebeneinander.}
\label{fig:cycle}
\end{figure}

```

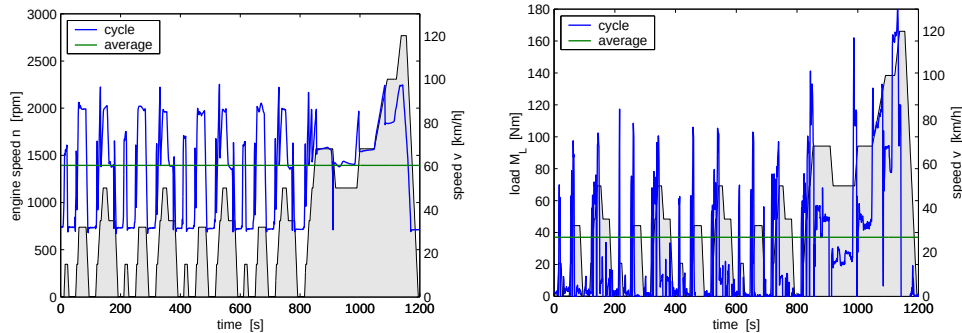


Figure 1.3: Zwei Bilder nebeneinander

1.5.1 Positionierung von Bildern

Der Zusatz [h!] nach `\begin{figure}` sorgt für die exakte Positionierung an der Stelle im Code. Wir nur der Zusatz [h] verwendet, platziert Overleaf das Bild nur ungefähr am gleichen Ort. Jedes Bild und jede Tabelle soll ausserdem eine aussagekräftige Bildunterschrift (caption) haben. Diese liegt immer *unter* dem Bild.

1.5.2 Verlinkung von Bildern

Jedes Bild und jede Tabelle muss im Text verlinkt werden. Dies wird durch eine Information in Klammern gemacht (see fig. 1.3). Dazu wird der Befehl `\see \cref{.}` verwendet. Schreibt keine Verlinkungen von Hand und achtet darauf, dass das Bild *nach* der Verlinkung folgt. Um die Bilder und Tabellen zu verlinken müssen diese im Sourcecode gelabelt werden. Dies macht man mit `\label{fig:section_titel}`

1.6 Mathematische Formeln

Einfache mathematische Formeln werden mit der equation-Umgebung erzeugt und sind immer im Text zentriert:

$$p_{me0f}(T_e, \omega_e) = k_1(T_e) \cdot (k_2 + k_3 S^2 \omega_e^2) \cdot \Pi_{\max} \cdot \sqrt{\frac{k_4}{B}}. \quad (1.1)$$

Der Code dazu lautet:

```

\begin{equation}
p_{me0f}(T_e, \omega_e) \ = \ k_1(T_e) \ \cdot \ (k_2 + k_3 \ S^2 \ \omega_e^2) \ \cdot \ \Pi_{\max} \ \cdot \ \sqrt{\frac{k_4}{B}} \ , \ .
\end{equation}

```

Mathematische Ausdrücke im Text werden mit `$formel$` erzeugt (z.B.: $a^2 + b^2 = c^2$).

Vektoren und Matrizen werden mit den Befehlen `\vec{.}` und `\mat{.}` erzeugt (z.B. $\mathbf{v}$, $\mathbf{M}$).

Herleitungen oder eine Auflistung von Variablen werden nicht zentriert, sondern mit einem Einzug dargestellt. Beispiel:

$$\begin{aligned} \Rightarrow dM(a,b) &= dF_R \cdot l \geq \mu_H \cdot F_g \cdot l(a,b) \\ \Rightarrow dM(a,b) &\geq \mu_H \cdot m \cdot g \cdot l(a,b) \end{aligned}$$

oder

$$\begin{aligned} \alpha &= 2^\circ = 0.035\text{rad} \\ l_1 &= 1\text{m} \\ l_2 &= \tan(\alpha) \cdot l_1 = 3.5\text{cm} \end{aligned}$$

Diese werden dargestellt durch einen `\hspace`-Befehl und `$formel$`.

```
\hspace*{1cm}$\rightarrow dM(a,b) = dF_R\cdot l \geq \mu_H\cdot F_g\cdot l(a,b)$ \newline
\hspace*{1cm}$\rightarrow dM(a,b) \geq \mu_H\cdot m\cdot g\cdot l(a,b)$ \newline
oder \newline
\hspace*{1cm}$\alpha = 2\text{\textdegree} = 0.035\text{rad}$\
\hspace*{1cm}$l_1 = 1\text{m}$\
\hspace*{1cm}$l_2 = \tan(\alpha)\cdot l_1 = 3.5\text{cm}$\
```

1.7 Hervorhebungen und Einheiten

Hervorhebungen im Text sehen so aus: *hervorgehoben*. Erzeugt werden sie mit dem `\epmh{.}` Befehl. Verwendet Hervorhebungen für Fremdworte, Firmennamen und Typenbezeichnungen.

Einheiten werden mit den Befehlen `\unit[1]{m}` (z.B. 1 m) und `\unitfrac[1]{m}{s}` (z.B. 1 m/s) gesetzt. Bitte benutzt den Einheitenbefehl, damit die Darstellung von Einheiten konsistent bleibt.

1.8 Section

Nach der Sectiondeklaration beschreiben wir weshalb das Kapitel von Bedeutung ist, die sogenannte *motivation* / das *Why?*. Zusätzlich wird auch das Endprodukt dieses Kapitels in knappen 3 bis 4 Sätzen zusammengefasst, sodass der Leser weiss wohin dieses Kapitel führt (ein ergänzendes Bild dazu ist ebenfalls sehr hilfreich).

1.8.1 Subsection

Falls du welche hast, ist es wichtig deine Subsections ebenfalls kurz und knapp zu beschreiben (2-4 Sätze).

Hier ein kurzes Beispiel um herauszufinden wann du eine Subsection verwenden sollst und wann nicht:

1. Introduction
2. Mechanical Design
 - 2.1. Platform
 - 2.2. Steering
 - 2.3. Weeding tools
 - 2.3.1. Active Weeding Tool
 - 2.3.2. Passive Weeding Tool
3. ...

Hier sieht man, dass die Sections *Platform* und *Steering* nicht weiter unterteilt werden. In diesen Fällen werden keine Subsections verwendet!

Die Section *Weeding tools* wird in zwei weitere Kapitel aufgeteilt: *Active Weeding Tool* und *Passive Weeding Tool*. Diese beiden letztgenannten Kapitel werden als Subsections aufgelistet.

Subsubsection

Der Final Report orientiert sich stark an nachstehendem Schema. Dieses Schema kann in Subsubsections aufgeteilt werden, sodass unser Report kongruent wirkt.

1. What? Requirements, Options, Research & Evaluation
What is required? What options suit the requirements? Why was the final option chosen?
2. How? Realization
How did it get implemented? How should it work?
3. Now? Testing, Review, Conclusion & Outlook
Did it work as planned? What could be learned? What has to be adjusted in the future?

PS: Das Why? sollte bereits nach der Section-/Subsectionangabe gemacht worden sein.

Subsubsubsection aka emph

Da es in Latex keine subsubsubsections gibt verwenden wir anstededessen `\emph{.}`. Diese `\emph{.}`'s werden verwendet um verschiedene Aspekte der Subsubsection zu unterteilen.

Abstract

To create a viable alternative to the frequent but conventional use of harmful pesticides and herbicides in agriculture, the focus project Rowesys aims to develop a robotic solution suitable for organic farming in sugar beet fields.

In order to achieve this goal, Rowesys has carried out a research phase to investigate the various possible areas of application of robotic systems in agriculture.

The result of this investigation and development process is a prototype of a robotic weeding system for sugar beets called *Rosie*. Cameras provide information to enable the autonomous row following and crop detection, which controls the movement and weeding behaviour of the system. The robot's four wheel drive, individual suspension and swerve steering allow for moving in rough agricultural terrain and versatile maneuvers. The weeds are removed with the help of an actuated as well as a rigid weeding tool, making use of goosefoot sweepers operating mechanically without the use of herbicides. The various parts of the casing can be easily removed to allow access to the robot's subsystems.

The robot already reliably masters the intended functions such as autonomous row recognition, end of row detection and weed recognition. However, the robot is still pending development work to achieve greater robustness until it is able to autonomously process whole fields.

Preface

The focus project Rowesys involved the design, implementation and programming of a functional, autonomous weeding robot in the time frame of nine months. The achievement of this work would not have been possible without the help of many external supporters.

At this point, we, team Rowesys, would like to express our thanks to the following partners:

- Prof. Dr. Marco Hutter for providing us with infrastructure and resources, the insightful feedback and the time spent to attend our meetings and follow the development of the system.
- Dhionis Sako, Giorgio Valsecchi, Giuseppe Rizzi, Lennart Nachtigall, Lukas Schaupp and Malin Siegwart for the countless hours spent in meetings and reviews, the detailed feedback on our ideas, designs and codes, the great interest in our system and the appreciation of our work.
- Prof. Dr. Achim Walter, Dr. Simone Nanzer, Dr. Frank Liebisch, Hansueli Zellweger, Dr. Andreas Hund and everyone else from the ETH Crop Sciences for welcoming us in Eschikon and providing us with the necessary test facilities and their advice.
- Roland Haas, Maria Trodella, Konrad Meyer, Martin Schütz, Jean-Luc Emery and Christian Schorno for the logistical and technical support.
- Angela Guyer and the Strickhof in Lindau for letting us ~~destroy~~ use their sugar beet field during data gathering and testing.
- Gerhard Aebi, Martin Bauer and Hansueli Dierauer for sharing their agricultural knowledge.
- The families Lieberherr and Wagner for providing us with workplaces and also for their hospitality during the Covid-19 lockdown.
- Michael Hetzer from Elobau and the entire HelloSolution team for their special support and trust.
- All our sponsors and patrons for their trust and support.

Symbols

Symbols

v_x	robot speed in x-direction
v_y	robot speed in y-direction
ϕ	angular velocity of robot
$\mathbf{v}_1$	speed vector wheel front left
$\mathbf{v}_2$	speed vector wheel front right
$\mathbf{v}_3$	speed vector wheel back right
$\mathbf{v}_4$	speed vector wheel back left
ϕ_1	absolute steering position wheel front left
ϕ_2	absolute steering position wheel front right
ϕ_3	absolute steering position wheel back right
ϕ_4	absolute steering position wheel back left
d_l	distance between left front wheel and left back wheel
d_w	distance between left front and right front wheel

Indices

x	x axis
y	y axis
z	z axis

Acronyms and Abbreviations

ASL	Autonomous Systems Lab
BAFU	Swiss Federal Office for the Environment
BFH	Technical College Bern
BMS	Battery Management System
CAD	Computer-Aided Design
CAN	Controller Area Network
COG	Center of Gravity
COR	Center of Rotation
COVID-19	Coronavirus Disease 2019
CPU	Central Processing Unit
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
DC	Direct Current
DHCP	Dynamic Host Configuration Protocol
E-Stop	Emergency Stop
EAS	Elmo Application Studio
ECU	Electronic Control Unit
ETH	Swiss Federal Institute of Technology Zurich
EtherCAT	Ethernet for Control Automation Technology
FHNW	University of Applied Sciences of Northwest Switzerland
FOV	Field of View
FP16	Half-precision floating-point format
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GPU	Graphics Processing Unit
G-SOLTWI	Gold Solo Twitter
HLC	High Level Controller
HSV	Hue, Saturation, Value Color Space
HWI	Hardware Interface
IP	Internet Protocol
ITX	Information Technology eXtended
IMU	Inertial Measurement Unit
LFP	Lithium Iron Phosphate
LiDAR	Light Detection and Ranging Sensor
LTE	Long Term Evolution
MAC	Media Access Control
MOSFET	Metal Oxide Semiconductor Field-Effect Transistor
MVP	Minimum Viable Product
NIR	Near-Infrared
NMC	Lithium nickel manganese cobalt oxides
NMT	Network Management Protocols
NUC	Next Unit of Computing
PC	Personal Computer
PCB	Printed Circuit Board
PI	Proportional Integral

RGB	Red, Green, Blue Color Space
ROI	Region of Interest
ROS	Robot Operating System
Rosie	Name of the Rowesys Robot
RSL	Robotic Systems Lab
RTK	Real Time Kinematic
RV	Recreational Vehicle
SAE	Society of Automotive Engineers
SDK	Software Development Kit
SDO	Service Data Object
SMD	Surface-Mounted Device
SRS	System Requirements Specifications
SSR	Solid State Relay
STO	Safety Torque Off
TFLOPS	Terra Floating Point Operations per Second
URDF	Unified Robot Description Format
USB	Universal Serial Bus
VIO	Visual Inertial Odometry

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Chapter 2

Introduction

2.1 Vision and Motivation

The widespread use of pesticides worldwide and in Switzerland causes considerable damage to people [4] and the environment. According to the water research institute EAWAG, 71 of 89 active substances were found per site in the five Swiss rivers tested. A total of 145 different substances were found, with 66 of them exceeding the limit concentrations [5].

However, pesticides are not only found in rivers, but also in the air [6], in food, and, as revealed by a study of the Swiss Federal Institute of Environment, even in groundwater [7] as shown in fig. 2.1. The size of the circle shows the amount of measured pesticides. Concentrations exceeding the allowed limit are marked purple.

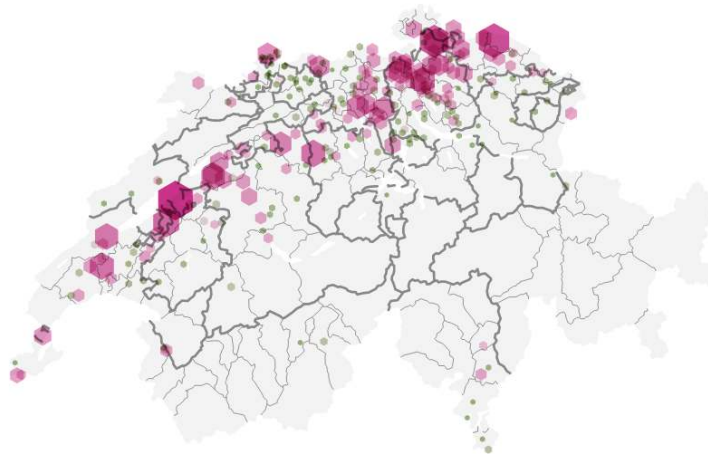


Figure 2.1: Measurement of pesticides in ground water

The public was shocked by those revelations and the use of pesticides is more and more seen as something to be condemned.

Team Rowsys agrees that the widespread and intense application of pesticides needs to be replaced by a more sustainable solution. But so far, organic farming can not compete with conventional agriculture: On average the yield is 20% lower [8] and the workload for the farmers is increased. This results in a relatively high price. As companies have recognized the problem, in the last few years a new approach has appeared: reducing cost through the use of weeding robots. Human-assisted

weeding systems, such as Blue River's "See & Spray" (see fig. 2.3(a)), are providing an effective solution for sustainable weeding. However, these technologies can only be used in large agricultural farming since they are too expensive for regular farmers. Human-assistance puts an additional pull on the cost for using this technology which is why there needs to be an autonomous solution for organic farmers. The complexity of autonomous farming is a big obstacle for this now developing young industry. So far, none of the completely autonomous approaches have been able to prevail. Common issues with current systems are the price tag, getting stuck in the field and imprecise plant recognition.

In our project, we aim to be part of the solution to a pesticide-free future in agriculture by attacking some of the most common problems that current autonomous weeding robots still have.

2.2 Introduction into Weeding in Agriculture

In this section, the reader will be given an introduction into weeding in agriculture. Weeding is the removal of unwanted plants. Crops and weeds compete against each other in terms of nutrition, sunlight and water. Since many weeds grow fast and aggressively, they can easily out compete the crop, reducing yield or even leading to the death of crops. Therefore, the weed has to be removed especially at the beginning when the crop is small and weak compared to a fast growing weed. An effective weed control management is crucial for a high yield. Thus, it is a top priority for farmers. The market offers different machinery and methods to remove the weed. Based on conversations with farmers, three major requirements from a customer point of view for a weeding method were identified. The requirements are:

- The weeding method should have low costs per hectare and application.
- The weeding method should be effective and reliable meaning that all the weed is removed and that it preferably does not regrow during the growth of the crop.
- The weeding method should be automated and include as little (hand) labour as possible

The first requirement listed is secondary for the development of the robot since Rowsys' goal is to develop a prototype, where costs are less important than function. The second and third requirement are important for the development of the robot. A detailed list and explanation of the requirements is given in section 2.6.2.

2.3 State of the Art of Weeding in Agriculture

In this section, an overview of different weeding methods is given alongside with their advantages and disadvantages.

2.3.1 Herbicide Spraying with Tractor

Spraying herbicide with a module that can be attached to a tractor (shown in fig. 2.2) is the most used method. It has the lowest cost per hectare and is the most effective option. Because the weed is removed everywhere plus it does not regrow for a longer time period such that herbicides only need to be applied 2–3 times during a growth cycle depending on the crop. According to the "Deckungsbeitragskatalog"¹

¹Lists costs and time effort for each task during a growth cycle of the plant, released by the German government

(see appendix C.3), costs for this method account to 350 €/ha for an entire growth cycle of sugar beets.



Figure 2.2: Herbicide spraying module attached to a tractor

2.3.2 Precise Herbicide Spraying

In recent years, precise herbicide spraying methods have emerged. Precise herbicide spraying means that herbicide is only sprayed on the weed and not everywhere. This allows to reduce the usage of herbicide by up to 90%². This method can be used as a module attached to a tractor. Blue River³ is a manufacturer of such a precise spraying module (see fig. 2.3(a)). Startups like Ecorobotix are offering this method also in combination with an autonomous robot (see fig. 2.3(b)).



(a) Blue River



(b) Ecorobotix

Figure 2.3: Two different methods for precise herbicide spraying

2.3.3 Mechanical Weeding Methods

The two presented methods rely on the use of herbicides. Since organically grown crops cannot be treated with herbicides, there are herbicide-free approaches. The mechanical approach of chopping with goose feet is widely used. Little plows are pulled through the soil in a depth of approximately 2–5 cm. This way, they cut off the weed which then dries out. This approach is available as a module attachable to a tractor shown in fig. 2.4(a). In recent years, this technology has also been implemented on autonomous field robots e.g. by *Naïo Technologies*⁴ as shown in fig. 2.4(b). The mechanical methods are not able to remove the weed completely. The section in the row (blue section in fig. 3.30) still remains untouched since the

² according to www.ecorobotix.com

³ www.bluerivertechnology.com

⁴ www.naio-technologies.com/

goose feet cannot till the ground in the row without damaging the crops. Thus, hand labour is required which makes these methods expensive. According to the "Deckungsbeitragkatalog" (see appendix C.3), the costs with this method attached to a tractor account to 2000 €/ha for an entire growth cycle of sugar beets.



Figure 2.4: Two different methods for mechanical weeding

2.4 Problem Statement

2.4.1 Herbicide Overuse

Even though spraying herbicide is the most effective and cheapest method, it comes with a downside. These toxic chemicals end up in rivers, lakes and ground water. In Switzerland according to a report from BAFU [7], 50% of the measurements have contained herbicide or metabolite products of herbicides. In areas with dense agriculture, even 90% of the measurements contained herbicide or its metabolic products. Furthermore, Austria has banned the most used herbicide named Glyphosat by the beginning of 2020. Considering the political situation with the two political initiatives⁵ coming in fall 2020 it is likely that also Switzerland will ban herbicides sooner than later. These facts highlight the need for an alternative to herbicide.

2.4.2 The Dilemma of the Farmer

Farmers are in a dilemma. They can either use cheap and effective herbicides, which contaminate the fresh water resources, or they can rely on mechanical approaches requiring expensive hand labour for in row weeding.

The mechanical approach performs the weeding organically. Provided the previous steps were also organic, the sugar beet plant can be sold as an organically grown plant and a higher price results. However, the increased price does not make up for the additional labour costs. As a result, from a financial point of view, there is no motivation for the farmer to grow sugar beet organically. To sum up, besides the environmental damage herbicides cause, there is no point for the farmer to refrain from the use of herbicides.

2.5 Task and Project Objectives

Rowesys has identified this dilemma of the farmers and wants to contribute to an alternative to herbicides. The alternative should perform a sustainable weeding without the need for herbicides or intensive hand labour. With such a solution it

⁵www.initiative-sauberes-trinkwasser.ch and www.lebenstattgift.ch/

would also become financially attractive to grow sugar beets organic, since no large costs for hand labour nullify the higher selling price.

Based on the farmers dilemma, Rowesys developed an autonomous weeding robot for sugar beets which removes the weed mechanically. No use of herbicide is required and autonomy ensures no additional labour is needed.

2.6 Concept

2.6.1 Concept Development

In this section, two core decisions which affect the requirements will be explained. These are namely the decision concerning the selection of the test plant as well as the number of rows the robot covers at a time. Furthermore, the research and development focus will be discussed. Afterwards the requirements for the system are presented.

Test Plant

To be able to start developing the robot, it had to be decided which crop or which type of crops would be mainly considered as the mechanical design and software of the robot depends on the choice of the plant. The mechanical design is influenced by the selection of the plant since the row distance can vary from plant to plant. Therefore, choosing a plant implies the track width of the robot. The navigation software also depends on the chosen plant since a different plant has a different colour and shape. Thus, the navigation and crop detection software must be adapted to the phenotype of the plant.

The first criterion for the selection of the plant were the possibilities for testing, namely infrastructure and growth cycle. As testing at *ETH Crop Sciences* in Eschikon was the most convenient option, it meant that likely a crop that could be grown on their fields would be chosen. Also, it was important to choose a crop that is in the right growth stage for testing in the months of March, April and May, in which it was planned to test the robot on the field.

The second criterion was the value the robot adds to the farmer when used in the chosen crop type. Sugar beet, depicted in fig. 2.5(a) and fig. 2.5(b), requires a lot of manual labor by organic farmers: typically, 150 hours of manual labor are needed per hectare for one crop cycle according to the "Deckungsbeitragskatalog" attached in appendix C.3. Whereas lettuce only requires 40 – 80 hours of hand labour. This is owing to the slow growth of sugar beet in its early stages, which makes it prone to being out-competed by faster growing weeds.



(a) Fully grown



(b) Early stage

Figure 2.5: Sugar beet

Since sugar beets grow during the testing phase of the project from March to April, can be grown at *ETH Crop Sciences* and an autonomous weeding robot adds value to the farmer, it was decided to choose sugar beet as a test plant.

Another advantage of sugar beet is that its leaves have quite a distinct shape, which is beneficial for plant recognition. Furthermore, it is planted in rows, meaning that a robot which works in a sugar beet field might also be used in other row crop fields when a few changes are applied.

Number of Rows

Since sugar beet grows in rows with a row distance of 50 cm, the team had to decide on how many rows the robot covers at a time. Based on the following arguments, it was decided that the robot drives over one row at a time.

- Covering one row means the robot would still fit through doors wider than 100 cm. This enables the team to have the robot in their office which simplifies the logistics and speeds up the development process since everything can happen at one place.
- A robot with a width smaller than 100 cm fits in a van. This makes transportation easier since no special driving license is required for this type of vehicle. Furthermore, the team could use the van from RSL to transport the robot. Thus, no special transportation vehicle or trailer needs to be organised.
- The robot is considered to be a research platform. Therefore, it is sufficient to cover one row for a proof of concept. Furthermore, costs are also reduced since components like the weeding tool only need to be implemented for one row. An extension to multiple rows can still be done later once the concept has proven to work for one row.

Research and Development Focus

As developing a complete weeding robot and implementing innovative changes to all functions and features would be out of scope for this project, the team decided to lay its research and development focus on locomotion and navigation.

Locomotion

As mentioned in section 3.2, current weeding robots are lacking reliable locomotion concepts to operate in agricultural terrain. Therefore, the team decided to focus on a locomotion concept that allows reliable operation on agricultural terrain. Reliable operation is specified in section 2.6.2.

Navigation

The second focus is set on navigation, which in the case of Rowesys means following a sugar beet row as well as entering the next row. The robot should be able to perform this tasks without the use of a previously made GPS map from seeding and without the use of a GPS RTK system. Creating the seeding map requires additional equipment during the seeding and using the GPS RTK necessitates according antennas in close proximity. Current agricultural machinery and robots usually rely on such GPS maps. The team decided to develop a solution which is capable to manage the navigation only based on vision without a premade GPS map or GPS RTK system. This can facilitate universal use of autonomous weeding robots. The robot is easier to be used since no map needs to be prepared and no additional infrastructure is needed.

Weeding

The team considered the time available as too little to also do extensive research and development on the weeding tool besides locomotion and navigation. Therefore, it was decided to implement existing tools to save time.

2.6.2 System Requirement Specifications

For Rowsys, the solution to the problem of herbicide usage is a sustainable weeding robot, that protects the crops and makes the farmer's life easier.

The requirements deduced from this, are stated below.

1. Sustainable

According to the vision (see chapter 2), the team decided that the system and also the project should be sustainable.

1.1. Herbicide free weeding

1.1.1. Alternative weeding tool

The complete field should be weeded by a tool that does not depend on chemical weed killers.

1.2. Ecological power source

The system must be powered by a sustainable, ecological power source.

2. Protect the Crops

Damaging the leaves or the root of the crops can compromise their growth and decrease the yield. Therefore, no crops must be harmed by the robot.

2.1. No soil compression

Soil compression occurs when heavy machinery drives over the field. Muddy or even moist soil enhances this effect. Compressed soil obstructs the flow of nutrients to the roots of the crops. The pressure on the soil must not exceed 49 kPa (See appendix B.9.1).

2.2. No impact on leaves

Leaves can be damaged upon impact and impair the crops growth.

2.2.1. Clearance

The system has to provide clearance for sugar beet plants up to the 6 leaf stadium. This corresponds to a volume of 20 cm width and 25 cm height along the row of crops (see fig. 2.6), where no parts of the robot, except for the weeding tools, must be placed. If the sugar beet crops are larger than this, they are strong enough to compete with weeds and weeding is not needed anymore.

2.2.2. Robust row following

The autonomous row following has to be precise enough, to prevent the robot from accidentally driving over the crops or damaging them with the weeding tools. Less than 1/100 crops must be damaged.

3. No Extra Labour for Farmers

Applying pesticides is simple and cheap but most alternative weeding solutions imply additional work for the farmers. Rowsys does not want to add extra labour for the farmers.

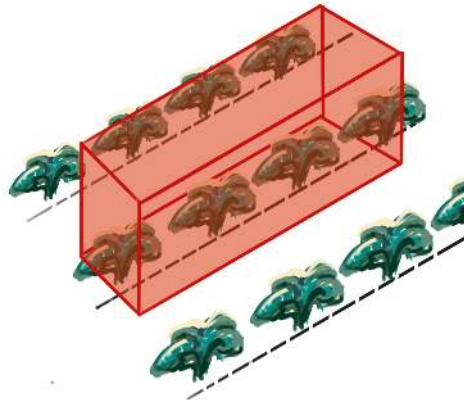


Figure 2.6: Clearance zone (red) around the sugar beet plants

3.1. Autonomy

The system has to operate as autonomously as possible and require as few user inputs as possible.

3.1.1. End of row turn

The system has to identify the end of a row, turn around and enter the next row without the help of manual inputs.

3.1.2. Robust Locomotion

The locomotion concept must enable maneuvering on rough agricultural soil without the robot getting stuck. The propulsion system has to be strong enough to allow driving up hills of a slope of up to 15° . Furthermore, traction even on moist soil has to be ensured.

3.2. Surface power of 1 ha/day

As agreed with farmers and supporters from agriculture, the surface power of 1 ha/day is reasonable to be considered for agricultural reasons. If the surface power is lower, one system could not tackle the further growing weeds fast enough on an average sized field in Switzerland.

3.2.1. Speed

By driving over only one row, the speed of the system required to tackle 1 ha/day is 0.2 m/s . The robot should be able to drive up to 3.0 m/s when driving on the road.

3.3. No additional infrastructure

The farmer should not have to be dependent on additional, expensive infrastructure.

3.3.1. No GPS RTK

To make use of GPS RTK, at least two antennas in close proximity (20 km) are needed. The system should not rely on the presence of these antennas to work on any agricultural field.

3.3.2. No seeding data

Many weeding robots depend on the GPS data collected during the seeding process. The system should not rely on those kind of information and seeding machines with according software.

4. Realizable within Focus Project

4.1. Transportation

4.1.1. Fit through doors

The robot has to fit through the doors of the team's workplace (≤ 100 cm) in order to transport the robot without needing to dismount parts.

4.2. Testing

4.2.1. Energy Storage

To provide a reasonable testing time without needing to refuel energy, the energy storage of the robot should provide a run time of at least 5 h.

5. Safety

5.1. User's safety

The user's safety is first priority when it comes to operating the robot. Different emergency switches must provide the possibility to turn off the system immediately when triggered.

5.1.1. Remote control

The system must be remotely controllable from a distance of at least 15 m.

5.1.2. Remote STO

A remote safety torque off must allow the operator to turn off the power for all the motors from a distance of at least 15 m.

5.1.3. Emergency switches on robot

Additional emergency switches on the robot must enable turning off the system completely. They have to be in arm's reach from every side of the robot.

5.1.4. Braking

In order to ensure safe operation, the robot must be able to brake and come to a full stop within 0.5 m from maximal speed of 3 m/s.

5.2. Commercial safety

5.2.1. Not considered

Safety features which would allow a commercial use of the robot are not considered since commercial use is not feasible in the scope of this project.

5.3. System safety

The system must be protected during tests under agricultural circumstances. This includes light rain (2.5 mm/h), dust and sunlight.

5.3.1. Basic water and dust protection of electrical components

Drives, controllers, computers and cabling should be protected by IP54 standard.

5.3.2. Casing protection from sunlight and impacts

A casing should protect the electrical components from sunlight and impacts from people's movement or falling tools.

2.6.3 The Weeding Robot

After the decisions concerning the main concept of the robot were taken and the requirements were listed, the foundation for the mechanical development is laid. Before getting into the details of the mechanical design, this section gives an overview

of the robot. This helps the reader to better understand the following chapters. The robot was named *Rosie* by the team and its name will also be used in this report. Below, it is mentioned in which chapters the respective subsystems are discussed. The following numbers in parentheses correspond to the subsystem in fig. 2.7. The platform (1), the suspension (2), the wheels (3), the steering (4), the weeding tools (5) and the casing (6) are part of the mechanical design chapter 3, where first a concept for the respective subsystem is elaborated and then the mechanical engineering is discussed. The electronics box (7) and the batteries (8) are part of the electrical design chapter 4. The first section of the software chapter 5 is about the system status. Afterwards, the communication and the control of the motors are discussed in the sections actuator level control section 5.1 and actuator communication section 5.2. After that, the simulation of the robot is explained in section 5.3. The software chapter continues with the steering controllers in section 5.4, followed by the manual navigation in section 5.5. The last four sections in the software chapter are about the state machine and the planner in section 5.6, the state estimator in section 5.8, the row following in section 5.9 and the crop detection in section 5.10. For a better understanding of the following chapters, the robot's coordinate frame is introduced in fig. 2.7. The x-axis points in the driving direction of the robot and the z-axis points upwards.

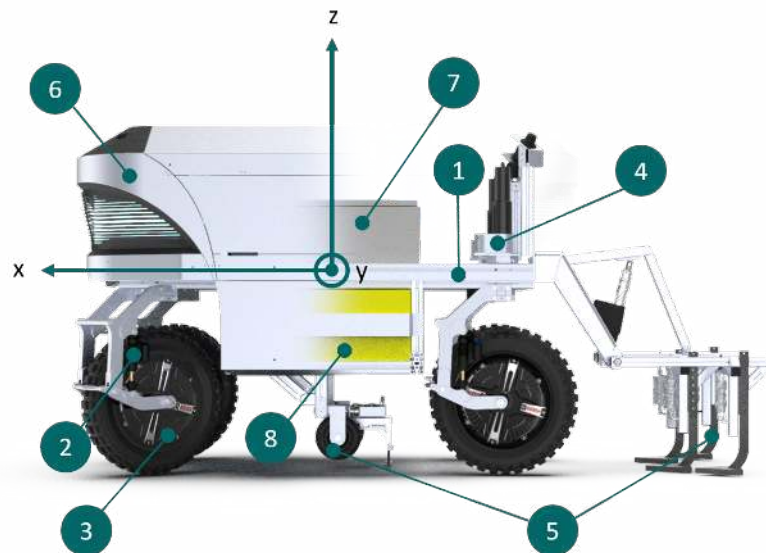


Figure 2.7: Overview of Rosie - the weeding robot

2.7 Project Management

2.7.1 Team and Support

Focus project Rowesys consists of 10 bachelor students: 6 mechanical engineering students (ME) of ETH Zurich, two electrical engineering students (EE) of ETH Zurich and two industrial designers (ID) from FHNW in Basel. The project was performed at the Robotic Systems Lab (RSL) at ETH Zurich and is supervised by Prof. Dr. Marco Hutter. The team was coached and supported by Dhionis Sako, Giorgio Valsecchi, Giuseppe Rizzi, Lennart Nachtigall and Malin Siegwart. For

further assistance concerning technical matters, Rowesys was additionally supported by other members of the RSL. From the *ETH Crop Sciences* in Eschikon, the team has received support concerning agricultural matters including a sugar beet test field. The students are presented in fig. 2.8.



Figure 2.8: The Rowesys student team. L.t.r.: Pascal Lieberherr (ME), Laurin Baumann (ID), Markus Wagner (ME), Anna Bossard (ME), Nico Burger (ME), Silvan Häsli (ID), Timo Schöneegg (EE), Andrea Cavelti (EE), Manuel Knecht (ME), Gian Erni (ME).

2.7.2 Organisation

In order to allow people to focus on specific tasks and subsystems of the project, the team was split up into subgroups. Due to the different nature of the two semesters of project Rowesys, each semester had its own organizational structure tailored to the tasks involved during the corresponding semester. The different sub-teams during the first semester can be found in fig. 2.9.

In the first semester, the mechanical engineering students were all working on the hardware while the electrical engineering students designed the electrical part of the robot. In the second semester, two mechanical engineering students kept working on the hardware while all the other mechanical and electrical engineering students were developing the software. The sub-teams of the second semester can be found in fig. 2.10.

Besides that, each team member was assigned to a certain sub-team, he or she was also assigned a certain administrative task or lead function. The different functions and the responsible persons are listed below.

Project Management The project manager coordinates different sub-teams and keeps an eye on the overall project. He ensures smooth communication between all members of the group to create consensus at all times over the course of the project. He plans milestones and functions as the timekeeper and evaluates whether there is a need for change on the prototype. Furthermore, he identifies needed resources

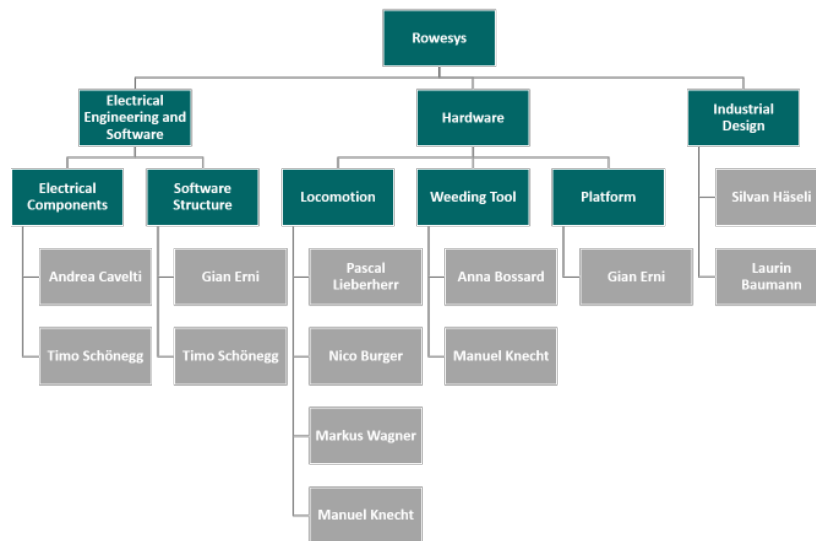


Figure 2.9: Organigram based on the different sub-teams in the first semester

such that the project can meet the goals that have been set. (Gian Erni and Pascal Lieberherr)

System Engineering His role is to have an overview of the status of the system design, to look at risks and to manage problems ahead of time. He ensures that the interfaces between the subsystems are well thought-out and controls the work done. He is responsible for every part that goes into production. (Pascal Lieberherr)

CAD Superuser The CAD superuser helps team members when CAD problems arise. He also manages a reasonable structure in *TeamCenter*. (Manuel Knecht)

Electrical Lead He is responsible for the electrical components of the robot. The electrical lead manages the cabling of the batteries, the sensors, the motors and is responsible for an efficient energy supply. (Andrea Cavelti)

Software Lead The software lead is responsible for the whole software of the robot. The task includes keeping the code well structured and stable. (Timo Schöneegg)

Sponsoring The person responsible for sponsoring initiates contact with potential sponsors and is responsible for maintaining communication as well as setting up contracts with confirmed sponsors. (Nicola Burger)

Finances His task is to make a budget plan and control the expenses accordingly. He is also responsible for orders and payments. (Markus Wagner)

Manufacturing Two people keep an overview over all of the producers and know what is in production. They are also responsible for material and storage. (Pascal Lieberherr, Markus Wagner)

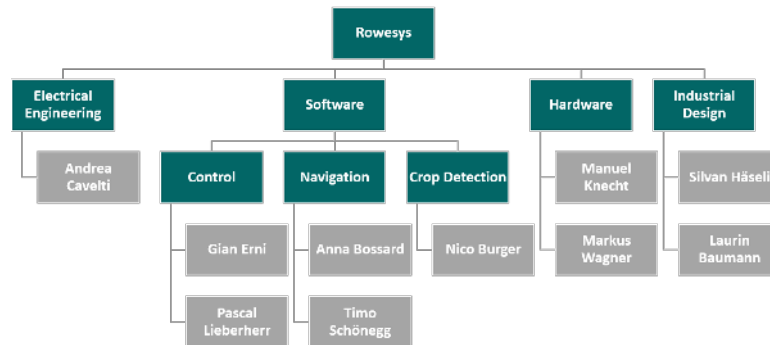


Figure 2.10: Organigram based on the different sub-teams in the second semester

Social Media Presence The social media manager maintains the Rowsys webpage and regularly posts on the Instagram account. (Silvan Häseli, Laurin Baumann)

Corporate Identity The two industrial designers take care of tasks that fall under the scope of corporate identity, such as designing the logo, setting up and designing the webpage, create flyers and presentation templates. (Laurin Baumann and Silvan Häseli)

2.7.3 Time Management

The project started in September 2019 and lasted nine months. These nine months were subdivided into two major parts by the semester break in December. Since building an entire robot from scratch requires to start with the body of the system, the first half of the project was strongly focused on the hardware development of the system. Doing so resulted in the opportunity to produce most parts during the break and enabled to start the assembly and to test after the break in February. After that, the second half of the project was highly focused on software development and testing.

Since, the hardware was not a fast prototyping product the first half of the project needed longer iterations than the software development phase. This led to the decision to subdivide the first semester into three milestones, as can be seen in table 2.1.

The goal at the end of the third milestone was to send all parts into production. During the first semester, the time plan that had been drawn up at the beginning of the project proved to be too optimistic. This is why some features of the robot were not designed in the first semester. They were either designed in the second semester or outsourced to Bachelor Theses. The best example for this is the active weeding tool, which first was intended to be produced and implemented in the project but had to be left aside since the time available was not sufficient to properly design it in the first semester and implement it in the second semester of the project.

After the break, the focus was switched to software development and testing, which allowed an agile time plan with shorter iterations. The team used a Scrum framework, making sprints of the duration of three weeks. Each sprint consisted of

software and hardware development followed by testing of those new components. The sprints can be subdivided into three main types:

1. *Unit tests* are checks of all the single components that are implemented on Rosie separately and controlling whether they work properly.
2. *Integration tests* are performed when grouping a few components together and verifying whether they interact as planned.
3. *System tests* are meant to make sure that the system is working as a whole and that all the components interact together the way they are meant to.

	Milestone	Description	Deadline
1 st Semester	1	System Requirement	20.10.2019
	2	Concept milestone	17.11.2019
	3	Production milestone	18.12.2019
2 nd Semester	4	Unit Test Milestone	13.03.2020
	5	Subsystem MVP Milestone	17.04.2020
	6	Integration Test Milestone	08.05.2020
	7	Global MVP Milestone	29.05.2020

Table 2.1: Milestone list throughout the whole project

2.7.4 Sponsoring and Finances

Project Rowsys has been supported with technical knowledge and infrastructure from ETH Zurich and financed by sponsors from the private sector. The team could not only profit from financial and material support but also from the know-how of industrial partners.

Sponsoring

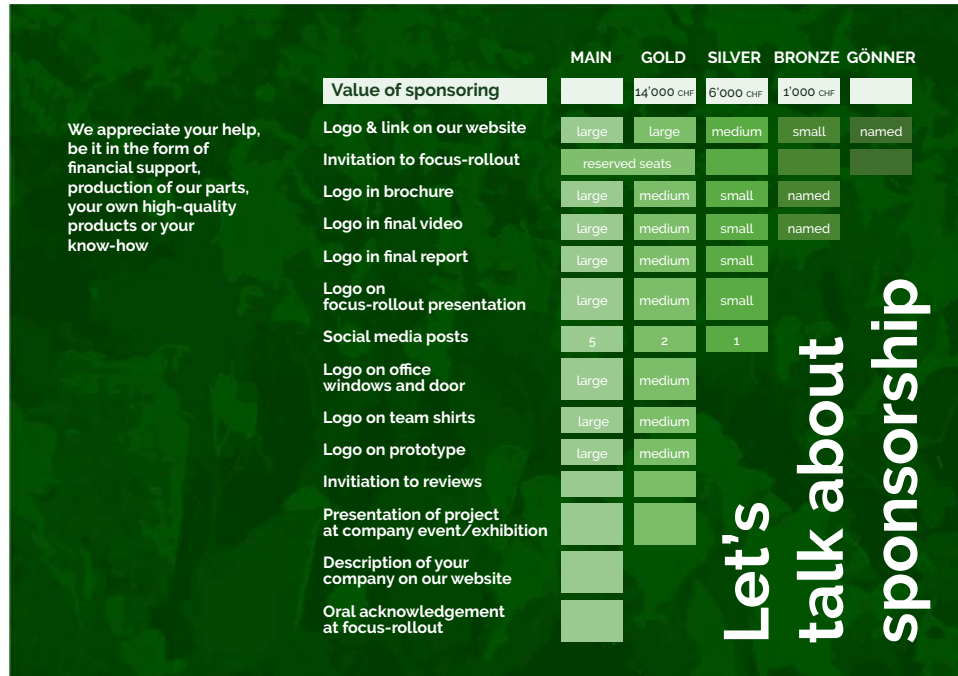
In order to make a possible sponsorship attractive to companies, different options for possible compensations were listed and finally a sponsoring brochure with different sponsoring categories was created.

For the designation of different sponsorship categories, the team oriented itself on experience from previous focus projects. In order to ensure a certain exclusivity in the respective categories, the number of sponsors was limited for the top three categories (see table 2.2). This was based on the budget planning: with the contributions of the first three completely filled sponsoring categories, the expected expenses could be covered.

Sponsoring category	As of (CHF)	Max. number of sponsorships
Main Sponsor	30000	1
Gold Sponsor	14000	4
Silver Sponsor	6000	8
Bronze Sponsor	1000	unlimited
Patron	> 0	unlimited

Table 2.2: Sponsoring Categories

With the sponsoring package, each sponsorship category is offered respective compensation. An arrangement of these benefits with respect to each category can be seen in fig. 2.11.



The flyer features a green background with a grid of sponsorship benefits. On the left, a text box reads: "We appreciate your help, be it in the form of financial support, production of our parts, your own high-quality products or your know-how". On the right, large white text says "Let's talk about sponsorship". The table below details the benefits for five categories: MAIN, GOLD, SILVER, BRONZE, and GÖNNER.

	MAIN	GOLD	SILVER	BRONZE	GÖNNER
Value of sponsoring		14'000 CHF	6'000 CHF	1'000 CHF	
Logo & link on our website	large	large	medium	small	named
Invitation to focus-rollout	reserved seats				
Logo in brochure	large	medium	small	named	
Logo in final video	large	medium	small	named	
Logo in final report	large	medium	small		
Logo on focus-rollout presentation	large	medium	small		
Social media posts	5	2	1		
Logo on office windows and door	large	medium			
Logo on team shirts	large	medium			
Logo on prototype	large	medium			
Invitation to reviews					
Presentation of project at company event/exhibition					
Description of your company on our website					
Oral acknowledgement at focus-rollout					

Figure 2.11: Flyer of the Sponsoring Packages

Budget

Project Rowsys was able to raise a total of 64'000 CHF. Most of the budget was used for the electronics and the casing. This left the project with a surplus of approximately 10'000 CHF which will be used for future works on the robot as discussed in section 6.2. A detailed partitioning of the expenses may be found in fig. 2.12. Furthermore, the project received sponsored parts worth about 69'000 CHF such that the material value of the robot sums up to approximately 123'000 CHF.

The team agreed upon the rule that every expense exceeding 100 CHF has to be discussed and permitted by the team. This value has then been raised to 500 CHF due to the large number of orderings and the amount of permits needed. Additionally, every team member had access to the order list, which kept track of every expense, the sponsored material and all the delivery dates of the project.

2.7.5 Infrastructure

Until the lockdown due to Covid-19, Rowsys was working in offices located at ETH Zentrum. During the lockdown all team members were working remotely from their home offices. After the lockdown, testing was done at the *ETH Crop Sciences* in Eschikon where the team had the opportunity to leave the robot at the station and use the test fields.

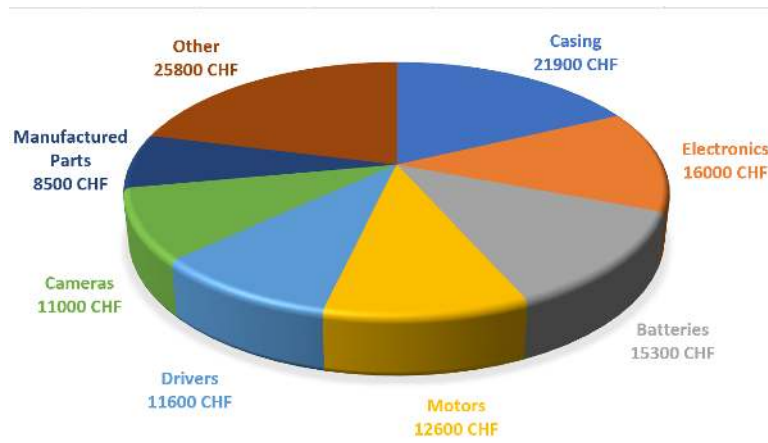


Figure 2.12: The Final Expenses

Chapter 3

Mechanical Design

The complete robot is shown in fig. 3.1. The mechanical features are presented in the following chapters. The platform (1) in section 3.1, the suspension (2) in section 3.2, the locomotion and propulsion system (3) in section 3.3, the steering mechanism (4) in section 3.4, the weeding tools (5) in section 3.5 and the casing (6) in section 3.6.



Figure 3.1: The Rowsys Robot and its mechanical key features

3.1 Platform

Motivation

The robot consists of multiple subsystems that need to be connected to one robust body. At the same time, multiple devices inside the aforementioned body need to be accessible and easily movable. The platform was conceived with the goal of giving a stable basis, where all the components could be mounted on, while keeping a high degree of variability.

Overview

The platform consists of aluminum profiles manufactured by the company *Item*. These pipes are connected by a flexible linkage system. The dimensions of the platform resulted to be 1300 mm in length, 900 mm in width and 600 mm in height. As the platform represents the core of the system, it had to be fully designed early on and sent to production over the winter break in order to enable starting to test the different subsystems as soon as possible. The history of the different platform designs is shown in fig. 3.2.



Figure 3.2: History of the platform

Requirements

1. Robustness

The platform has to withstand dynamic forces that are created by the system itself, which moves on potentially very uneven ground. Bending of the platform might result in different orientations and positions of the sensors and tools compromising their functionality, which is why the platform needs to be stiff enough to ensure viable and consistent sensor data.

2. Suitable for transport

Rosie has to be moved to and from the fields and to different buildings while testing. According to the SRS (see section 2.6.2), the size of the robot should allow easy management, and the entire system needs to fit into a transportation vehicle with dimensions of 1250 mmx1250 mmx2300 mm. It also has to pass through doors without risking damage on the system or the environment. These constraints allow the robot to have a maximal width of 985 mm and a maximal height of 1250 mm.

3. Ground clearance

While the robot is weeding, Rosie moves on the field and over crops. It is essential that no part of the robot touches the crops because this would risk damaging them. Hence, a minimum clearance in height from the soil of 250 mm was defined to ensure enough distance to the crops even in later growth stages.

4. Modularity

The final product is a first prototype of a weeding robot, which is why multiple modifications and hardware updates will be performed on the platform after validation. The platform has to provide the possibility of adapting during development without needing to redesign the entire base.

5. Weight

The total weight of the robot should not exceed 300 kg in order to not cause soil compression. The platform offers the possibility to optimize weight through geometry and design.

Research

Width Changing mechanism

The robot is intended to be used for sugar beets, which are row crops planted in rows with a distance of 50 cm between each other. Since different crops are normally planted with different row spacings, it was considered making the width of the platform variable to allow for using the robot for other plants in future projects as well. However, since the fields, which are treated by Rosie, have a constant row width and since the focus of this project was set on one crop, the ability to vary the width of the robot would not result in any advantages and would only be an additional feature.

Platform Shape

There were three possible platform shapes that were evaluated, taking the aforementioned platform's requirements into consideration, these shapes are shown in fig. 3.3:

1. The idea of a simple plate as base was rejected because it did not give enough room for components and it would be harder to attach components spatially separated in the third dimension. Furthermore, a flat surface would need a certain thickness to be able to absorb all the forces resulting in significant additional weight (number 1 in fig. 3.3)
2. Non-orthogonal truss structures can absorb large forces while being relatively lightweight. Because these truss structures are made of angles different from 90°, the implementation of multiple levels faces more challenges. Furthermore, the modularity is decreased since single components of the structure cannot be moved as required and the ability of devices is compromised due to cross struts (number 2 in fig. 3.3).
3. A quadratic profile structure of only right angles, is very flexible in regards of modularity and allows for full accessibility. Because of the high second moment of inertia with respect to the weight of its profiles, the structure can also be quite robust with little diameters allowing to optimize the total weight of the robot without running the risk of not fulfilling the stiffness requirements (number 3 in fig. 3.3).

Profile structure specifications

Based on the requirements of the platform, two different types of profile structures were compared:

1. Welded quadratic pipes: using common quadratic pipes for the platform allows for a lightweight, and relatively cheap design. By welding the pipes at the joints a stable and stiff structure that withstands high forces would be ensured. The main downside of this type of platform is that welding results

in a definitive shape, thus making further development and changes more challenging.

2. Standard profiles: those are tube like rods that can be attached to each other with different types of flexible, screw based junctions allowing a high degree of modularity even after production.

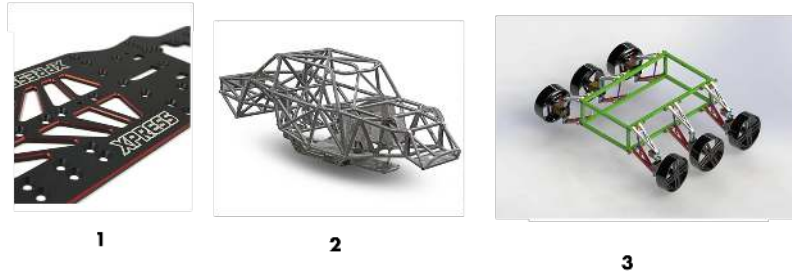


Figure 3.3: Different platform shapes

At the end of the evaluation, the standard profiles made out of aluminum were chosen because of its remarkable flexibility which provides a huge advantage in comparison to a welded structure. The loss in stiffness and the additional weight were insignificant.

Implementation

The shape of the platform goes hand in hand with the location of the single components and their respective requirements.

The biggest constraints come from the height clearance from the ground of 250 mm, which is needed to drive over the crops without touching them, the row width of 500 mm, given by the sugar beet, and from avoiding the risk of tipping over.

The latter implies finding a trade off between keeping the center of gravity as close to the ground as possible and the height clearance.

Furthermore, all of the components have to be arranged in a way that the weeding camera can be placed in front of the active weeding tool (see section 3.5) and can have the field of view in the size of the row dimensions. Hence, 150 mm in width without any obstruction in the field of view is needed. To keep the center of gravity low, it was decided to put the batteries (section 4.2) - those are some of the heaviest components inside Rosie - as low as possible, which meant putting them on the sides. The final version of the platform can be analyzed in fig. 3.4.

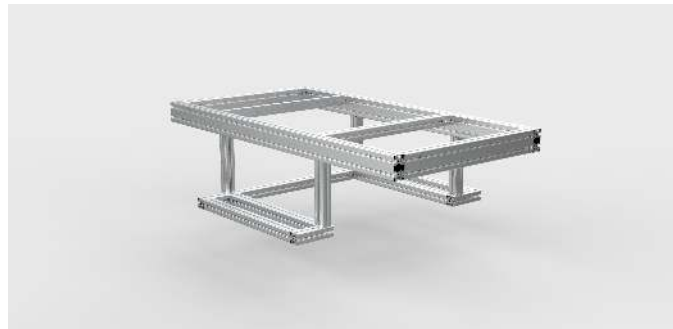


Figure 3.4: Final Platform model

Force Absorption

As already mentioned above, platform is the structure, where all the forces of the robot come together. Those were not only the gravitational forces but also the ones caused by the active and the passive tool as well as by the suspension and the steering subsystem. A FEM analysis has shown that the peak stresses amount to about 50 MPa as can be seen in fig. 3.5. Therefore, the profiles absorb the imposed stresses without any plastic deformation since the yield point of the utilized aluminum is at 250 MPa. The large displacements in fig. 3.5 are for the sake of better visualization and do not show the real deformations.

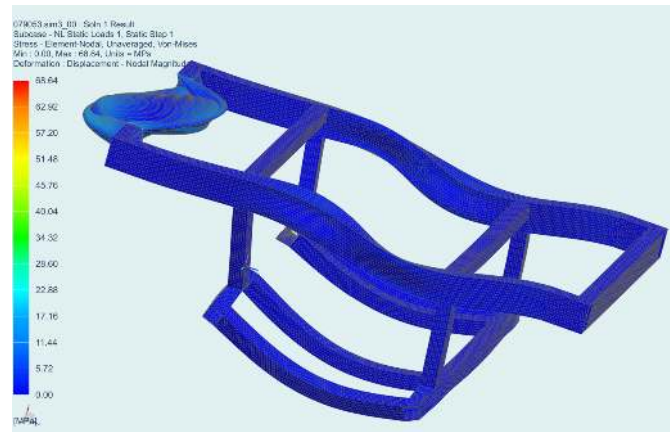


Figure 3.5: FEM Analysis of the platform

Risks

Tipping over

A considerable risk in the design of the platform is the risk of tipping over. In order to be able to iteratively check whether the design is safe throughout the development phase, a mathematical model that represented the robot was made and exposed to different environmental and internal situations. The worst case was calculated to check whether the system would tip over, this calculation can be found in appendix B.2.

The situation used is having the robot on a declined plane, driving perpendicularly to the gradient of the decline and turning with an angular acceleration. Those situations are depicted in fig. 3.6 and fig. 3.7.

By adding some noise to the model, a strong correlation of the height of the center of gravity (COG) with the stability was noticed. Therefore, the effort of lowering the COG as much as possible when designing the robot was strongly aimed at.

If at some point it had not been possible to further lower the COG, there would also have been the possibility to lower the performance of the robot in order to always have a safe robot to operate with.

Review

Working behaviour

The platform worked as intended. No bending was visible and the platform could withstand all the forces when Rosie was completely assembled. The transportability of the system was fully made use of since the circumstances caused by COVID-19 resulted in more changes of location than initially planned. The modularity of the

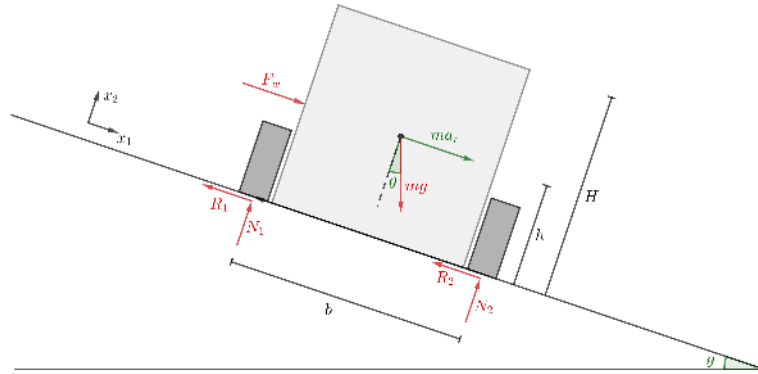


Figure 3.6: The model of the robot's worst case situation for tipping over

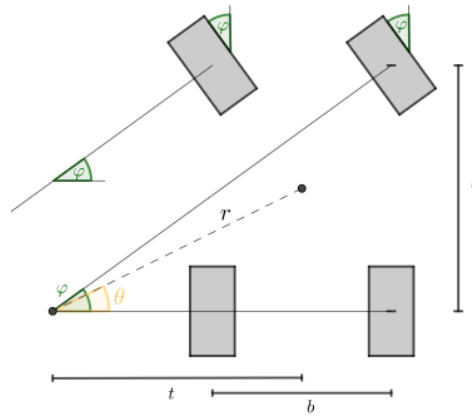


Figure 3.7: Car model that was used to calculate radial acceleration

platform proved to be very useful since multiple adaptations have been made in the second semester. The Aquila weeding camera (section 5.10), for example, was integrated in the system only in the second semester, which led to the need of changing the position of some cross braces, due to the camera's large size. Furthermore, the choice of the platform's profiles allows future additions on hardware side to be integrated easily.

The required profile types and junctions had to be designed in *Item's* own design tool, which required learning a new CAD tool and which made iterative changes more cumbersome.

3.2 Suspension

A suspension is a system that connects the platform of a robot with its wheels while a relative motion between wheel and platform is possible. Usually, this system consists of linkages, springs and dampers. Its goal is to isolate the platform from impacts. In this chapter the reader will be guided through the development process of the suspension.

Motivation of Suspension

Most of the robots used in agriculture have been developed in recent years. Farmers and agricultural machinery retailers mentioned the fact that existing robots can get stuck in agricultural terrain due to stones or holes in the field, which restricts their usability. An example for this is the robot model Dino produced by Naïo Technologies shown in Figure 3.8.



Figure 3.8: Dino from Naïo Technologies

The reason this weeding robot has trouble overcoming obstacles lies in the rigid suspension of its wheels. As soon as a bigger hole or stone is encountered, the robot loses ground contact with one or two wheels. As a result, the robot can not continue driving. The same is true for the weeding robot from *EcoRobotix*¹. The agricultural robotic platform from *Thorvald*² is an example of a robot equipped with a suspension. Having detected the challenge of overcoming obstacles on agricultural terrain, it was decided to equip Rosie with a suspension which allows locomotion in rough agricultural terrain without getting stuck due to holes and stones. Furthermore, the test plant sugar beet as well as many other plants are capable of growing in rough terrain, e.g. in fields with a lot of stones. This makes it even more important to have a good locomotion concept to master challenging agricultural terrain.

Requirements for Suspension

Several requirements for the suspension were specified which are first derived and then summed up in the table 3.1. The requirements can be divided into 3 subsections.

No plant shall be damaged

First of all, the suspension shall not damage the crops while the robot drives over a row. Hence, no part of the suspension is allowed to enter the clearance zone specified in System Requirements Specifications section 2.6.2.

Overcome Obstacles

Secondly, Rosie shall also be capable to overcome obstacles. Overcoming obstacles also means Rosie ensures ground contact with all four wheels while driving over the obstacle. In order to find the maximum size of obstacle Rosie shall be capable to overcome, the team asked farmers and visited sugar beet fields. Based on these investigations, the obstacles were modeled with a step input of +7 cm (stones) and -7 cm (holes). Figure 3.9 shows a common sugar beet field with stones.

¹www.ecorobotix.com

²www.sagarobotics.com/pages/thorvald-platform



Figure 3.9: Sugar beet field of Biohof Trottengarten in the canton of Zürich

Stable Platform

In addition, the goosefoot sweepers should always be in touch with the ground. If the platform tilts to the side around the x-Axis the outer goosefoot sweepers could come out of the ground. As a consequence, the weed would not be cut off anymore. Based on the calculation in appendix B.1 this results in a maximum tilt angle of $\alpha = 3.12^\circ$ of the platform around the x-axis.

For a better overview the table 3.1 sums up the requirements for the suspension.

Requirement	Description	Numerical value
No plant shall be damaged	No part inside clearance zone	width = 20 cm, height = 25 cm
Overcome obstacles	Obstacles modeled as step input	± 7 cm
Stable platform	tilt angle α around x-Axis	$\alpha \leq 3.12^\circ$

Table 3.1: Requirement overview of suspension

Concept of Suspension

The team evaluated five different suspension concepts to account for those requirements. Hereafter, each option is quickly described and the reason the team decided to pursue or discard it is mentioned as well.

Arrangement

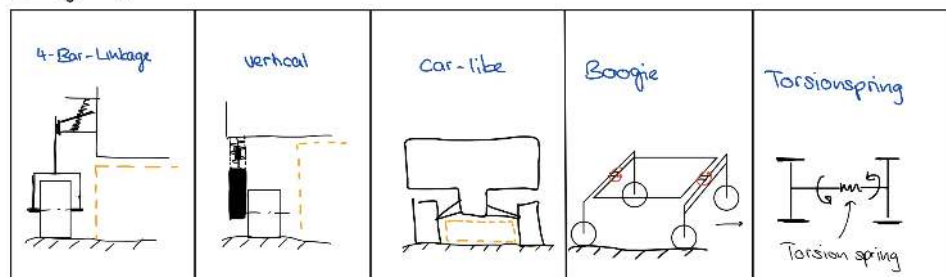


Figure 3.10: Different suspension concepts

4-Bar-Linkage

This arrangement (see figure 3.10, Box "4-Bar-Linkage") fulfills all requirements listed above. There is enough clearance, it should be possible to overcome the specified obstacles and should ensure a stable platform. Nevertheless, the biggest disadvantage is the space needed to design this uplifted-car-like suspension. With a width of our robot of approximately 50 cm, the suspension would use up the complete width. Thus, there would barely be room for different components at the front and at the back of Rosie. This is why the team decided to reject the 4-Bar-Linkage concept.

Vertical

A vertical arrangement as used in bicycles fulfills the requirements listed above. Also in terms of space this arrangement would fit the needs since it leaves enough space on the platform for other components such as batteries or electrical components. Therefore, it was decided to further pursue this option.

Car-Like

In this case the suspension would mainly be located at the inside of the wheel where the clearance zone is, compare to fig. 3.10 box *car-like*. Thus, this option was cancelled.

Boogie

A different approach is the floating platform. The idea is to have the two front and the two back wheels connected and mounted at only one point. This way the front wheels and the back wheels could rotate around the mounting. This option was not further pursued because a stable platform is needed for the goosefoot sweepers. The Requirement "Stable Platform" is not met.

Torsion Spring

The torsion spring concept was dropped because the platform itself would then also have to rotate. As mentioned above, it is required to have a stable platform to put the payload on.

Hence, the team ended up with a vertical arrangement for the suspension. Therefore, different variants for vertical suspensions, which can be found in the figure 3.11, were examined.

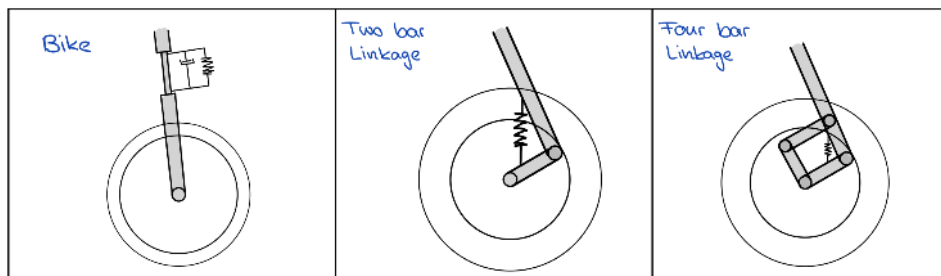


Figure 3.11: Different versions of vertical suspensions

Evaluation of vertical suspensions

The bike fork was one of the favorites in the beginning. It was eventually rejected since off the shelf bike forks did not fit onto the wheels because the latter were too wide and the wheel diameter too small. In general, the team was looking for a

suspension concept that does not require a lot of additional engineering since time was short. Therefore, a rotational vertical spring such as the two bar and four bar linkage as shown in fig. 3.11 came in handy. Because there a rear shock absorber from bikes could be used (evaluated in section 3.2) which means the team only has to design the linkages such that they can handle the load and that that the shock absorber can be mounted. Hence, the options left are the two bar linkage and the four bar linkage. Those two options were chosen to be further pursued.

Suspension Design

In order to have a starting point to design a linkage fork for Rosie, a comparison to the rear shock of a bike was drawn. At that time, the estimated weight of the robot was $m_R = 200\text{kg}$. Based on this the static load per wheel was calculated.

$$m_w = \frac{m_R}{4} = 50\text{ kg} \quad (3.1)$$

For a bike, the static load that applies to the rear shock is approximately $\frac{2}{3}$ of the rider's weight. Assuming the weight of the rider is $m_{Rider} = 80\text{ kg}$, the static load is equal to

$$m_{rearshock} = \frac{2}{3} \cdot m_{Rider} \cong 53\text{ kg} \quad (3.2)$$

Bikes usually have a transmission of $i_{Bike} = 2.2 \dots 3$. This means the static load that acts on the rear shock is

$$F_{rearshock} = i_{Bike} \cdot m_{rearshock} \cdot g \cong 1144\text{N} \dots 1560\text{N} \quad (3.3)$$

The transmission in this case is defined as the ratio of the force acting on the wheel F_{ext} and the force acting on the spring-damper-system F_{SD} .

$$i = \frac{F_{SD}}{F_{ext}} \quad (3.4)$$

By choosing a transmission rate of $i_{robot} = 2.3 \dots 3.2$ the same static loads apply to our spring-damper-systems as it is the case for rear shocks of bikes ($F_{rearshock} = F_{robot}$). Since bike rear shock absorber fit the load case of the robot and the design space available based on the defined clearance zone in section 2.6.2, the team decided to go for bike rear shock absorbers. Furthermore, they are quickly implemented since spring and damper are already mounted.

$$F_{robot} = i_{robot} \cdot m_w \cdot g = 2.3 \dots 3.2 \cdot 50\text{kg} \cdot 9.81 \frac{\text{m}}{\text{s}^2} \cong 1144\text{N} \dots 1560\text{N} \quad (3.5)$$

A transmission rate of $i_{robot} = 3$ was chosen. Any value between 2.3 and 3.2 would have been feasible. After the transmission rate was set, the lengths and pivot points for the two bar linkage suspension needed to be chosen. At that point, another requirement needs to be taken into account which it is outlined in the following section.

Constant, Progressive or Degressive Transmission

When the rear shock is compressed, which means β gets smaller (see 3.12), the transmission can change. It can either be influenced by the selection of the spring constant (constant, progressive, degressive) or by the linkage design of the linkage fork. It was decided to have at least a constant or a progressive transmission. A degressive transmission would easily cause a punch on the platform since less force is required to deflect the rear shock towards the end of the deflection length of the spring. In this chapter the optimal design parameters of the two bar linkage fork are investigated such that at least a constant transmission ratio is achieved.

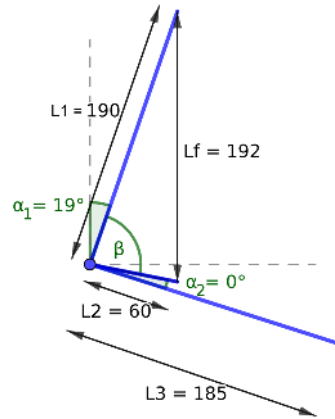


Figure 3.12: Suspension sketch

Figure 3.12 shows the pivot points, the lengths and the angles. α_1 , α_2 , L_1 , L_2 and L_3 are the design parameters of the two bar linkage. Provided they are chosen correctly, at least a constant or progressive transmission should result. With the help of a Geogebra file the most suitable set of parameters was investigated.

Figure 3.13 shows a suitable set of parameters. The vertical red lines show the shock absorber in the fully extended (R_u : $\beta_u = 129.5^\circ$) as well as in the fully compressed condition (R_o : $\beta_o = 57.9^\circ$). Between those two lines one can see the transmission ratio which is apparently constant. Thus, this set ($\alpha_1 = 19^\circ$, $\alpha_2 = 0^\circ$, $L_1 = 190$ mm, $L_2 = 60$ mm and $L_3 = 185$ mm) of parameters meets the specs.



Figure 3.13: Geogebra plot shows the transmission i_{robot} on the y-Axis as a function of β on the x-Axis in radian

Dimensioning of Suspension

Fig. 3.15 shows the mechanical design of the suspension. As it can be seen in the figure, the team decided to go for a two bar linkage. The two bar linkage was sufficient because the only difference between a two and four bar linkage was higher torsion stiffness of the four bar linkage compared to the two bar linkage. Since this stiffness is not required, it was decided to go for a two bar linkage.

As shown in fig. 3.15, rear shocks from *FOX* were only used on one side. With rear shocks on both sides there would not be enough load per rear shock and suitable bike rear shocks which are designed for half the load were not available. Other shock absorber available for half of the load were rejected since they would have required additional effort and redesigning to integrate in the system. In order to take this asymmetries into account, the linkage fork was appropriately designed. In order to choose the bearing size and the cross-section of the linkages a free body diagram was analyzed. Figure 3.14 shows the free body diagram.

F_{ext} : Impact force when the wheel hits an obstacle

M_S : Maximum steering torque

M_M : Maximum torque from the motor according to data sheet

F_{SD} : Force from shock absorber

$F_{ext} = 3 \text{ kN}$, estimated

$M_S = 75 \text{ Nm}$, see chapter "Steering Requirements" 3.4

$M_M = 60 \text{ Nm}$, from motor data sheet

The detailed calculations leading to the resulting bearing force

$$F_{res} = \sqrt{A_y^2 + A_z^2} = 6.5 \text{ kN} \quad (3.6)$$

can be found in the appendix B.8.1.

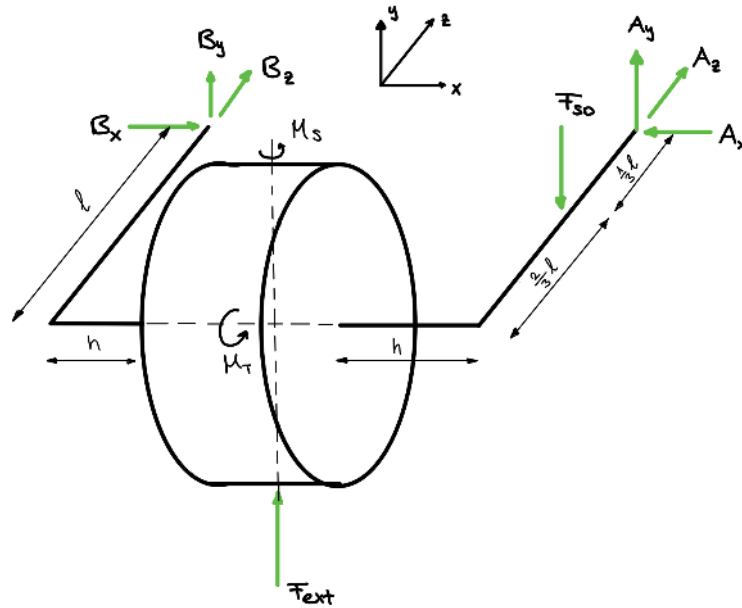


Figure 3.14: Free Body Diagram Suspension



Figure 3.15: Suspension Assembly

Thus, a needle bearing was selected which is capable of taking the radial load, because the resulting radial force F_{res} is smaller than the max. static load of the bearing $C_{0r} = 8300 \text{ N}$. The data sheet of the bearing can be found in the appendix C.2. The outer diameter of the bearing was limited because it was decided to be able to rotate the wheel 360° around the steering axis. If the outer diameter of the bearing was too big, no 360° turning would be possible due to collision with the other wheel. Hence, it was decided to use a needle bearing with an integrated axial ball bearing in order to take some axial loads that might occur due to the asymmetric design of the suspension. An explosion view of the bearing concept is shown in fig. 3.16.



Figure 3.16: Explosion view of the bearing concept

Spring and Damper selection

As described above, rear shocks from bikes shall be used. The team decided to use the DHX2 from *FOX*, see Figure 3.17. The damper constant can be adjusted. Hence, the best setup for the damper will be found on the field. The spring constant was chosen such that a sag of $\Delta x_{sag} = 7\text{ cm}$ is reached. With $m_w = 50\text{ kg}$ and $i_{robot} = 3$ the force acting on the spring when the robot is put on the ground, is $F_{spring} = i_{robot} \cdot m_w \cdot g = 1471.5\text{ N}$. Thus,

$$k_{spring} = \frac{F_{spring}}{\frac{\Delta x_{sag}}{i_{robot}}} = \frac{1471.5\text{ N}}{\frac{0.07\text{ m}}{3}} = 63064\text{ N/m} \quad (3.7)$$

A spring with a constant of 62500 N/m was ordered. Since it was the one closest to 63064 N/m available from *FOX*.

The chosen *FOX* shock absorber has a stroke of 5.33 cm . With $i_{robot} = 3$ this results in a total stroke of the suspension of $5.33\text{ cm} \cdot 3 = 16\text{ cm}$. With a sag of 7 cm this results in 9 cm available for compression and 7 cm available for extension.



Figure 3.17: Rear Shock Fox DHX2

Testing and Review of Suspension

After the robot was assembled, the weight of the robot was measured. The measured weight is $m_{robot} = 300\text{ kg}$. The calculation for the suspension was done for an estimated weight of the robot of 200 kg . Due to the difference between the measured weight and estimated weight, the suspension did not behave as expected. Namely,

- the sag was not as expected 7 cm and
- the robot tilted towards one side (see fig. 3.18(b)).

The robot tilted to the side because the center of gravity was not perfectly in the middle of the robot. This led to a compression of the springs on one side and an extension on the other side. The originally chosen spring constants were too weak to compensate for this.

Therefore, the spring constant was recalculated with the measured weight of 300 kg . This led to a required $k_{spring} = 94600\text{ N/m}$. The next bigger spring available from *FOX* was chosen which is $k_{spring\text{chosen}} = 116000\text{ N/m}$. After installing this spring

- the sag was as expected 7 cm and
- the robot did not tilt towards one side anymore.

Review of Suspension Requirements

The first requirement was *no plant shall be damaged*. Due to the compact design of the suspension no part of the suspension is inside the clearance zone specified in

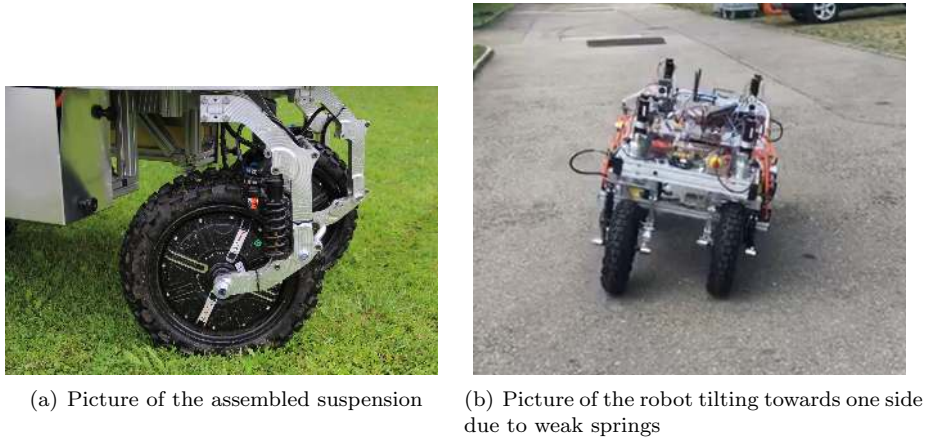


Figure 3.18: Suspension

section 2.6.2 (see fig. 3.19, $27\text{ cm} > 20\text{ cm}$). Thus, this requirement is fulfilled. It was possible to overcome obstacles of $+7\text{ cm}$ and -7 cm while ensuring ground contact with all four wheels (see fig. 3.19). Therefore, the requirement *overcome obstacles* is fulfilled. The third requirement *stable platform* could not be achieved when the robot is overcoming obstacles since the platform tilted towards one side for more than 10° (see fig. 3.19). When the robot was driving through agricultural terrain all four goosefoot sweepers were in touch with the ground which means the platform did not tilt more than 3.12° . Hence, the requirement could only be fulfilled if there were no obstacles.



Figure 3.19: Rosie on an obstacle with a height of 7 cm

3.3 Locomotion and Propulsion

Motivation behind Locomotion and Propulsion

Since agricultural terrain is rather challenging for vehicles with its rough surface structure, a robust locomotion is needed. The suspension subsystem ensures that there is the possibility for the robot to move over obstacles, while the required power and torque to overcome those is provided by an appropriate locomotion and propulsion method which will be discussed in the following sections.

3.3.1 Locomotion Type

Overview

The locomotion type is the subsystem which ensures the movement of the robot and goes hand in hand with the suspension described in section 3.2. Together with the suspension, the locomotion type ensures that the robot does not get stuck on rough agricultural terrain. The main question in the locomotion type subsystem was whether to rely on tires or on tracks, which are both commonly used in agriculture. In the end, it was found that tires are a more suitable solution for this robot.

Requirements

The requirements for the locomotion type arise from the demands of the farmers, the constraints imposed by the nature of the fields and the needs of the computer vision subsystem to operate under constant conditions. According to the SRS (see section 2.6.2) those are:

1. Ensure that no crops are harmed while driving.
2. Ensure that the pressure on the soil is less than $p_{soilcompaction} < 49050Pa$ such that no soil compression occurs. Therefore, the minimum required contact area per wheel is $0.015 m^2$.³
3. Little mechanical complexity.
4. Ensure a constant driving speed of $0.2 m/s$ while on the field on humid as well as on uneven ground.
5. Ensure that the power applied by the motors may be transmitted to the ground. The power requirements are discussed in section 3.3.2.

Evaluation

As mentioned above, there were two main concepts considered for the locomotion type; wheels and tracks. Both are widely used in agriculture. A comparison of the main points of the two concepts can be found in table 3.2. The scores are allocated with respect to each other and the weights represent the importance for meeting the requirements outlined in section 2.6.2. For more detail on the evaluation refer to appendix B.9.2.

Criterion	Wheels	Tracks	Weight (1-5)
Surface Area	3/10 ($0.015 m^2$)	9/10 ($0.06 m^2$)	3
Traction	8/10	10/10	5
Little Complexity	10/10	3/10	4
Personal Interest	5/10	8/10	1
Continuous Force Distribution	10/10	2/10	5
Availability	10/10	4/10	2
Total	164	115	

Table 3.2: Track and Wheel Evaluation

³Little soil compression is desirable for the crops to flourish and for the weeding tools to be pulled behind as they only work in non-compact soil. A thorough explanation of soil compaction may be found in the appendix appendix B.9.1.

Soil Compaction

As shown, the most important criterion against tracks and for wheels was the peak forces - and therefore the peak pressures - onto the ground. This is because, as explained in section 2.6.2, the crops cannot flourish and the weeding tools described in section 3.5 cannot be pulled behind the robot in compact soil.

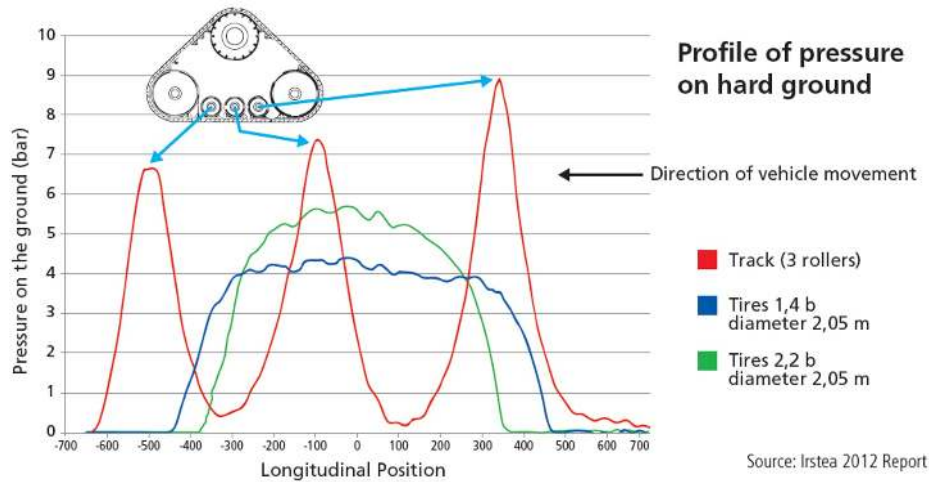


Figure 3.20: Soil Compression of Tracks and Tires

As can be seen in fig. 3.20 the peak pressure under the pulleys of the tracks is more than 10x higher than between them and more than double the pressure of tires at 1.4 bar. Since soil compaction is determined by the maximum pressure applied on the ground, the advantage of a larger surface area with tracks is weakened. The contact area of the tires, on the other hand, increases when they are deflated.

Complexity

Furthermore, tracks are much more complex than tires since they consist of at least 6 individually moving parts⁴, and wheels are made out of only one piece, which rotates around its center.

Availability

Moreover, tires are broadly available since they are used in almost every vehicle. Tracks, which are suitable for the requirements and size of the robot, on the other hand, are hard to find as they are rarely produced and implemented in only a limited amount of machines.

Conclusion

Hence, tires were chosen as the locomotion type since they are not only a better solution for matching the critical requirement of soil compaction but also because they are less complex and more available.

Design and Review

Since the locomotion type has strong ties to the propulsion method, the design of both of them will be discussed and reviewed after the next section.

⁴3 main pulleys and at least 3 supportive pulleys as suggested by Wong in [9].

3.3.2 Propulsion Type

Overview

Based on the choice of using wheels for the locomotion subsystem, it was decided to use hub motors directly integrated into the rim for the propulsion. Out of the evaluated options, they reduce the complexity to a minimum and lower the center of mass. A picture of the motor eventually implemented can be found in the subsection Locomotion Type Design below in fig. 3.21.

Requirements

The requirements for the propulsion are the following based on the SRS:

1. No interference with the clearance zone around the crops.
2. Provide the possibility of backwards driving for the sake of agility and mobility especially while turning (see section 3.4).
3. Ensure a maximal speed of 3 m/s while moving from or to deployment area.
4. Combinable with other subsystems without major adjustments.⁵
5. Brake and come to a full stop within 0.5 m from maximal speed of 3 m/s.
6. Little mechanical complexity.
7. Keep center of gravity low.
8. Provide torque and power needed as summarized in table 3.3 to pull the weeding tools and to overcome obstacles as defined in section 3.2.

Power	700 W
Torque	121.6 Nm (for 17")
	89.6 Nm (for 12")
	76.8 Nm (for 10")
	64 Nm (for 8")

Table 3.3: Motor Requirements

The maximum diameter for the wheels is 43 cm (17") due to space constraints, and no smaller wheels than 8" were considered because they do not offer enough contact area to prevent soil compaction (see appendix B.9.3 for detailed calculation).

Evaluation

There are several different concepts on how to transmit power and torque from the motor to the wheels. The three most important concepts (locations of motors), which were evaluated, are discussed below and in table 3.4 [10].

The critical requirement for the choice of the propulsion type was whether they interfered with the clearance zone around the crops as defined in section 2.6.2 or not and whether they were combinable without major adjustments with other subsystems such as the steering mechanism which allows for swerve drive (section 3.4). This left the rim motors as the only viable solution even though they did not score best in the other categories mentioned in table 3.4.

⁵This requirement arose because the propulsion method was decided on after most of the other subsystems such as the platform or the steering had already been designed.

Table 3.4: Different Propulsion Concepts

Criterion	Indirect Drive ¹	Direct Drive ²	Rim Drive ³
No Interference with Clearance Zone	✓	✗	✓
Combinable with Other Subsystems	✗	✓	✓
Little Complexity	3/10	8/10	9/10
Availability	10/10	7/10	3/10
Weight	3/10 (10-15 kg)	4/10 (10-12 kg)	5/10 (8-10 kg)
Height of Center of Mass	1/10	7/10	7/10

¹ Indirect drive means that the motor is not directly connected to the wheels but via a cardan shaft, a chain, or gears.

² Direct drive describes a concept where the motors is directly attached to the tires or to the tracks. This excludes motors integrated into the rim of the motor.

³ Rim drives are actuators, which are directly integrated into the rim as shown in fig. 3.21. They are also referred to as hub drives.

Locomotion Type Design

Propulsion Motors

Different hub motors, which are listed in appendix B.9.4, were evaluated, but it was only possible to find one single reputable manufacturer which fulfilled the two requirements listed in table 3.3. The largest motor and therefore the largest wheel with the lowest power and peak torque available was chosen, which still fulfills the aforementioned requirements; the largest motor in order to maximize the contact area to the ground, and the lowest power and peak torque in order to save energy. This is the *Bosch Hub Drive N1S* with 1500 W and 120 Nm, which is depicted in fig. 3.21. Normally, the hub motor is used in electronic scooters, can, however, be implemented in different systems using the drives explained in section 4.6.



Figure 3.21: The Chosen Hub Motor

Tires

The size of the hub motor and hence the rim narrowed the range of available tires already to some few ones. After comparing them, the offroad tire depicted in fig. 3.21, as it is used for scooters, was chosen. It fulfills all the requirements mentioned in section 3.3.1 and section 3.3.2, has a profile, which is suitable for driving on soil and sustains the weight of the robot. Furthermore, it can be deflated below the critical soil pressure as illustrated in fig. 3.20.

Review

The propulsion and locomotion type subsystem were tested during every deployment of the robot at different places, under various weather conditions and on different ground structures.

The main problem with the rim motors was that it was not possible to receive a data sheet from the manufacturer because of the corona-related lockdown in China and communication problems with the supplier of the motors. Therefore, there was only little information about the peak current the motor can take and how the maximal torque could be reached around. Nevertheless, the motors mostly provided sufficient torque and power during testing. They only failed when the robot needed to drive upwards and several goosefoot sweepers hit an obstacle at once or were pulled through hard soil. With more detailed information, the propulsion system could be further optimized and the current limitation discussed in section 5.1.1 could be increased in order to receive more power and torque from the motors.

The tires worked as intended and no adjustments needed to be made. The profile proved to be suitable for the agricultural soil, and the wheels never started to slip due to a lack of grip during the time tested. The wheels could be further deflated in order to optimize soil compaction and increase grip.

3.4 Steering

In order for the system to be able to steer and perform manoeuvres, a steering mechanism was designed. The steering concept was chosen to be *swerve drive*, which allows for flexible steering and can be implemented by simply allowing each wheel to orient itself independently. Therefore, each wheel is equipped with the same steering mechanism which can be seen integrated onto the suspension through a shaft (see fig. 3.22).

Motivation behind Swerve Drive

The first decision to be made for the steering mechanism was to find the most suitable concept for our requirements. Four potential options were evaluated. A benefit analysis showed that swerve drive was the most suitable concept. The determining factors were the high ability for turning, but also the personal preference of the team (see appendix B.4).

Swerve drive describes a steering mechanism which allows each wheel to be steered separately. Hence, if all wheels are set to 65.5° , facing the neighbouring wheel, rotation on spot is possible. This can be used to change the rows (see fig. 3.23).

$$\begin{aligned} \text{length } l &= 1.12 \text{ m} \\ \text{width } w &= 0.51 \text{ m} \\ \gamma &= \arctan\left(\frac{l}{w}\right) = 65.52^\circ \end{aligned}$$



Figure 3.22: Steering mechanism

As a consequence, each wheel requires individual propulsion motors (see more detailed information on the propulsion motors in section 3.3.2).

Steering Requirements

Once the steering concept had been decided on, suitable components were chosen and designed. The four crucial requirements for the steering mechanism are listed according to their priority and explained below.

Steering torque in relation to the weight of the system

In order to steer a wheel, the required torque is dependent on the system's weight, tyre and ground conditions. To evaluate this number, the expected torque was calculated as well as compared to experimentally tested values. The minimal required torque was calculated as follows (see fig. 3.24). The moment of inertia was neglected for simplicity.

width of wheel-ground contact: $a = 13 \text{ cm}$
length of wheel-ground contact: $b = 30 \text{ cm}$
Static friction coefficient: $\mu_H = 0.45$
Force of gravity: $F_G = m \cdot g$
Moment: $M = F \cdot l$

For rotating a wheel the following equation must be satisfied:

$$\begin{aligned} dF_R &\geq \mu_H \cdot F_g \\ \Rightarrow dM(a, b) = dF_R \cdot l &\geq \mu_H \cdot F_g \cdot l(a, b) \\ \Rightarrow dM(a, b) &\geq \mu_H \cdot m \cdot g \cdot l(a, b) \end{aligned}$$

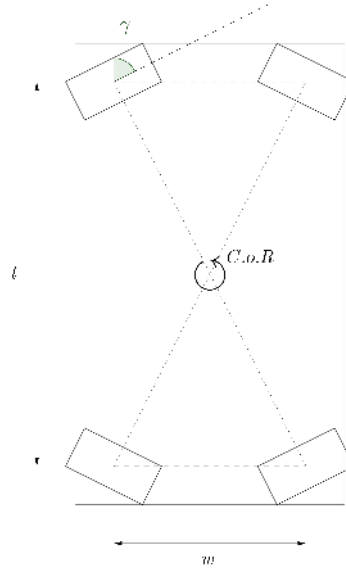


Figure 3.23: Rotating on spot

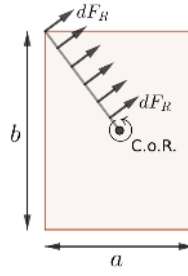


Figure 3.24: Required torque calculation

$$M \geq \frac{\mu_H \cdot m \cdot g}{a \cdot b} \cdot \int_{-b/2}^{b/2} \int_{-a/2}^{a/2} \sqrt{x^2 + y^2} \, dx \, dy$$

The numerical solution to this equation, including our expected weight of 75 kg onto each wheel, led to a minimal required torque of $M \geq 28.52 \text{ Nm}$.

The tested values were obtained by adding weights onto a wheel and rotating it up to 95° while measuring the applied torque with a force gauge at different rotation velocities. The tests were executed on slightly moist acre ground. The most conclusive and important tested values are shown in table 3.5 with the following parameters: $\dot{\psi} = 2.5 \text{ rpm}, 7.5 \text{ rpm}$ and mass $m = 75 \text{ kg}$.

The calculation helped to establish the main relationship between the required force with respect to surface area and weight of the system. However, the tested values were more meaningful as they contain all the neglected variables from the calculation. Next up the values were compared, estimation of the errors (e.g. due to ground conditions, test setup, tire profile, etc) and application of standard variance on the estimated value were done. With these adjustments we ended up with the following requirement: intermittent torque $M_{\text{intermittent}} \geq 58.3 \text{ Nm}$.

Steering speed

The required steering speed was evaluated by setting the following requirement: the

Test	75 kg at 2.5 rpm	75 kg at 7.5 rpm
1	40.75	36.81
2	35.49	42.07
3	42.07	35.49
4	36.81	40.75
5	44.69	40.75
6	34.18	39.44
7	39.82	47.32

Table 3.5: Required torque for rotating wheel in [Nm]

system shall readjust its position within the field from an undesired orientation to the correct one within 1.5 s and a covered ground distance of 30 cm (see fig. 3.25). In order to achieve this, the rotational velocity must be greater than $44.5^\circ/\text{s}$ which is why we set a requirement of: $\dot{\psi} = 45^\circ/\text{s}$ (see appendix B.3 for the calculation of this requirement).

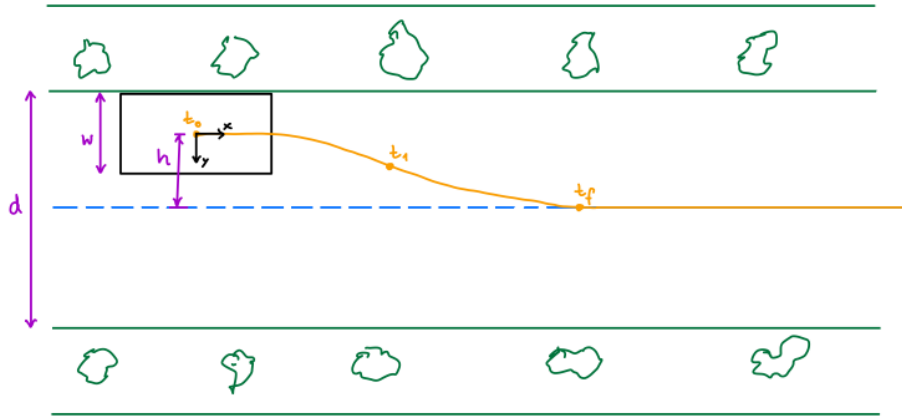


Figure 3.25: Readjusting position

Allowed backlash

In order to reach the required torque, a gearbox is needed in combination with a motor. Each gearbox has a certain backlash which has to be taken into consideration. Since the tight schedule of the project did not allow testing the tolerated backlash, the team asked two manufacturers of farming robots, *Naïo* and *Sagarobotics*, about the backlash of their robots. Both have very similar systems to ours. *Naïo's Dino* has a backlash of about 2° while *Sagarobotic's Thorvald* reassured us that a backlash of up to 2° has not been a problem for them.

From an analytical point of view, as the rotational velocity is at least 7.5 rpm ($45^\circ/\text{s}$), the backlash will be compensated for within 0.044 s ($2^\circ/0.044 \text{ s}$). Also, with an offset of 2° while driving straight without steering, this would result in an offset of 3.5 cm after 1 m (see fig. 3.26).

$$\begin{aligned}\alpha &= 2^\circ = 0.035 \text{ rad} \\ l_1 &= 1 \text{ m} \\ l_2 &= \tan(\alpha) * l_1 = 3.5 \text{ cm}\end{aligned}$$

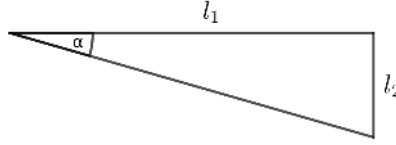


Figure 3.26: Readjusting position

The lateral forces acting on the wheel caused by the backlash are negligible compared to the maximum forces to expect. The forces acting laterally on the wheel with, for instance, the two front wheels offset to each other would lead to a force of 11 N per wheel which is way less than what we have to expect laterally (776 N) for dimensioning our system.

$$\begin{aligned} \text{From motor calculations: } F_{\text{acceleration}} &= 321 \text{ N} \\ F_{\text{offset}} &= \tan(\alpha) * F_{\text{acceleration}} = 11.21 \text{ N} \end{aligned}$$

The analytical information as well as the experience of people working on similar steering concepts results in the conclusion that a backlash of 2° is allowable.

Weight

Since a light system reduces soil compaction, it was tried to avoid unnecessarily heavy parts. However, the surface area of the wheels, which is the limiting factor for the allowed weight, was not yet decided on while this subsystem was designed. Therefore, an estimated goal weight of 8 kg per wheel was set.

Implementation

For the design of the steering box, the external forces on the mechanism were calculated. Additionally, the motor- as well as gearbox-selection were decided upon with the torque requirement in mind ($M_{\text{intermittent}} \geq 58.3 \text{ Nm}$). For an overview of all the mechanical components used in the steering mechanism, an illustrative explosion view is provided in fig. 3.27.

Absorbed Forces

To compensate for the forces expected while driving on the field (0.2 m/s) as well as on concrete (3 m/s), a steering box was designed. The steering box absorbs external forces through two bearings which mount the steering shaft to our system. Hence, the motor and the gearbox are spared from the forces absorbed by the steering box. The material used for this box ended up being the aluminium alloy *AlSi1MgMn* ($R_{p0.2} = 240\text{--}260 \text{ MPa}$) as it withstands the expected forces and is a material which allows for easy machining during manufacturing. The forces on the steering box were analyzed with FEM (see fig. 3.28).

Further parts designed to compensate or transmit forces/torque:

- Shaft
The shaft was designed to withstand bending torques caused by forces acting from the suspension. It was designed with a predetermined breaking point on the outside of the steering box, where breaking leads to less damage than at other locations within the steering box.
- Coupling
The coupling transmits the required 75 Nm provided by the output shaft of the gearbox.

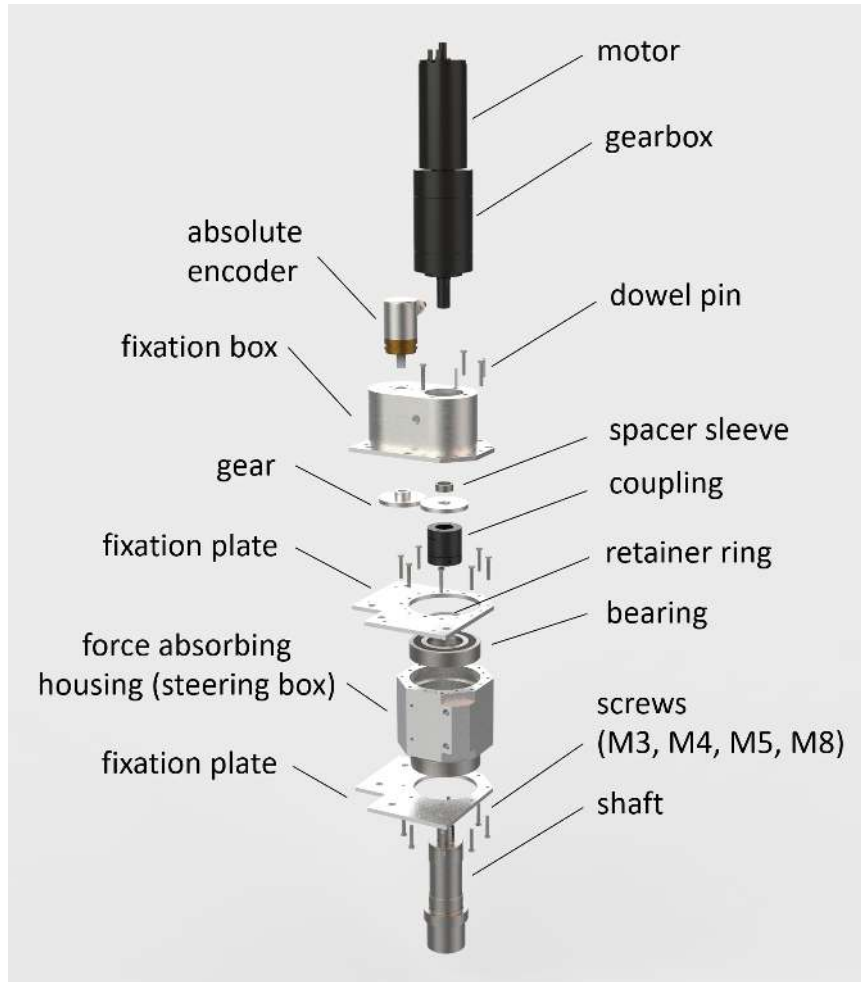


Figure 3.27: steering mechanism explosion view

Motor and gearbox selection

Maxon Group was chosen as the motor and gearbox supplier since RSL has a lot of experience in using these motors and can therefore support the team well. The motor in combination with the gearbox have to allow for a maximum intermittent torque higher than our maximum expected peak torque of 58.3 Nm. After finding the correct motor/gearbox combinations with the maxon selection manual, the major differences in options came from the gearboxes rather than the motors. The specifications of the possible gearboxes are illustrated in fig. 3.29.

According to *maxon*, the heavy duty gearboxes would be costly (up to 5000 CHF, depending on the gear ratio) and *GP 81* would lead to overshooting the weight goal of the steering mechanism. Hence, it was decided on using the *GP 62 A* (ratio 236:1, part number 110508) gearbox in combination with a *RE 50* (150 Watt, part number 578298) motor (data-sheets can be found in appendix C.1.1).

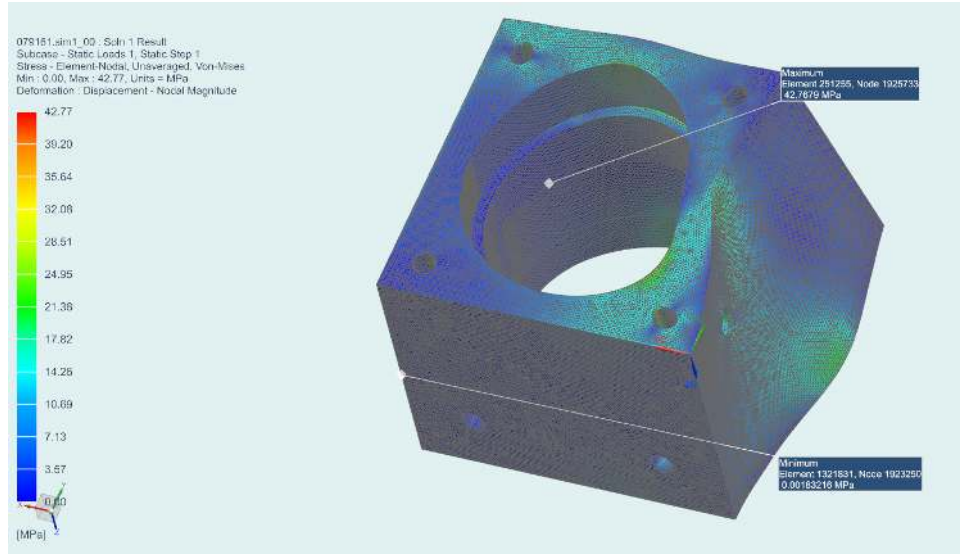


Figure 3.28: FEM of steering box

Gear	max. torque (Nm)	Ratios	input speed (Rpm)	Weight (kg)	Av. backlash (")	Price (CHF)
GP 42 HD (HD = heavy duty) Planetary Gearhead	60	150:1 - 676:1	<8000	0.78	1	>1000
GP 42 HD (HD = heavy duty) Planetary Gearhead	75	150:1 - 676:1	<8000	0.78	1	>1000
GP 62 A Planetary Gearhead	75	71:1 - 236:1	3000	1.54	2	553.4
GP 81 A Planetary Gearhead	180	14:1 - 308:1	3000	3.00 - 3.70	0.55 - 0.60	424 - 504.3

Figure 3.29: Gearbox selection process

Absolute Encoders

In order to steer each wheel, an integrated incremental as well as an external absolute encoder are used. The absolute encoder allows the system to be used without readjusting the direction of each wheel before usage. This was highly recommended by *Sagarobotics*, who have tested a similar system for 4 years now. In order to implement an absolute encoder, on-axis and off-axis encoders were evaluated. Since off-axis encoders are normally of smaller diameter than the diameter of the output shaft of the chosen gearbox (140 mm) and the considered ones that would fit were extremely expensive, an on-axis encoder in parallel with the output shaft (connected by gears) was chosen.

The requirement set for evaluating the accuracy of the encoder was the following: there may be a maximal lateral offset of 1 cm while driving at 0.2 m/s for 5 s (reasoning that the navigation software would not provide feedback for 5 s). Therefore an accuracy of $\phi \leq 0.673^\circ$ must be fulfilled:

$$\text{lateral offset } x \leq 0.01 \text{ m}$$

$$\Delta\phi \leq \arcsin \frac{x}{0.2 \text{ m/s} \cdot 5 \text{ s}} = 0.673$$

The absolute encoder chosen counts 1024 bits, which means it has an accuracy of 0.35° and therefore fulfills our requirements.

3.5 Weeding Tools

Motivation

In order to execute the weed control, several tools are used. Generally, the weeding can be divided into two categories (see fig. 3.30). The passive weeding will take care of the weeds *between the rows* i.e. in the area where the wheels of the robot will be driving. The passive weeding can be accomplished on a large area of 35 cm width. The active weeding will destroy weeds *in the row* i.e. in the line of crops. Therefore the active tool has to be precise and dodge the crops to not harm them. The active tool will work on a width of 15 cm, centered around the line of crops. The differentiation of active and passive tool allows the robot to operate at a higher speed by reducing the area that has to be covered by an active, moving tool.



Figure 3.30: Ranges of the passive (orange) and the active tool (blue)

3.5.1 Choice of Weeding Method

Motivation

According to the SRS (see section 2.6.2) Rowesys aims for a herbicide free weeding solution. The choice of a weeding tool also influenced the requirements and design of the weeding mechanisms.

Research and Evaluation

The team consulted the results from the tests of the *Technical College Bern* (BFH) [11] and own ideas. The possible options were compared considering the *complexity* (challenge in design, actuation and control), *effectiveness* (long-term weed destruction) and *efficiency / speed* (amount of tackled weeds per time period). The options were rated good (+), average (0), bad (-) or unknown (?) in every criteria as shown in table 3.6. The ratings were based on test results from the BFH and own testings of the team.

Weeding method	Comp.	Effec.	Effic.	Comments
Spot flaming	0	+	+	buring hazard
Mini oxyhydrogen explosion	-	?	?	
Capacitor discharge	-	0	+	damage to soil life?
Laser	-	0	-	too slow
Water jet	+	?	+	high energy consumption additional substance needed
Shock wave	+	?	+	energy consumption?
Concentrated solar light	0	0	-	too slow
high-frequency heating	-	?	?	energy consumption?
Hot air	+	+	0	high energy consumption
Hot water	0	+	0	high energy consumption additional substance needed
Pulling out	0	+	-	too slow
Mowing / mulching	+	-	+	only short-term effect
Drilling / Milling	+	+	0	slow
Plowing / Cutting	+	+	+	

Table 3.6: Evaluation of possible weeding methods

Chosen Weeding Method

The team decided to implement the mechanical weeding method of *Plowing / Cutting*. The risk of implementing a lesser known and tested solution, with unproven effect was too big, especially since the task of weeding is no research or development focus (see section 2.6.1). Other options were rejected mainly due to a high power consumption, which was not considered suitable for a mobile system. Both the *Plowing / Cutting* and the *Drilling / Milling* options proved promising. But due to the lower speed and higher complexity because of the precise actuation of the tool, the former was rejected for the *Plowing / Cutting* option. The main advantages of the chosen option is the simple and proven working principle without complex actuation. Additionally, it works great in combination with the chosen development focus (see section 2.6.1), since it relies on a robust locomotion.

3.5.2 Passive Weeding Tool

Overview

The basic concept that is implemented can be taken from fig. 3.31. A more detailed discussion of the requirements and the design follows in the sections after. The passive tool consists of an arrangement of so-called goosefoot shares (1).

Working position and lifting the frame

The complete array, consisting of four goosefoot shares with overlapping trajectories, needs to be lifted for turning or driving on roads. Therefore, a linear motor (2) is attached to the base frame (3) which can lift up the complete passive tool with all the shares at once. The linear motor moves the frame via a four-bar linkage (4) that reduces the force on the motor. In the weeding process, the frame is held in a fixed, horizontal position.



Figure 3.31: The passive weeding tool with goosefoot shares and linear motor

Spring arrangement

The goosefoot sweepers are mounted to vertical bars (5) that can each rotate upwards independent of the others. The bars are preloaded by individual tension springs (6) and lay on the base frame (3) in resting position. The springs allow each shear to move up individually to react to higher traction forces.

Removable

The complete array is mounted to the platform on the back of the robot, behind its rear wheels. The four bars (5) of the tools are attached to a shaft (7) that is then connected to the platform by two mounts (8). The whole passive weeding tool can therefore be removed by 16 screws.

Requirements

The requirements of the passive tool are based on the SRS and empirical values from the team's contacts in agriculture and robotics.

1. Mechanical weeding

According to the team's vision (see chapter 2), the weeding process has to be completely independent of herbicides.

2. Area of tillage

The passive tool is going to cover 35 cm of soil. Since no crops will be planted in its working area, every plant can be removed.

3. Simplicity

The passive tool is going to be pulled by the robot and should only use a minimal amount of electronics.

4. Lifiable

The passive tool needs to be lifted out of the ground at least 15 cm to rotate the robot, drive on a road or go over a curbstone

5. Smooth operation

The design of the passive tool shall enable smooth operation of the robot and include a mechanism to prevent getting stuck in hard soil or when hitting stones. The maximum size of stones to be encountered is assumed to be 7 cm. The tool shall compromise steering as little as possible.

6. Installation space

As the tool is not a central function, the other subsystems have priority when it comes to installation space. The passive tool has to be placed behind the rear wheels, to loosen the soil compressed by the wheels.

7. Removable

The tool has to be removable with basic tools (wrench or hex key) and within three minutes by two persons, such that the system can also be tested without the tools attached. Also, the tools are designed to be an extension of the robot, not a central, fixed part.

Research and Evaluation

Goosefoot sweepers

After researching and comparing existing farming tools, it was found that goosefoot sweepers (see fig. 3.32(a)) are frequently used in sugar beet and are a good option for most soil conditions. They have the ability to on the one hand cut the weed under the surface and on the other hand loosen the ground. Loosening the ground brings in air and promotes crop growth. The basic working principle is as follows: the arrow-shaped sweepers are pulled through the ground and cut the weeds under the surface.

Harrow

The option of using a harrow on wet soil was also evaluated since in this case, according to Hansueli Zellwenger from *ETH Crop Sciences*, goosefoot sweepers are not as effective and can in contrary lead to soil puddling, which has a negative effect on the growth of the crops. The harrow has an advantage in wet soil because its tines are quite thin (see fig. 3.32(b)). But the undesired puddling effect can also happen in case of high moisture.

Conclusion

To keep the robot as flexible as possible, an interchangeable weeding tool would be best to tackle the task of weeding under every weather and soil condition. In the scope of this project, however, it was sufficient to implement one tool to prove the robustness of the locomotion system. Since weeding in moist soil also increases the effects of soil compression and is therefore only possible up to a certain moisture level, the team decided, also upon suggestion from farmers, to go with the goosefoot sweepers as the passive weeding tool.

Implementation

Goosefoot sweepers

Goosefoot sweepers are usually used in combination with tines in the shape of a torsional spring that introduce vibrations to the sweeper (see fig. 3.32(a)). Those

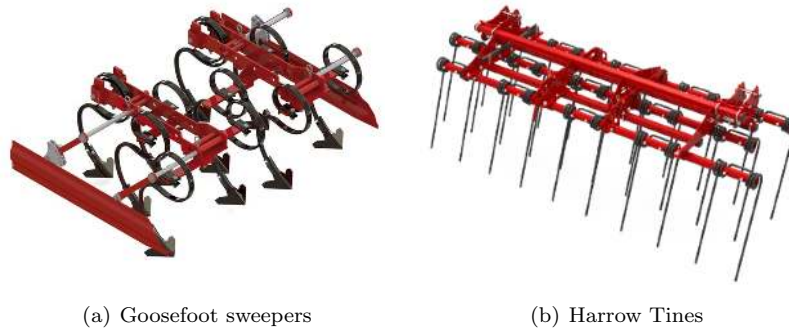


Figure 3.32: Different passive weeding tools

small displacements loosen the ground, smoothen impacts caused by obstacles and lower the traction needed to pull the loads. Despite those beneficial properties, the goosefoot sweepers are implemented without the tines. They increase the weight of one goosefoot significantly (from 1.2 kg to 3.0 kg) and the attachment lies approx. 15 cm higher, which increases the torque due to the force of ground resistance.

To attach the sweepers to the structure, holes were drilled into the goosefeet to mount them with screws to groove stones. The regular working depth of goosefoot sweepers is between 2 to 5 cm and can be adjusted by the position of the groove stones. In total four goosefoot sweepers were used (two on each side of the row of crops) with an overlap of 15 mm to ensure a complete treatment of the soil.

Based on suggestions from farmers and considering own measurements, the maximum force for the goosefeet in the front was defined to 300 N and 200 N for the sweepers in the back. This number is only an assumption and could still not be verified, since it depends on the soils composition and consistency and the speed in which the tools are used.

Framing structure

Each sweeper is individually mounted to a bar (red in fig. 3.33 and fig. 3.34) that are connected to a shaft. To create the bars, the same kind of aluminum profiles as used for the platform (see section 3.1) were implemented but with a width of 30 mm. Since the use of ball bearings, to connect the bars to the shaft, would need an additional connection piece, sleeve bearings (*Igus GFM-1618-16*) ((1) in fig. 3.34) were used. To protect them from the edges of the aluminum profile, they were inserted into a stainless steel shell and over the shaft ((2) and (3) in fig. 3.34).

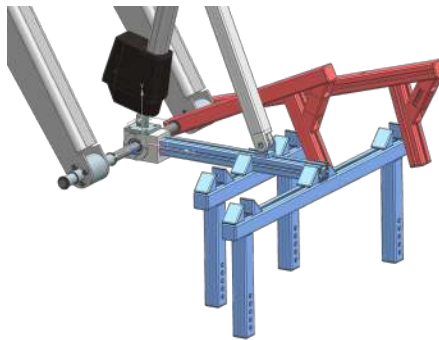


Figure 3.33: The framework of the passive tool: baseframe (blue) and bars for the goosefeet (red)

The baseframe (blue in fig. 3.33) provides a support for the four bars, is fixed by the position of the linear motor and thereby allows the lifting of all four goosefeet simultaneously. It is made out of the same aluminum profiles, but due to higher forces, a milled connection piece, housing two ball bearings, is used to connect the baseframe to the shaft. It also provides the interfaces to connect the springs with the bars and to attach the bars of the four bar linkage of the lifting mechanism. To connect the passive tool to the robot, two bent and welded aluminum mounts are rigidly connected to the shaft and can be screwed to the robot's platform. They are designed to keep the passive tool as close as possible to the robot, without interfering with the rear wheels.

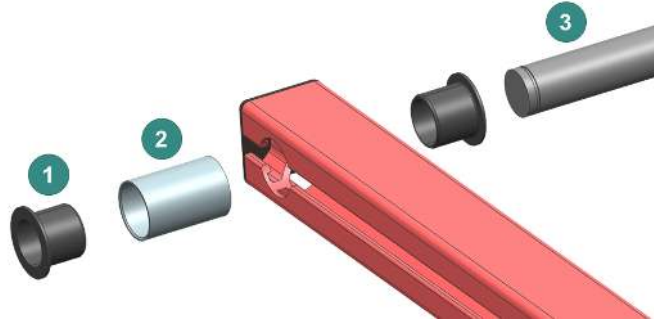


Figure 3.34: Bearing concept for the bars of the goosefoot sweepers

Springs

The traction force of the sweepers create a torque around the shaft, that pushes the tools out of the soil. This force is counteracted by tension springs, which at the same time provide an overload protection (see fig. 3.35). When the forces on the goosefeet increase (e.g. when hitting an obstacle like a stone) the springs deflect and the sweepers are moved out of the soil. The springs are chosen such that the sweepers move up by 5 cm (the maximum regular working depth) when $\frac{2}{3}$ of the maximum estimated force (300 N resp. 200 N) is reached and a total of at least 7 cm to dodge the obstacles, mentioned in the requirements. The calculation to evaluate the appropriate springs can be found in appendix B.10.1.

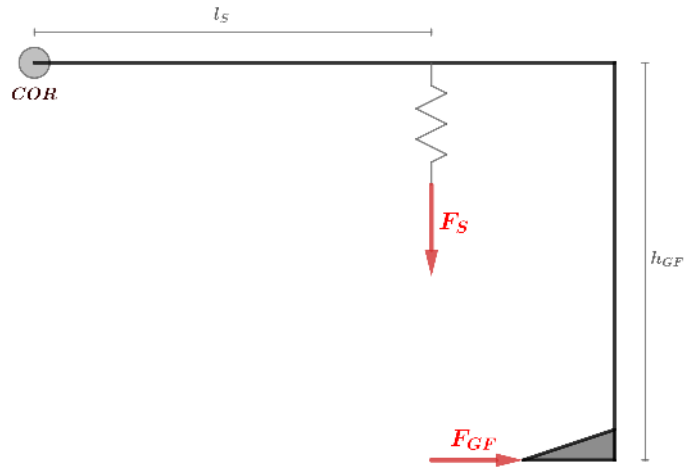


Figure 3.35: Spring force counteracting the dragging force

The springs are mounted by eye-bolts connected to the bar of the goosefoot on top and on the bottom to a vertical extension of the baseframe. The eye-bolts on top are held in place by nuts and the tension of the spring. By adjusting the screwed length of the eye-bolt, the pretension of each spring can be adjusted.

Torsional and compression springs were also considered since they don't need additional vertical bars to connect the baseframe and the bar of the goosefeet. No torsional springs could be found that provide the necessary forces and have a reasonable size to implement them to the structure. Compression springs rely on a guide which was considered a more complex solution than to add the vertical extensions on the baseframe.

Lifting mechanism

The linear motor defines the position of the baseframe and therefore also takes up the forces on the goosefeet. During normal operation the linear motor does not need to move. Since those forces also act only during normal operation, a linear motor with a self-locking property was chosen. The forces to lift and lower the baseframe are mainly caused by the weight of the system and are significantly smaller than the traction forces. The magnitude of the self-locking force was therefore the most critical property. Considering the maximum estimated force on all four goosefoot sweepers, the *LINAK LA25* with a maximum self-locking force of 1500N and a stroke of 200 mm was chosen. (See appendix B.10.2 for derivation)

Due to space reasons, a direct connection of the linear motor from the platform to the baseframe was not possible. The chosen four-bar linkage is optimized to reduce the forces on the linear motor and therefore on the shaft, by choosing the lengths of the two upper bars accordingly.

Review

The passive tool was assembled and mounted to the robot and tested at *ETH Crop Sciences* in Eschikon on dry soil. The result can be seen in fig. 3.36.



Figure 3.36: The passive weeding tool mounted on the robot

Working behaviour

The passive tool worked as intended. No precise adjustment of the preload was needed to allow a proper working behaviour. The overload protection proved effective and allowed the goosefeet to move out of the soil to react to increasing

traction forces. The framework turned out to be less rigid as expected, but having all four goosefoot sweepers in the ground stabilizes the passive tool significantly.

Flaws

The chosen linear motor takes close to a minute to move its full stroke. The speed of the motor was chosen, to enable the small installation space and the high self-locking force. It is no problem during the weeding process, but delays the turning at the end of the row. With more precise information on the maximum acting forces, a different motor with higher speeds could be implemented.

The assembly of the framework did not work as intended. The corner brackets to connect two orthogonal profiles are not meant to connect crossing profiles. This issue could be bypassed by filing off parts of the brackets. This resulted in a slight backlash in the tight fit of the brackets which changed the shape of the tool slightly, but did not interfere with its working behaviour.

3.5.3 Active Weeding Tool

According to fig. 3.30, the active tool tackles weeds in the row of crops and therefore completes the weeding area of the robot. Since the active tool could not be implemented in the scope of the focus project due to time reasons, it is being developed as bachelor theses. However a quick overview is given here.

Requirements

The requirements of the active tool are similar, even mostly identical with the requirements of the passive weeding tool. For further explanation see section 3.5.2.

1. Mechanical weeding

See section 3.5.2

2. Area of tillage

The active tool is going to cover 15 cm of soil, centered around the crops.

3. Lifiable

See section 3.5.2

4. Constant working depth

Since the platforms height can change due to the suspension, the active weeding tool needs an additional way of keeping a constant working depth.

5. Dodging crops

The active tool needs to be able to dodge the crops which are located within the area of tillage. The crops diameter is considered to be between 4 and 8 cm. The active tool also needs to be able to adjust to the crops lateral position on ± 5 cm, given by the precision of the row following.

6. Overload protection

The active tool needs a protection from high traction forces to prevent damaging its components.

Overview

Connection to the robot The basic concept that is going to be implemented is shown in fig. 3.37. The active tool is mounted beneath the platform. The weeding mechanism (1) that holds the actuators and the shares is explained in the following sections. To ensure a constant working depth, even if the robots platform is moving vertically, the weeding tool is attached to the platform via a four-bar linkage (2) with tension springs (3) that provide the necessary preload to keep the wheels (4) connected to the ground at all times. The wheels are rigidly connected to the weeding mechanism and thus adjust its height to the local level of the soil. As for the passive weeding tool, the active tool has to be lifted up, if the robots drives on the street or turns at the end of the row as explained in section 5.9.4. A linear motor (5) (*LA25* from *LINAK*) lifts the system, that is connected by a steel wire (6) to the platform of the tool.

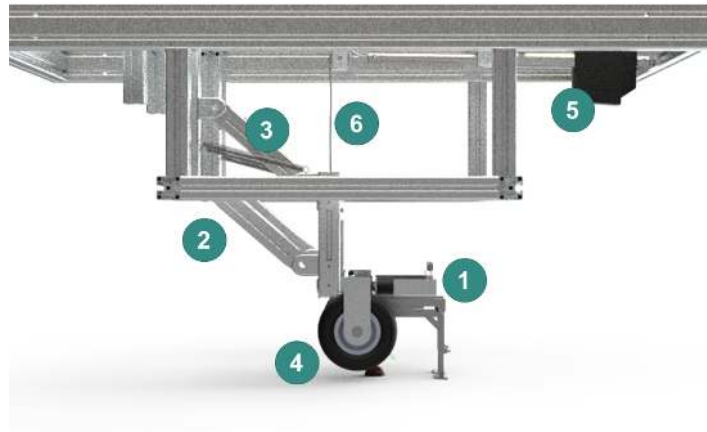


Figure 3.37: The connection of the active tool to the platform of the robot

Active weeding mechanism

The concept of the weeding mechanism is shown in fig. 3.38. The two shares (1) are made of sharpened steel plates and work similarly to the goosefoot sweepers of the passive tool. They can move sideways, powered by two *DCX 32* motors (2) by *maxon motors*. They allow the active tool to dodge up to 2.5 crops/s which results in a possible speed of the robot of 0.5 m/s . The worm drive (3) between the motors and the holding bars (4) allows a self-locking property in one direction. Therefore, the motors do not need to counteract lateral forces in the ground to keep the sweepers in its position. On top of the worm drives, end switches and a rod on the output shaft on the drive serve as homing hardware (5) to control the absolute position of the motors for the shears. The active weeding mechanism is equipped with an overload protection (6). Magnets hold the shears in place, and detach when the forces on the shears exceed a certain value. This value can be adjusted by the number of magnets and their mounting positions. A torsional spring mounted around the pivot of the shears, provides the retaining force to put the system back in its regular position. All parts are mounted to a bent base plate (7) that also provides connections to the four-bar linkage and the wheels.

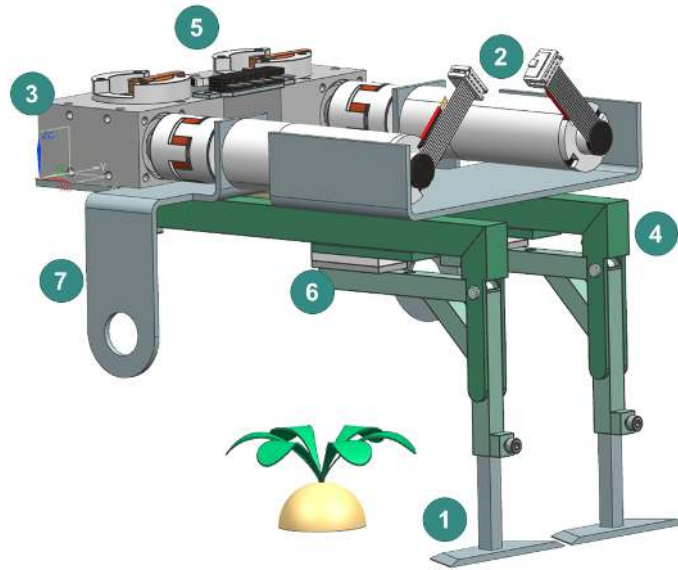


Figure 3.38: The active tool's weeding mechanism

Current State and Outlook

The active weeding mechanism and the holder to the platform are part of the bachelor theses of Manuel Knecht and Markus Wagner and will therefore not be discussed further. By the time this report is written, the design is finished and the parts are in production. The assembly, the motor control and the implementation to the robot will follow after the end of the focus project.

3.6 Casing

Motivation

A Casing is needed to protect the system's electronics. In addition, the casing represents the spirit of the team and provides a memorable first impression of the robot.

Overview

The final casing consists of a main body made of aluminum sheets and can be seen in fig. 3.39. The corners and the front and back of the casing are made of a carbon reinforced 3D-printed structure. Emergency stop switches are mounted in the corners, and the hatches on both sides allow access to the electronics and the steering units. The shell is designed to be quickly dismountable and its larger parts, such as the main cover, can be removed without the use of any tools in case an intervention to the electronics is needed. The 3D-printed lamellar structure in the front and back of the robot is backed by an acrylic glass plate illuminated by LEDs, which can be used as status lights of the robot (see section 5.7).

Requirements

Protection

According to the SRS (see section 2.6.2), the casing has to serve as protection from



Figure 3.39: The final casing mounted on the robot

sunlight and mechanical impacts. The team decided to extend this requirement to protect from the following:

- Environmental impacts (sunlight, wind, light rain)
- Invasion of larger objects (stones, leaves, twigs > 2 cm)
- Mechanical impacts (people, dropping tools)

Accessibility

The casing must not obstruct the access to the robot's subsystems and electronics. A main focus lies on the following electrical parts:

- Status Lights
The status lights of the robot have to be seen during daylight and from all sides of the robot without removing parts of the casing.
- Emergency Stops
The electrical emergency switches must within arm's reach from every side of the robot in every composition of the casing.
- Electrical Box
The electrical box (see section 4.7) has to be accessible by opening or removing parts of the casing. Detaching the necessary parts must not require special tools.

Design Requirement: Benefits of a Designed Casing

The casing should not only have an aesthetic function but also direct the focus from the individual components to a complete system. Therefore the casing must have a consistent and attractive design.

Design Research

In order to collect inspiration and to develop a desired language of form, the industrial design team started analysing images of products and machines from various

areas, that appealed to them. The considered areas were wide ranging for example Autonomous Agriculture or Medical Design. The analysis aimed at working out which design features represent the intended image.

Conclusion of the Picture-Based Product / Inspiration Analysis

Common aspects of the design concepts dealt with in the analysis are:

- The autonomous impression is often created by Symmetry, clean surface structure and the elimination of the clearly visible manual control centre or cockpit
- Most of the analyzed images show the interplay of hard, clean, geometric forms and rounded, soft, dynamic and organic lines
- The majority of the examples show a symmetrical and clear basic form
- Almost all examples are accentuated by bright signal colors
- One of the features, which emphasizes modern appearance, is the covering/disguising of technical aspects by a designed shell

Design Keywords

Based on product analysis and research, four keywords for our design were chosen and served as orientation in the ideation process:

Autonomous – Powerful – Dynamic – Simple

Based on these four terms, the team tried to generate design specific form inferences shown in fig. 3.40. Those were constantly applied to the new designs.

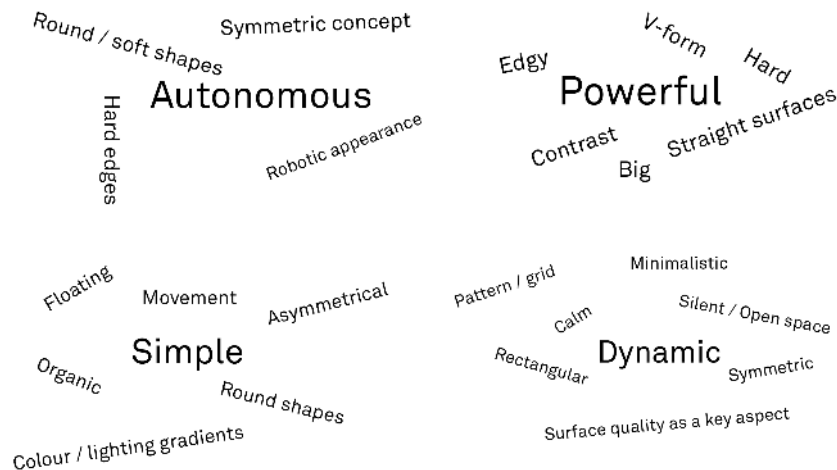


Figure 3.40: The four design keywords and according properties

More information about this process can be found in the bachelor thesis of Laurin Baumann [12].

Implementation

Design Principles

In order to create a consistent design, the shell of the robot is practice-oriented. The following principles lead the design:

1. Protect parts that need protection.
2. Leave parts that do not need casing or even put them in the limelight.
3. Design the casing as an overall concept and in a second step divide it into the various segments.
4. Make aspects that must be easily accessible.

Production Process and Materials

During the research and design process, it was decided to build the casing out of aluminum sheets in combination with carbon reinforced 3D-printed structures. 3D-printing allows the implementation of free forms to achieve various effects. To reduce the high material costs for the 3D-printed parts, the majority of the straight surfaces are made of aluminum sheet metal. It is cheap, easy to machine (bending, rolling, cutting) and light weight.

Possible Casing Designs

Many sketches, designed mockups and refined ideas were created during the process of brainstorming. They were evaluated and rated based on the design keywords mentioned above and their form conclusions. Upon reduction, the team was presented with the variants in fig. 3.41.



Figure 3.41: The considered casing design variants

The team agreed on one variant during the discussion. This version (shown in fig. 3.42) was developed further within the framework of feasibility. The reasons that let the chosen version stand out against the other ideas, was that it is perceived as one united part. Also the transition between the middle section and the front and back running along a curved line and the structures that let the light shine through were appreciated by the team members. The selected version reflects the four design keywords and can fulfill all the requirements.

The chosen design was split into single, production friendly parts. A gap of 2 mm between adjacent pieces was chosen to account for tolerances in manufacturing. During this development, the focus was on perfecting the lines, finding the ideal curves and not losing sight of feasibility.

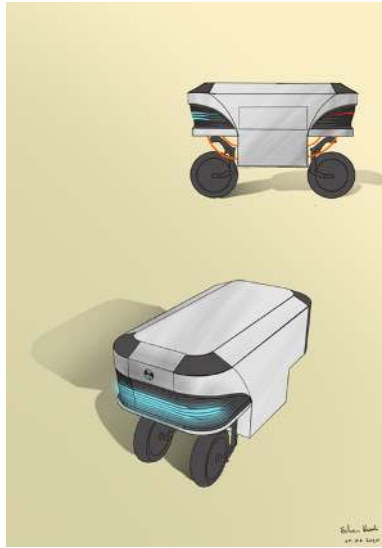


Figure 3.42: The chosen casing design concept

Technical Design

After the design was transmitted to CAD, it was connected to the platform, making use of bent sheet metal parts, screws and press nuts. The 3D-printed structure in the corners house the emergency switches (as shown in fig. 3.43), are screwed to the platform and connect to the metal sheets. The lamellar structure in the front and in the back lets the light from LEDs shine through. The lights serve as status lights of the robot and can be seen from all sides. The 3D-printed parts and the aluminum sheets in the front and the back are screwed to the platform and are not intended to be removed. The aluminum sheets that need to be removed to access the inside of the robot (e.g. the electrical box), are fixed with knurled screws, which allow removal without any tools. All sheets are connected to each other by aluminum plates, which are riveted or fixed by screws and press nuts. To enable a faster way to access the inside without needing to remove the top sheet, a side hatch was implemented on each side of the casing. It is held close by magnets and can be opened manually to serve as a table for keyboard and mouse.

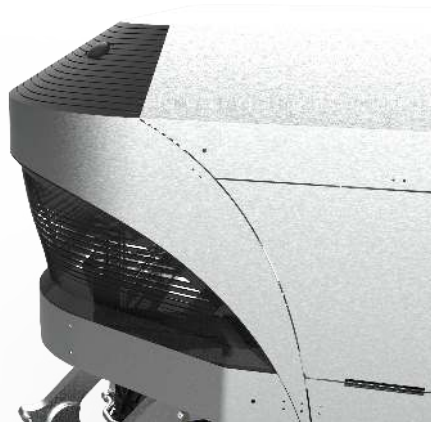


Figure 3.43: The casing made of 3D-printed parts (black) and aluminum sheets (silver)

Review

The casing meets all its requirements and received positive annotations.

There were problems in the production of the sheet metal parts. Almost all the parts were not bent to the expected precision. They had to be straightened by hand and were fixed to the 3D structure. The lamellar structure is very sensitive and can break quickly due to its thin gills. This design feature would need to be simplified to allow for the rough conditions in a long-term agricultural use.

Chapter 4

Electrical design

4.1 Overview

The electrical system of the robot is constructed on top of the mechanical system and allows the software to interact with the environment. As can be seen from fig. 4.1, the electronics of the robot consists mainly of the energy source, sensors, data processing units, connectivity and actuators. In addition, the individual systems have to be linked and the safe operation of the robot has to be guaranteed. Each part is described in the following chapters. The motors selected are discussed in the mechanical design sections (section 3.3 and section 3.4), and the sensors are discussed in the software part (see section 5.9.1). Most of the electronics is not water nor dust proof. Therefore, it was necessary to keep them in a sealed box. This will be described in section 4.7.

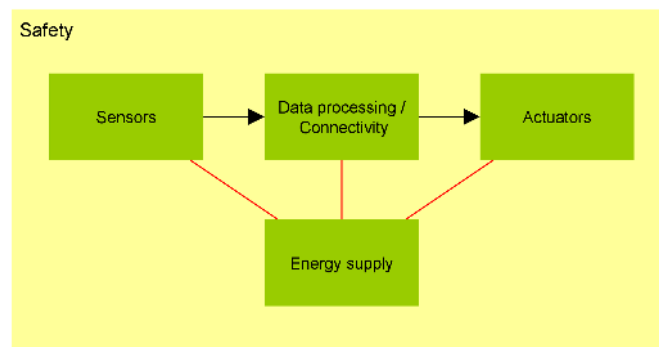
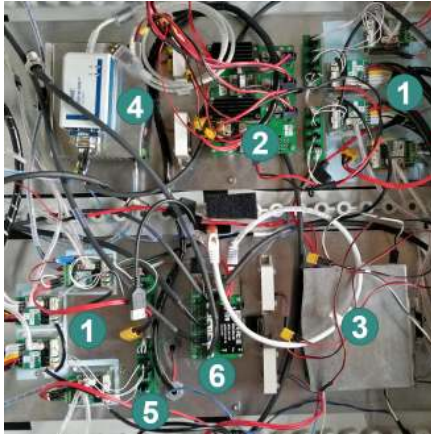


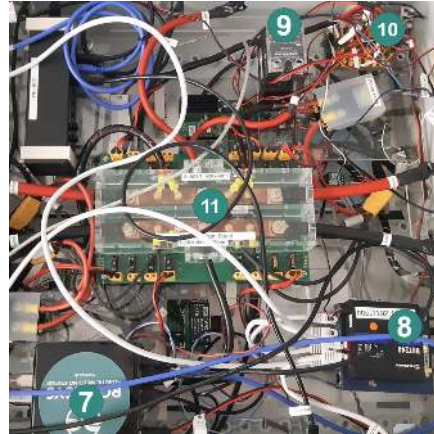
Figure 4.1: Highlevel overview

The following two pictures (fig. 4.2(a) and fig. 4.2(b)) depict the majority of the installed electrical components:

- The motor controllers (1) will be discussed in section 4.6.
- Following components will be explained section 4.3:
12 V DC/DC converter (2), 24 V DC/DC converter (3), 48 V DC/DC converter (4), 12 V distribution board (10) and 48 V distribution board (11).
- Following items will be discussed in section 4.4:
The STO PCB (5) and solid state relay (9).
- Following items will be described in section 4.5:
Raspberry Pi (6), *NUC* (7) and router/switch (8).



(a) Lower floor of the electronics box



(b) Upper floor of the electronics box

4.2 Energy Supply

Overview

Two lithium-ion batteries from *Ecovolta* have been chosen as the energy source. Each battery has a capacity of 5 kWh, resulting in a total capacity of 10 kWh. This should be sufficient for a theoretical driving time of 5 h on the field. The batteries are connected in parallel and offer a nominal voltage of 48 V. Each battery includes a BMS to monitor and balance each cell and also an emergency system which is able to switch off the voltage.

Current State of Energy Supply Concepts for Agricultural Robots

Most robots use an onboard battery for their energy supply. An example would be *Dino* from *Naïo technologies* [13] which uses interchangeable lithium ion battery packs. Under ideal conditions the *Dino* has a runtime of 8 h. Another example would be *avo* from *Ecorobotix*. This one has a 10'656 Wh battery in combination with a photovoltaic system which produces 1150 W [14]. Thanks to the solar panel and their low-energy weed control method of precision spraying, they claim that the robot can be used totally autonomously. So the robot can operate for 8 h without charging. This is probably only possible on a day with high solar radiation.

Requirements for Energy supply

1. *Environmentally friendly (SRS: 1.2):*

The energy supply should not produce environmentally harmful emission while the robot is driving over the field. It would be counteractive if the robot is able to weed without the use of herbicides but while doing his job, he is contaminating the crops with the robot's energy supply emissions.[15] A second goal is to minimize the ecologic footprint of the energy production and the grey energy.

2. *Run time and recharging (SRS: 4.2.1):*

The energy supply should be strong enough to supply the whole system for 5 h so that the robot can be tested for a reasonable time without major interruptions. The energy supply should be rechargeable. Recharging should be as easy as possible. That means only a few cables have to be plugged in and

then one switch or something similar has to be actuated. Furthermore, the robot should be able to be plugged in a power source and simultaneously be completely powered from it. This enables stationary testing while charging.

3. *Weight, size and placement:*

The gravimetric capacity should be as high as possible to hinder soil compaction. A high volumetric capacity is less important but desirable to enable more flexibility in the mechanical design of the robot. The energy supply should completely fit inside the robot, so no cord has to be pulled behind.

4. *Output:*

The output of the energy supply should be a DC voltage source. The voltage should be as high as possible to lower currents but also enable a low risk operation to gain a high safety standard without too many special precautions. So the preferable voltage is around 50 V.

5. *Easy and simple installment:*

The energy supply should be able to be bought as a complete package and be easily capable of being integrated into the whole robot. This reduces the workload, so that only the interface to the energy supply must be designed, and also increases the safety standard.

Research and Evaluation

To reach the goal of 5 h testing time a energy capacity of about 10 kWh is needed:

$$E_{req} = 5 \text{ h} \cdot P_{estimated} = 5 \text{ h} \cdot 2072 \text{ W} = 10,360 \text{ kWh} \quad (4.1)$$

The estimated power was calculated by summing the power of the individual electrical subsystems (see table 4.1). This table was created in the first phase of the project. Therefore, the initially planned arm as weeding tool and a multispectral camera are included. In addition, the *Aquila* (crop detection camera) and status lights were added. This adds a power consumption of about 110 W, but this cannot compensate for the power consumption of the missing arm.

Table 4.1: Power estimation

consumer	power [W]
switch	5
router	5
multispectral camera	16
IMU	0,1
GPS	0,4
PC	28
RGB camera	8
illumination	50
arm	800
tool	40
propulsion	1000
steering	100
goose feet	20
total	2072

The detailed calculation for the propulsion power can be found in appendix B.9.3.

Different possibilities were evaluated:

Internal combustion engine

A combustion engine seems to be good for mobile robotics, as it has a high gravimetric energy density and is therefore very lightweight. It can also be recharged very quickly. Propane is one of the greenest fuels for internal combustion engines, because there are less emissions than petroleum and it is not toxic for water and soil [16]. But as it is the goal to design the robot to be as sustainable as possible, the CO₂ equivalents for a battery and a combustion engine were compared. The calculations were done for a propane combustion engine and a 10 kWh lithium ion battery.

1. CO₂ Propane Combustion Engine:

The CO₂ emissions of a combustion engine are composed of the emissions from burning the propane and from the production of the propane. The emissions for the production of the combustion engine were assumed to be negligible. To produce 1 kWh energy, approximately 0.24 kg propane is needed [17]. The CO₂ emissions from burning 1 kg propane are 3.1 kg CO₂ [18]. The production of 1 kg propane releases 1.4 kg CO₂ [19]. In total, 1 kWh energy from a propane combustion engine is responsible for $(3.1 + 1.4) \cdot 0.24 = 1.08$ kg CO₂. Therefore, the produced CO₂ $c(n)$ after using n kWh energy with a propane combustion engine is

$$c(n) = 1.08 \cdot n \quad [\text{CO}_2]$$

2. CO₂ Lithium Ion Battery:

The CO₂ emissions for the production of the battery are approximately 175 kg per kWh [20]. For a 10 kWh battery this would lead to a total production emission of $10 \text{ kWh} \cdot 175 \text{ kg/kWh} = 1750$ kg CO₂ just for the production. In 2018, the generation of 1 kWh electrical energy produced 0.159 kg CO₂ in Switzerland [21]. Therefore, the produced CO₂ $b(n)$ after using n kWh energy with a battery is

$$b(n) = 1750 + 0.16 \cdot n \quad [\text{CO}_2]$$

To know at which point the battery is cleaner than the combustion engine, the following inequality has to be solved:

$$\begin{aligned} b(n) < c(n) & \iff 1750 + 0.16 \cdot n < 1.08 \cdot n \\ & \iff 1750 < 0.92 \cdot n \\ & \iff n > 1902 \end{aligned}$$

After using 1.9 MWh of energy, the battery is cleaner. This corresponds to 190 battery cycles. As the lifetime of the battery that would be used for the robot is 3000 battery cycles [22] and it is planned to use the robot over the years more than the 190 battery cycles, the lithium ion battery should be preferred over the propane combustion engine.

Battery

Three main groups of batteries were looked into: Lithium-ion, lithium-polymer and lead-acid.

1. *Lithium-polymer*: They have a very high volumetric capacity, but batteries with a capacity of 10 kWh are not available on the market. A self developed interconnection of individual batteries would have been possible, but this was decided against due to the increased time required and deteriorating safety standard.
2. *Lead-acid*: A second option would be a lead-acid battery which are characterized by a very high capacity per price. Because of the very low gravimetric capacity (about 3-4x lower than a lithium-ion battery) it is not suitable for this robot.
3. *Lithium-ion*: The last and best option are lithium-ion batteries. On the market there are many ready to use lithium-ion batteries available. Most of them are built for solar, marine and RV applications. Those batteries use LFP as the active cathode material, which makes them a good lead-acid battery replacement because of the similar output voltage[23].

Batteries from *Ecovolta* offered a more suitable product (see fig. 4.2). Their battery uses NMC as the active cathode material which is more preferable than LFP for electric powertrains. The *evoTractionBattery* from *Ecovolta* offers an easy installment energy storage solution for aftermarket electrification of internal combustion vehicles. The battery is offered in many different configurations and those can be connected in serial or in parallel. It also includes a configurable battery management system to allow high safety standards.

For the robot, two batteries (nominal voltage: 48 V, capacity: 5 kWh) and a charger (power: 4 kW) were chosen. The system allows charging while consuming power which enables uninterruptible testing. Two batteries were chosen instead of one to have a higher flexibility in the mechanical design of the robot. With the disadvantage, however, that the weight and volume is slightly higher, but due to the communication between the batteries and the voltage balancing made possible by this, the electrical properties remain unchanged. The chosen battery system has a continuous discharge current of 180 A which is clearly enough for the needed 43 A and also leaves enough headroom for uncertainties and future expansions of the robot. The IP67 ingress protection code also allows outdoor use without any further enclosures.

Solar Panel

Due to the dependence on the weather, solar cells alone are an unreliable energy source. However, together with a battery to increase the run time of the system it is an interesting approach. However, it is not a worthwhile solution for this system, as the energy produced is far too small compared to the power consumption of the system:

Best peak power density available on the market: 200 W/m² [24] (see fig. 4.3)
 Top surface of the robot: 0.91 m²

picture	Name	chemistry	Capacity [Wh]	Voltage	Volume	C/V [Wh/m³]	weight	C/W [Wh/kg]	IP	BMS
	MLI Ultra 12/2500	LiFePO4	2.500	13,2	0,025	100.697,1	31,00	80,65	65	☒
	evoTractionBattery	NMC	2.500	24,0	0,016	157.526,0	22,00	113,64	67	☒
	B-LFP12-100	LiFePO4	1.200	12	0,012	98.509,0	14,00	85,71		☒
		LiFePO4	2.560	25,6	0,012	206.003,1	28,50	89,82		☒
	CSX12100-BMS100	LiFePO4	1.280	12,8	0,012	109.593,7	10,50	121,90	65	☒
	URB121000	LiFePO4	1.280	12,8	0,012	105.453,9	13,90	92,09		☒
	BB10012	LiFePO4	1.200	12,0	0,013	95.850,6	14,06	85,35		☒
	SB12V210E-ZC		2.688	12,8	0,030	90.435,2	23,00	116,87		☒
	4 * 250Wh Lipo	Lipo	1.000	51,8	0,004	249.750,2	7,22	138,50		☐
	Lead Crystal 6-CNF Bleikristall		1.200	12,0	0,012	96.098,4	31,50	38,10		☐
	Lithium Superpack	LiFePO4	2.560	12,8	0,029	87.987,5	32,00	80,00	43	☒
	evoTractionBattery	NMC	10.000	24,000	0,036	281.075,7	56,40	177,30	67	☒
	evoTractionBattery	NMC	5.000	48,000	0,025	198.665,0	33,00	151,52		☒

Figure 4.2: Evaluation of different batteries

The peak power of the solar panel is based on a vertical irradiation of 1000 W/m^2 and a cell temperature of 25°C . At the upper limit of the atmosphere, the solar irradiation is at about 1352 W/m^2 . Due to the atmosphere, however, this is reduced by about a quarter to about 1000 W/m^2 . Since the sun is never perpendicular to the ground in Eschikon and therefore never to the surface of the robot, the surface power density must be further adjusted [25]. The maximum solar radiation in Eschikon (47.5°N) is:

$$I_{max} = 1000 \text{ W/m}^2 \cdot \sin(90^\circ - (47.5^\circ - 23.5^\circ)) = 913.19 \text{ W/m}^2 \quad (4.2)$$

Covering the whole top of the robot with solar cells would lead to the following power:

$$P_{produced} = 0.91 \text{ m}^2 \cdot 200 \text{ W/m}^2 \cdot \frac{913}{1000} = 166 \text{ W} \quad (4.3)$$

If the battery is large enough to fulfil the requirement of 5h run time alone, the energy produced by the solar panel would be:

$$E_{produced} = 5 \text{ h} \cdot 166 \text{ W} = 830 \text{ Wh} \quad (4.4)$$

This would increase the run time as follows:

$$E_{produced} = 830 \text{ Wh} / 2072 \text{ W} \approx 0.4 \text{ h} = 24 \text{ min} \quad (4.5)$$

So, the run time would be increased by about 8%. But this is only possible if it is in the middle of the summer, there are best weather conditions, cooled solar cells and also only if the robot is in use between 11am and 4pm. So under normal conditions the effect of a robot sized solar panel is negligible. One could increase the area of the solar cells but this would add a lot of weight at the highest point of the robot and thus reduce the stability of the robot, provide much larger wind attack surface and greatly increase the complexity of the system by a necessary fold in mechanism

for the overhanging solar panels.

For example with a target of 1 extra hour left of run time after 5 h the solar panels would need to produce 2072 Wh in 5 h. This leads to a needed area of 2.26 m² which is about 2.5x the top surface of the robot given the optimal weather conditions described before.

picture	Name	flexible?	Power (peak) [W]	Voltage	Max. po current	Weight [kg]	area [m^2]	Wp/m^2
	SolarWorld mono	no	100	12	5,62	7,2	0,63315	157,94
	SolarWorld flex	yes	100	12	5,55	2,7	0,5022	199,12
		yes	120	18	5,6		0,6968	172,22
	Flisom eMetal	yes	30	34	0,88	1,6	0,372288	80,58
	Victron Energy	no	215	24	6,3	15	1,27664	168,41
	SPR-E-FLEX 110	yes	110	18,8	6,3	2	0,64774	169,82
	Offgridtec ETFE	yes	110	18,56	5,97	2,5	0,5778	190,38

Figure 4.3: Evaluation of different solar panels

Review

1. *Environmentally friendly:*

To be more environmentally friendly than a propane generator with the chosen battery, the charging cycles will not be sufficient until the end of the project. However, since a long-term use of the robot is also planned, this goal can still be achieved. A big advantage is that the batteries are emission-free. Not only exhaust fumes free but also vibrations which would have disturbed the cameras.

2. *Run time and recharging:*

The estimation of the energy consumption is on the one hand too high, because the extent of the system has changed, and also the assumption that 5 h of testing means uninterrupted driving while pulling the goose feet in the ground is also exaggerated. On the other hand the capacity enables a full day of testing without the fear of running out of charge and it also leaves enough headroom for expansions which will already be needed after the end of the project for bachelor theses.

3. *Weight, size and placement:*

By choosing two single batteries, the centre of gravity could be set very low and still leave a large free space in the middle of the robot.

4. *Output:*

It turned out that the battery reacts quite slowly to changes in load. This resulted in power being drawn for the motors from capacitors inside a DC/DC converter (see section 4.3.2) for a short period of time. This caused the fuse for the DC/DC converter to blow although the converter was working correctly. This was fixed by placing capacitors near the motor drivers.

In order for the battery to switch on without high currents, it is necessary that as many loads as possible are turned off when the battery is switched on. This could be guaranteed for all loads except the motor drivers. Because this requirement only became apparent at a late stage.

5. *Easy and simple installment:*

The whole system was available from one manufacturer, unfortunately some of the cables were defective or delivered incorrectly. Additionally, the batteries did not function properly during the first operation. In one case even the BMS had to be replaced. Furthermore the documentation is very limited. Only a system is provided which disconnects the batteries when the emergency stop is pressed but does not switch off the charger. This functionality had to be added later to ensure a safe operation.

4.3 Power Distribution

Motivation

The power distribution forms the connection between the energy source and the consumers. On the robot are mainly three voltages available: 48V, 24V and 12V. For some of those a DC/DC converter or a distribution board is needed. An overview is shown in fig. 4.4 and fig. 4.7.

4.3.1 48V Distribution Board

Overview

The 48 V board is the distributor for all 48 V consumers. It is made of a PCB and two busbars. The following components are connected to it:

- 2 batteries and 1 charger
- 8 motor controllers
- 3 12 V DC/DC converter
- 1 24 V DC/DC converter
- 1 48 V DC/DC converter

Requirements

1. *Current-carrying capacity:*

In table 4.1 an average power consumption was calculated. This leads to the following average current:

$$I_{avg} = 2072 \text{ W} / 48 \text{ V} = 43.2 \text{ A} \quad (4.6)$$

Thus, the distributor must conduct at least 43.2 A continuously without heating up considerably during this time. Short-term currents can be significantly higher. This is the case, for example, when the robot encounters obstacles such as stones. The charger also delivers up to 80 A. This means that the energy supply to the batteries must withstand at least 80 A.

2. *Protection against accidental contact:*

With the nominal voltage of 48 V, the system is in the area of protective low voltage, which basically does not require contact protection. However, due to a fully charged battery, the voltage can come close to the 60 V at which a contact protection is necessary. The contact protection is not only intended to protect people, but also the robot itself from unwanted short circuits.

3. Overcurrent protection:

In order to minimize the consequences of electrical failures, consumers must be individually protected with a fuse. In addition, the fuse should be easily replaceable.

4. Voltage detection:

It should be easily visible if the system is turned on or off. This is especially useful to avoid accidental hot plugging (connecting or disconnecting a plug while the system is turned on).

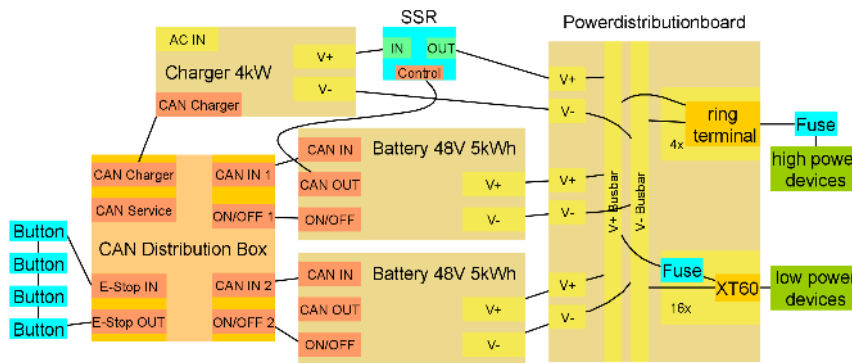


Figure 4.4: Powerdistribution and safety

Implementation

The easiest way to build such a power distribution would be to use copper bars and screw everything with ring cable lug on top of it. If one added a cover, the protection against contact is also guaranteed. The only way to add fuses for each individual load would be to mount them in the cable. However, this would be very cluttered. Furthermore, every time a load has to be unplugged, a screw has to be loosened.

Another possibility would be to design a PCB, which has connectors and fuse holders soldered on it. With an LED, it would also be easy to see if there is a voltage present. But for such high currents the copper thickness of the PCB would have to be very thick, which would increase the price immensely.

That is why the combination of those two was chosen. The high loads such as traction motors, charger and batteries are mounted directly onto the copper rails. The remaining loads are attached to the PCB via a connector. The only disadvantage is that the traction motors need a fuse attached to the cable. A simplified schematics can be seen in fig. 4.4. A picture of the final product is shown in fig. 4.5.

Review

The distribution board turned out to be reliable. The easily replaceable fuses have proven to be practical, especially in the initial testing phase. Only the footprint of the fuses was drawn a bit too small, so therefore the fuse must be removed carefully. Additionally the power distribution board needs a lot of space in the electric box. A better arrangement of the connections might have made a more compact installation possible.

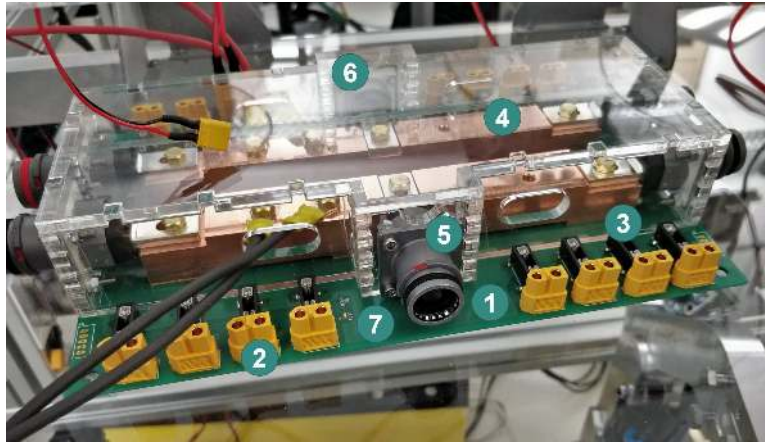


Figure 4.5: Powerdistribution board: PCB (1), connectors for low current devices (2), fuse holder (3), copper busbars (4), connector for charger (5), casing (6), power indication LED (7)

4.3.2 12V DC/DC Converter and Distribution

Motivation

On the robot are many 12V devices with slightly different requirements. The selection of the correct DC/DC converter is necessary for the reliable operation of the system.

Requirements

1. *PC*:

The *Intel NUC* has an input voltage between 12–19 V. To ensure reliable operation, the output of the DC/DC converter should be a bit above 12 V. The *Intel NUC* consumes up to 100 W of power.

2. *Router, switch, wireless E-stop, raspberry PI and USB hub*

The switch, USB hub and light strips controlled by the *raspberry pi* operate only at 12 V. The light strips consumes up to 80 W. The rest consumes at peak about 30 W.

3. *STO*:

The *STO* consumes about 4 W. The input voltage range is between 12–24 V. To ensure reliable operation the output of the DC/DC converter should be isolated and a bit above 12 V.

4. *Fans*:

The robot has 8 fans with an input voltage range of 7–13.8 V and each fan has a power consumption of 2 W.

Implementation

With these requirements there is no possibility to use only one 12 V DC/DC converter, because the *STO* and *PC* need more than 12 V but does not work with for example the switch. A PCB design to mount three DC/DC converter on top of it was provided by ASL. The compatible DC/DC converters are from the *Hammerhead* series manufactured by *ABB*. The output voltage is isolated and is either

3.3 V, 5 V or 12 V. It can be adjusted by $\pm 10\%$. The input voltage is between 18–75 V. The power is between 50 W to 225 W. On the final product the following DC/DC converter was used:

1. For the STO and the fans one 50 W isolated DC/DC converter (part number: *KHHD004A2B41Z*) trimmed to 13 V was used.
2. For the PC one 120 W isolated but with the same input and output ground potential DC/DC converter (part number: *EHHD010A0B41-HZ*) trimmed to 13 V was used.
3. For the router, switch, wireless E-stop, *raspberry pi* and USB hub one 120 W isolated but with the same input and output ground potential 12 V DC/DC converter (part number: *EHHD010A0B41-HZ*) was used.

The final product can be seen in fig. 4.2(a).

A small PCB was developed for distribution. Connectors for the respective devices are mounted on the front. On the back side SMD fuses for the respective connectors are attached. The PCB can be seen in fig. 4.6. The connectors are yellow. Each group of connectors are labeled with its respective function.

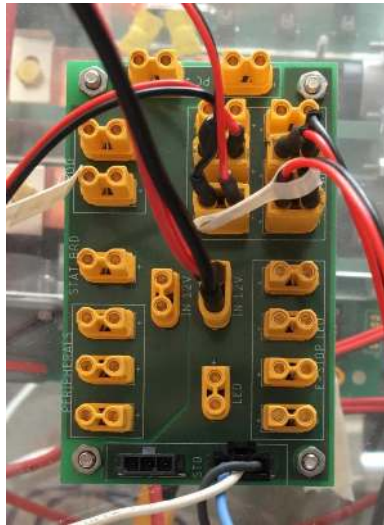


Figure 4.6: 12V distribution board

Review

12V DC/DC converter

The PCB was designed for a 24 V input voltage. Therefore, capacitors with a 50 V rating were used. The charged batteries, however, are well above 50 V. Therefore new suitable capacitors with a higher rating had to be found. This became clear only at a late stage due to a misunderstanding and led to a delay in production. Since the DC/DC converters were put into operation they have worked flawlessly. All devices were successfully supplied with power.

12V distribution board

The connected components have not shown any malfunction. For this reason, no fuses have yet had to be replaced. However, by plugging in the wrong place, fuses have already been triggered. Therefore it would have been worthwhile to use different connectors for different devices. To change a fuse one also needs a soldering iron which is very time consuming when for example testing the robot in the field.

4.3.3 48V and 24V DC/DC Converter

Motivation

The linear motor is the only motor for which no 48V variant could be found. In addition, the voltage change of the battery during discharge is too high for the *Aquila*. Therefore two additional DC/DC converters are necessary.

Requirements

1. *Aquila*:

The nominal voltage of the *aquila* and of the batteries are both 48 V but the output voltage range of the batteries is larger than the input voltage range of the *Aquila*. This means the *Aquila* cannot be directly connected to the batteries.

2. Linak LA25:

The input voltage of the *Linak LA25* is 24 V. It consumes up to 60 W of power. The voltage should be isolated because the Linak and the battery share one CAN network.

Implementation

48 V DC/DC converter

For this application the *DDR-120C-48* was used from *MEAN WELL USA*. With an input voltage range of 33.6 V to 67.2 V, it is fully compatible with the output voltage of the battery. This DC/DC converter delivers up to 120 W of power, leaving even a little room for expansion. The output voltage is 48 V isolated. Only the power density is not very high. However, this is subject to the argument that it can easily be put into operation and used without a self-developed and assembled PCB. This DC/DC converter can be seen in the top left corner in fig. 4.2(b).

24 V DC/DC converter

For this application the *DDR-240C-24* was used from *MEAN WELL USA*. It has the same features as *DDR-120C-48* but the output voltage is only 24 V and the output power is 240 W. Although the *Linak* motor consumes only 60 W, it was decided to use the 240 W model instead of the 120 W model. The reason for this is that a second *Linak* motor is planned (see section 3.5.3). This would increase the total consumption to about 120 W. To be on the safe side and to be able to have future expansions possible, the 240 W model has been chosen. This DC/DC converter can be seen in the bottom right corner in fig. 4.2(a).

Review

Only at a late stage, it became apparent that it was necessary to be able to switch the DC/DC converters on and off. Everything except the 48 V DC/DC converter

already had this functionality. So it is planned to replace it. To do so one can simply upgrade it with the 240 W version, which has this feature.

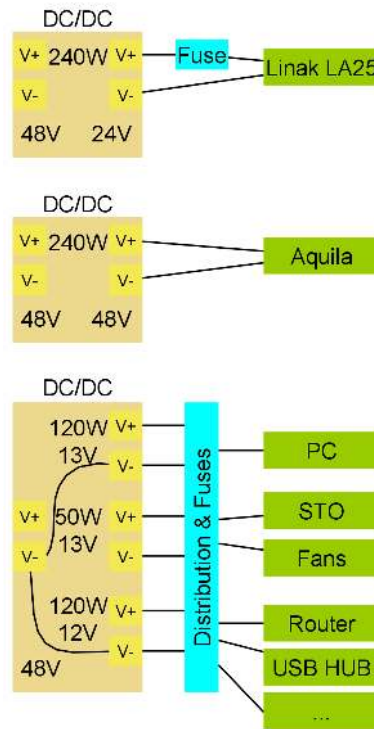


Figure 4.7: DC/DC converter setup

4.4 Safety

There are three main electrical safety features:

Emergency off switch (SRS: 5.1.3)

In case of a severe malfunction the robot can be stopped by 4 emergency stop switches, which are located in each corner of the robot (see fig. 4.8). As you can see in fig. 4.4, these buttons (turquoise) are directly connected to the batteries via the CAN distribution box. As soon as a button is pressed, the robot is completely disconnected from the power supply. This feature is provided by the battery. Unfortunately, the charger is not switched off, so the robot will still be switched on when the emergency stop is pressed and the charger is connected. To fix this, a solid state relay (SSR, turquoise) has been connected between the charger and the distribution board. This will disconnect the robot from the power supply but the safety is not as high as with a mechanical relay. A mechanical relay was not possible as this would consume too much power from the 12 V battery supply.

STO (SRS: 5.1.2)

The Elmo motor drivers have two Safety torque off (STO) inputs. This means that as long as the voltage on these inputs is below 12 V, the motor is not powered. The STO inputs of all eight motors are connected in parallel. One of the inputs can be



Figure 4.8: E-stop button and STO button

switched on by a button located on the wall of the electronics box and the other one is connected to a radio receiver which can be switched on and off by remote control. The wiring of the STO (turquoise) is shown in fig. 4.9. This safety feature has proven to be particularly useful during testing. It allows you to get closer to the robot without the risk of it moving unexpectedly. For the parallel connection a small PCB was developed. It contains 4 connectors: 2 connectors for connecting two Elmos and two connectors for connecting them in series. It also contains two LEDs to indicate the current state of each STO input.

Fuses

In order to minimize damage in the event of a malfunction that leads to an over-current, each load is individually protected (see fig. 4.4). In addition every load, which is connected to a DC/DC converter is also protected (see fig. 4.7). In order to simplify exchangeability, automotive fuses were used for the most part.

4.5 Data Processing and Connectivity

4.5.1 Data Processing

Motivation

To breathe intelligence into the robot, a sophisticated control unit is required. The selection for the correct computing unit is explained in the following section.

Overview

As a computer the *Intel NUC8I7BEH2* was chosen. It is equipped with an i7-8559U processor (up to 4.5 GHz), 512 GB SSD and 8 GB RAM.

Requirements

1. *Computing power:*

The computing power of the system should be as high as possible, because at the time of decision, the algorithms that will be used later were not known and due to the tight schedule, as little time as possible should be wasted on

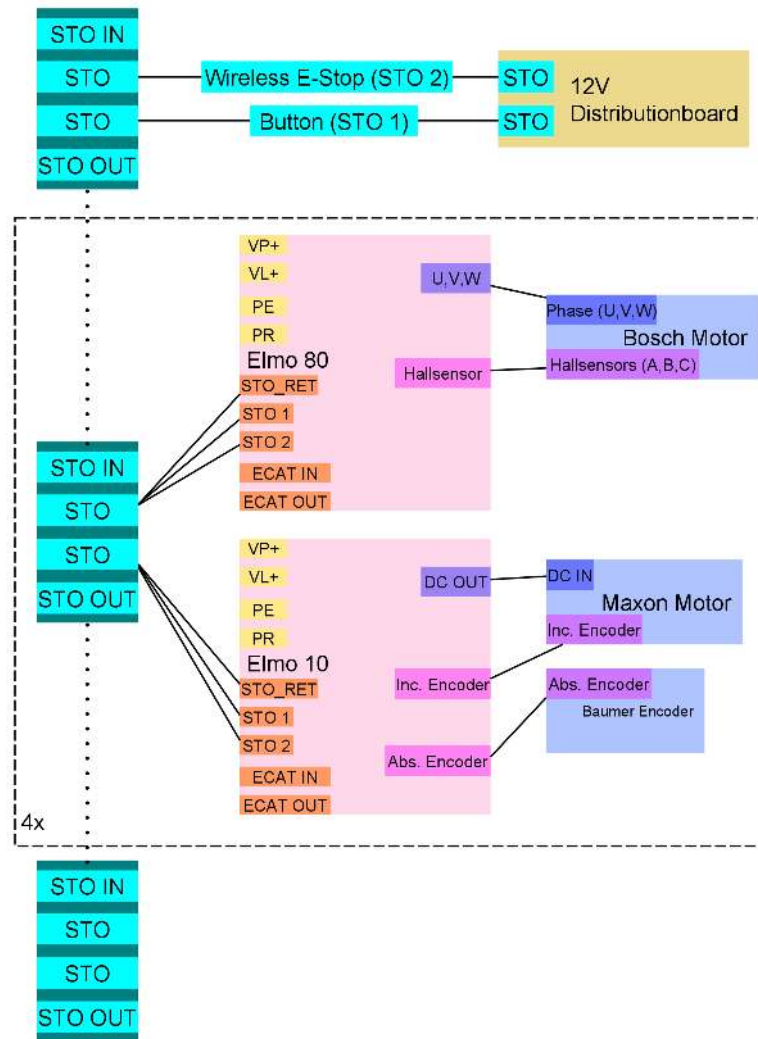


Figure 4.9: Safety torque off setup

optimizing the code. The goal is that the computer is not a bottleneck of the robot.

2. Inference:

For row detection, image processing is used. This can often be accelerated by dedicated graphics cards. Therefore, it is an advantage if the machine is equipped with this feature.

3. Energy efficiency:

To allow the robot to run as long as possible, the computer should have a high energy efficiency. The computer should account for only a small part of the energy consumption of the system. This means no more than about 100 W continuously.

4. Size and weight:

The computer should be light and compact. The focus is on a small size, as the space available for non-waterproof components is limited and the weight savings are minimal.

5. *Connectivity*:

The computer should have many and different connection possibilities to enable the control of actuators and the readout of sensors. At least one ethernet port and 2-3 USB ports are needed.

Research and Evaluation

Different possibilities were evaluated:

Mini ITX

Mini ITX is a designation for the mainboard size. It is the smallest mainboard size which is standardized [26]. The smallest systems have a volume of about 4-7l [27] and are therefore still very large compared to the other options (about 6x larger than a *NUC* or a *Jetson*). There is the possibility to extend the system with a graphics card. By using a strong desktop processor, the computer should be powerful enough. However, such systems are not designed for energy efficiency at all, for example the *Intel i7-7700*, which is a desktop class CPU with a similar performance as the *Intel i7-8559U*, which is used for instance in a *Intel NUC*. [28] However, the *NUC* processor name ends with a U which means ultra low energy consumption. [29]

Nvidia Jetson [30]

The Nvidia Jetson series offers computers tailored for autonomous machines. This is especially noticeable by the powerful graphics card (up to 11 TFLOPS (FP16)) in combination with the high energy efficiency (30 W). The computer is similarly small as the *NUC* with 0.71l. The biggest weakness is that the processor has an ARM architecture, which means that the instructions the CPU can execute are not the same as used for an *Intel* CPU. Therefore most of the software online available does not work on this architecture or only after a lot of effort.

Intel NUC [31]

Intel offers with its *NUC* line-up very compact (about 0.7l) computers with a high energy efficiency despite a powerful processor. There are also *NUCs* with a dedicated graphics card from *AMD* which would be the ideal solution, but this graphics card is not compatible with GPU acceleration using *openCV*.

The *Intel NUC* offers the best option because mini-ITX systems are too big and the *Nvidia Jetson* delivers a high risk with incompatible software. If a GPU acceleration proves necessary, there is the possibility to upgrade the system with a *Jetson*.

The *Intel NUC8I7BEH2* was chosen because it is one of the most powerful *NUCs*. It is equipped with an i7-8559U processor (up to 4.5 GHz), 512 GB SSD and 8 GB RAM. The only *Intel NUC* with a similar performance would be the *Intel NUC8i7HVK*. This *NUC* is not as power efficient as the other one and is also nearly double in size.

Review

1. *Computing power*:

The computer is powerful enough. Only for the real time critical EtherCAT, small optimizations were necessary.

2. *Inference*:

It was not necessary to accelerate the image recognition with a GPU. The CPU was already sufficient.

3. *Energy efficiency:*

Energy efficiency has turned out to be less important than expected. Because no active tool with high energy consumption was implemented, the batteries have a great safety factor in terms of capacity.

4. *Size and weight:*

The space in the waterproof box turned out to be very limited, a bigger computer would not have had room.

5. *Connectivity:*

The available connections on the computer are relatively limited. A second ethernet connector would have been advantageous, one for the EtherCAT and a second for the internet. There are also quite a few USB devices connected, but this was answered by a USB hub. The current setup is just sufficient, so there is little room for expansion.

4.5.2 Connectivity and Feedback

Motivation

Some additional hardware is required to enable communication between the individual computers. It is also important to get feedback from the robot during the test. This can be done via data transmission or a status indicator.

Router

To enable wireless debugging over WLAN or over the internet, a router is part of the system. In the future, it can also serve as an interface for a user. A *RUT240* from *Teltonika* was chosen. It offers 2.5 GHz WLAN and category 4 LTE. Furthermore, the router is characterized by a compact design and is intended for use in mobile industrial applications. [32]

Switch

The router is complemented by the *Netgear GS108GE* switch. It provides 8 Ethernet ports. Most ports are not used but provide easy system expandability. This switch was chosen because it can be operated at 12V like the router. [33]

ROS Network

To build a ROS network, the machines must have a bidirectional connection to each other. This is done by connecting them all to the switch. The slaves must also know the IP address of the master, so it is advantageous if all addresses do not change. This task is done by the router, which serves as a DHCP server and thanks to the constant MAC address, the router can always assign the same address to the *NUC*, the *Aquila* and the *Raspberry Pi*. The network setup can be seen in fig. 4.10.

Status Light

In order to recognize what the robot is doing, i.e. whether it is driving autonomously, remotely controlled, a fault has occurred or something similar, the robot has 3 light zones in the front and one in the back that can be seen by the operator (see section 5.7 and fig. 5.15). For this purpose RGB-LED strips are used, so that every colour can be displayed. The 4 RGB-LED stripes are controlled by a *raspberry pi*. On the *emphrasberry pi*, there is a custom PCB (see fig. 4.11). On

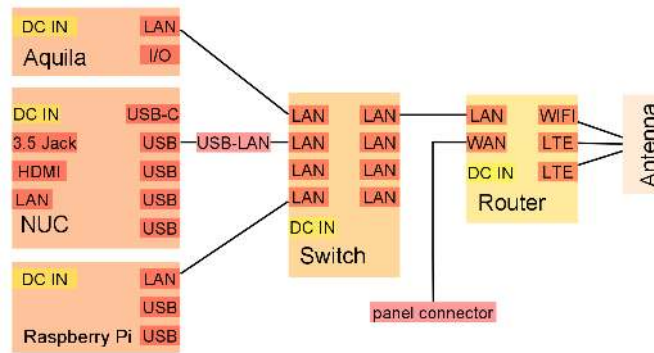


Figure 4.10: The network setup

this PCB, there are mostly mosfets, which control the 3 channels and can therefore display any colour.



Figure 4.11: Rasperry pi with a custom hat on top of it

4.6 Motor Controller

Overview

The motors need to be controlled. For that to happen robustly, it is important to have the proper hardware. Different types of motor controllers were considered, in the end *Elmo* controllers were chosen for the control of the locomotion actuation. The *LINAK* actuator, which is used for the passive tool (see section 3.5.2), has an embedded controller. Hence, no additional controller is needed.

Requirements

Motor

The motors for the locomotion were selected based on their respective, overriding requirements. The controllers then had to be chosen such that they are compatible with the chosen motors. The main constraint given by the motors is the maximal power of 2.4 kW for the hub motors and 200 W for the steering motors, which need to be transmitted by the controllers.

Control Frequency

In order to control the motors, a suitable control frequency is needed. The most critical components, which have to be controlled, are the steering motors since their control frequency has a direct effect on the accuracy of the navigation. The selected controllers and their respective communication protocol gives a range of possible frequencies.

Voltage

The controllers need to be compatible with the voltage provided by the batteries, which is in a range of 48-58 V.

Hub Motor Controller

The locomotion is based on two types of motors: the hub motors and the steering motors. The hub motors provide the thrust for moving the wheels forward (see section 3.3). In order to be sure that the motor-controller-encoder combination is compatible, the coupling had to be tested in practice since the chosen hub motors did not come with a data sheet with the required information. The hub motor was tested in the first semester in combination with an *Elmo* controller. They have an EtherCAT interface which allows a 100 Mbit/s bitrate, enabling an update rate higher than needed for the motors. Since the outcome of the tests was positive and the high bitrate allowed to connect multiple controllers on a single bus, those *Elmo* controllers were implemented for Rosie. The *G-SOLTWIR80/80EE1H* is the exact *Elmo* controller which has been chosen for the hub motors, because it works up to 80 A with 80 V. This specifications come directly from the voltage (section 4.6) and the motor section 4.6 requirements.

Steering Motor Controller*Overview*

Each wheel is equipped with a steering motor, that provides a torque about the z-axis to orientate the wheel to the desired direction (see section 3.4). To control the steering motors, two different controllers were taken into consideration. On one side, there was the possibility to use the controllers made by the same producer as the motor - a *maxon EPOS*. On the other side, an *Elmo* controller, which is also used for the hub motors, could be implemented.

EPOS

The *maxon EPOS* suits the steering motors perfectly since they were designed to control these motors. The *EPOS* is available with both an EtherCAT or a CAN interface. Furthermore, this controller also suits the sensors chosen for the motors. The only downside of the *EPOS* controller is that no previously built SDK was present and the whole lowlevel software would have needed to be built from scratch.

Elmo

The *Elmo* controllers also had the specifications to properly control the steering motors and this specific combination has - as stated by the manufacturer - successfully been done before. The *Elmo* controllers provide an interface to communicate through EtherCAT and could be interfaced to all chosen encoders. Furthermore, there is an SDK which has previously been built by the RSL which could be implemented without needing to build the whole communication software.

Decision

Based on the information stated above, the *Elmo* controllers were chosen because

the controllers can be used with a previously implemented SDK, which saves time that can be used for other tasks. Furthermore, one single software for tuning and testing the motor-controller-encoder setup can be used, namely the *Elmo Application Studio*. The specific controller chosen to control the steering motor is the *G-SOLTWI10/100EE7H* which works up to 10 A and 100 V.

Review

The decision to unify all the locomotion controllers paid off since only one new tuning software had to be learned. To tune the *Elmo* controllers, an external computer had to be connected to the hardware. This process could be widely optimized, because the same procedure could be repeated with the same plugs on all eight controllers.

Flaws

The power needed to control the motors was calculated by multiplying the current times the voltage. In the case of the *maxon* motors, which provide an exhaustive data sheet, the theoretically calculated value, which was used to select the controllers, proved to be correct in practice.

In the case of the hub motors, the power of the controllers seems to be under dimensioned. The calculation made to choose the controller for the hub motors was $80\text{ A} \cdot 50\text{ V} = 4\text{ kW}$. The information provided by the producer stated that the motor draws a continuous power of 1.5 kW.

The tests in the second semester, however, revealed that even though the power consumption of the motors is very low (500 W), the motor draws high currents (80 A). This means that the peak power of the motor cannot be reached while driving slowly. This was not clear during the selection of the controller since the motor does not provide a data sheet and there was not enough time for testing the motor before choosing the right controller in the first semester.

More tests and inquiries have to be done in the future to find out the correct specifications and the limits of the hub motors.

4.7 Electrical Box

The electrical box, which is shown in fig. 4.12, is the cabinet which contains all the electrical elements needed to operate the different subsystems. It must provide shelter from dust and water (at least IP 54) for the electronics and allow the easy exchange of components. This was reached by organizing all the components on two floors and connecting them through different plugs with the outside. The lower level contains all the heat producing parts such as the DC/DC converters and the motor drivers. Cooling is ensured by using ventilated cooling elements on the bottom outer surface of the box. The upper level contains the components, which often need to be reached from the outside. The lid of the box is made out of transparent acrylic glass such that the important status lights of the heat producing motor drivers are visible. A detailed schematic of the electrical box may be found in fig. 4.2(b) and fig. 4.2(a).



Figure 4.12: The Electrical Box

Chapter 5

Software

In the following chapter the software that is implemented in the project Rowesys is described and explained. First, there will be an overview of the software and of how the single software components are interconnected. In the second part of the chapter, the most important parts of the software will be analyzed following a bottom up process, starting from the lowest controls and ending with the visual sensor algorithms.

Requirements

The goal of the project was to build a robot able to robustly navigate in rough terrain and that is capable to autonomously weed a field using only visual sensors and without requiring to build a GNSS map beforehand. Rosie is not meant for commercial use but merely as a study platform intended to be used by the research group only. Detailed requirements will be further specified in each single software section separately.

Realization

The software structure can be subdivided into four main levels (see fig. 5.1):

1. Perception
2. Planning
3. High Level Controller
4. Actuator Level Controller

The visual sensing and the user interface level is responsible for taking external inputs such as joystick inputs or camera images, and emulating them in order to know the state the system is in and decide what the next goal of the system is.

The planning is given a certain state and then defines how the entire base has to move and which configuration it should be in.

The HLC takes the desired base motion provided by the planning level and computes the motion reference of each actuator such that the base performs the desired motion defined by the planner.

The actuator level control tracks the position and velocity commands provided by the HLC, using PID and cascaded control loops.

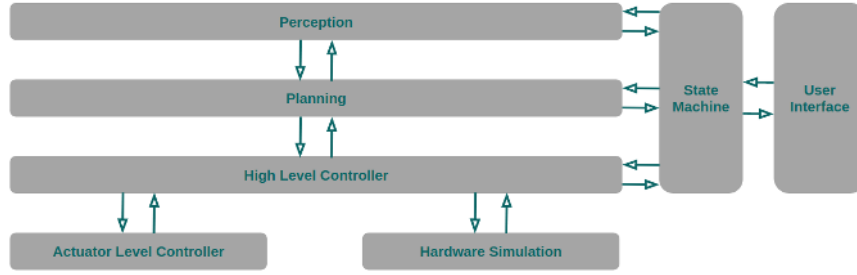


Figure 5.1: Software Structure Overview

Review

All the needed components have been implemented. Several unit and component testings have been carried out. Nevertheless, it is still not clear how well the robot's software performs. This is due to the fact that the whole system has not been extensively tested with all the components.

5.1 Actuator Level Control

Motivation and Overview

The section 5.4 are responsible for coordinating the movements of all actuators together. The actuator level controllers ensure that the respective actuators are properly tracking the position and velocity references. All eight locomotion actuators use an *Elmo G-SOLTWI* controller while the *LINAK actuator* uses its own position controller.

5.1.1 Elmo Control

Implementation

As previously mentioned, the hub motors and the steering motors are each controlled by an *Elmo* controller. Since the position is irrelevant to the function of the hub actuators, they are only controlled in velocity, which is already featured in the *Elmo* controller SDK provided by the RSL. The PI controller on the *Elmo* uses the data from a three phase hall sensor that is fed back to the controller.

The steering actuators on the other hand, require a position control. To accomplish this task, a cascaded control had to be implemented since the *Elmo EtherCAT* SDK provided by the RSL only supports velocity control. The inner loop of the cascaded controller is a PI velocity control loop that runs on the *Elmo G-SOLTWI*. The external loop is a proportional position control loop that runs on the main computer and feeds the velocity reference to the controller. The workflow is also shown in the block diagram visible in fig. 5.2. By using a cascaded control loop, no changes had to be made to the SDK.

Since each steering actuator possesses a gear box to allow a sufficient torque, two different sensors were implemented: An incremental encoder is used on the motor side of the gear box to have a proper velocity tracking of the actuator speed. On the load side of the gear box, an absolute encoder reads the position which is then fed back to the main computer enabling to track the exact position and compensate for the backlash caused by the gearbox. The setup is visible in fig. 5.3.

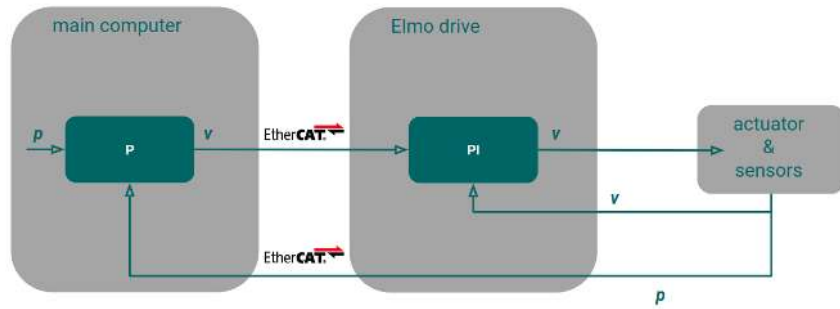


Figure 5.2: Cascaded controller for steering motor

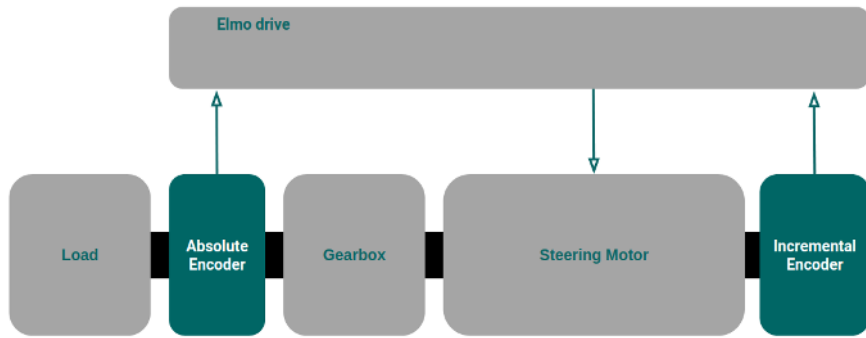


Figure 5.3: Steering motor setup

Tuning

To tune the inner PI controller of the cascaded control loop depicted in fig. 5.2 the software from *Elmo* called Elmo Application Studio (EAS) was used. To do so, the user first has to enter the specifications of the motor and sensors. In case of the steering this were the specifications of the *maxon* motor, the gearbox, the incremental encoder and the absolute encoder. The data sheets of these components can be found in the appendix appendix C.1.1. In case of the hub actuator the specifications of the motor and the the hall sensors which were already integrated in the hub motor have to be entered in the EAS. After entering all specifications, the EAS did the identification and commutation of the motor. Based on this identification the EAS calculated a plant for the current, velocity and position. Based on this plant the EAS directly suggested the P and I gains for the PI controller. After the suggested gains were adjusted such that the actual velocity and position were coincident with the velocity and position command as depicted in fig. 5.4 the PI controller was tuned properly.

Review

With the final choice of gains, the steady state error was negligible and the rise time fast enough for both the hub motor and steering motor setup.

Configuring the EAS correctly required several approaches until the identification was done properly and until the right gains of the PI controller were found. Furthermore, the 3 included hall sensors in the hub motor result in a very low resolution.

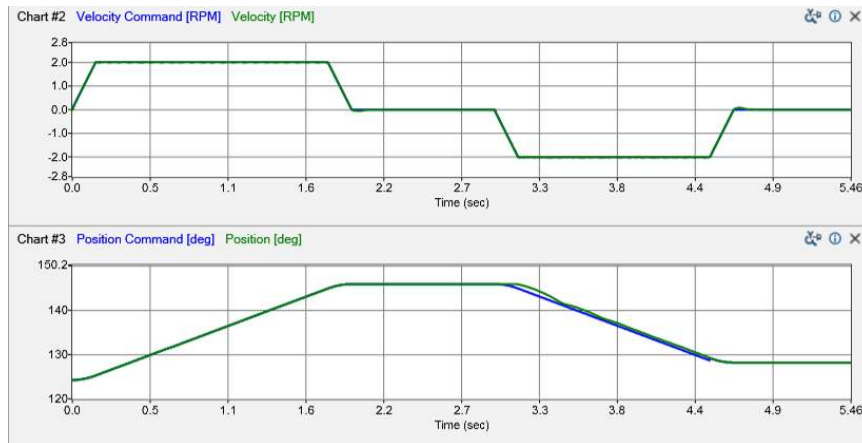


Figure 5.4: Plot in EAS after tuning the PI controller of velocity command and actual velocity as well as position command and actual position

Thus, the hub motor was not able to find a velocity and position plant which means the gains of the PI controller had to be found manually.

When turning on the robot, the absolute encoders used for the steering motors had to be manually set to zero since they were usually off by one turn. This problem could not be solved yet. To make it easy to set the absolute encoders to zero, hard switches were installed allowing to give the encoder the zero position when triggered.

5.2 Actuator Communication

Overview

The entire high level control for the robot is done on one central industrial computer. However, Rosie is composed of multiple subsystems, that each have to communicate with the central computer. To transport the data to and from the peripheral devices, a communication protocol supported by all communicating devices needs to be implemented. It is essential for the communication system to work robustly, since it marks the fundament for a working system. The robot is equipped with two different communication protocols: for all the locomotion actuators and hence time sensitive communication, an EtherCAT bus was used. For the battery and the passive tool actuator a CAN bus was implemented.

5.2.1 EtherCAT

The entire locomotion actuation is controlled by Elmo controllers - see section 4.6. The communication between the main computer and the locomotion controllers uses CANopen over EtherCAT. EtherCAT uses the Ethernet communication physical layer, hence an interface to the industrial computer is provided. This setup allows to communicate at a rate of 100 Mbit/s which gives the network a large bandwidth. The manufacturer guarantees support for up to 100 slaves. Since the messages are processed "on the fly", the time needed for them to be sent through the network and finally reach back to the master do not depend on the number of slaves. As such, the bottleneck for a high update frequency is not given by the EtherCAT for the amount of devices used.

The *Elmo* controllers support both EtherCAT and CAN protocols, but EtherCAT

was deliberately chosen because it provides a better performance than a CAN network and comes with a SDK from the RSL which can directly be used saving a lot of time.

5.2.2 CAN

The chosen Batteries have an ECU which exclusively uses CANopen over CAN for communication. Likewise, the *LINAK* linear actuator best suited for the passive tool offers only a CAN interface which uses the SAE J1939 protocol standard. Hence, it was necessary to have a CAN physical layer along with the aforementioned EtherCAT for communication with the BMS and the *LINAK* actuator. However, the main computer does not have a CAN hardware interface which is why an *IXXAT* USB-to-CAN adapter was used to enable a CAN communication. Although CAN communication does not support the same, high bitrate as EtherCAT, it did not create a problem, since the devices that use the CAN network are not time critical and as such do not need to satisfy a minimal update rate.

Implementation and Testing

The main challenge of the CAN network is that there are two different protocols on the same bus: SAE J1939 and CANopen.

SAE J1939 is a heavy duty vehicle standard that is widely used in agriculture. It communicates on 250 kbit/s rate and uses a 29-bit Identifier. CANopen is a protocol widely used in robotics. It allows data rates from 10 kbit/s up to 1000 kbit/s . The CANopen CAN-ID addressing schemata are based on the 11-bit CAN-ID but CANopen also supports 29-bit CAN-ID. By configuring the CANopen devices in such a way that they also run on 250 kbit/s and use the 29-bit Identifier, it is possible to use the two different standards on the same bus. In theory it is possible to run the devices on 250 kbit/s and keep the identifier at 11-bit by choosing the node address wisely, e.g. sending 0x1 with 11-bit is treated the same way as the 29 bit identifier.

A few more details have to be considered when running the two protocols on the same bus:

The SAE J1939 devices should not use the identifier value 0 which corresponds to the CANopen NMT and the identifiers for SDO. It is also important to rule out identifier conflicts when choosing the node address.

The two different protocol standards were implemented with their respective device separately and tested separately. The integration of them running on the same bus has been done but it has not been tested yet due to lack of time.

5.2.3 Review

The EtherCAT network has proven to work robustly. The only problem is that the *Elmo* controllers sometimes lose the correct position reading of the steering actuators. This might be due to an erroneous operational mode of the controller. It is still not clear when this happens and more testing to identify the source of the problem is needed. However whenever the problem occurs, it can be fixed by resetting the operational mode manually and hence this flaw did not affect the testing negatively.

The two CAN protocols have been successfully tested separately. However the integration of the two protocols has not been validated yet.

5.2.4 Hardware Interface

Motivation

For all the devices of Rosie to work properly, to coordinate the communication channels to the peripheral devices and to interface them to the ROS environment, an appropriate software is needed. An overview of the hardware interface is shown in fig. 5.5.

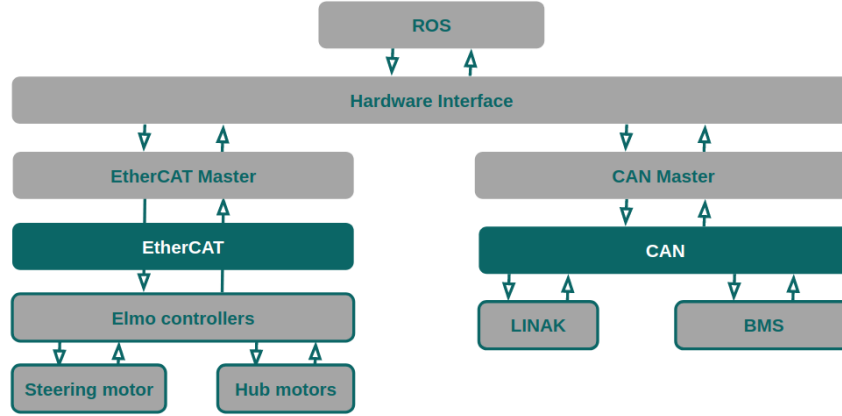


Figure 5.5: Hardware Interface

Implementation

The software responsible for the communication of all the actuators with the computer is the Hardware Interface. This core part of the software reads and writes data to the two networks asynchronously. The HWI also updates the HLC and stores the status of the actuators. The HWI is ROS.control compatible and inherits from its classes. By using this ROS compatible tools, it was possible to rely upon a strong community and to use open source libraries of software that have already been used many times, allowing to focus the limited time available on other tasks.

The HWI is able to communicate through the CAN and EtherCAT network by coordinating the EtherCAT Master and the CAN Master. The HWI on the EtherCAT network sends velocity commands to the *Elmo* controllers and reads position, torque and velocity states. The *LINAK* is position controlled and the HWI is able to read and handle errors of the actuator through CAN Master. The HWI also receives the battery status from the CAN network.

The states and commands of the actuator are stored by the HWI. All this stored data can be exchanged by the HWI with tools from the ROS environment.

5.3 Simulation

Overview

Rosie's software has many algorithms that depend on the software part responsible for the motion of the system. One goal for the project is that Rosie has to be autonomous, which had to be achieved in a short period of time. To satisfy the need to test the interaction of the motors, the different path planning algorithms and all the other overhead features of the robot, a simulation of the robot has been implemented (see fig. 5.6). By doing so, the whole software that makes the

motors move could be developed and tested at the same time. The simulation enables to test the features in parallel and facilitates future developments of new functionalities. Without the simulation the base software would have had to be perfected first and only in a second step the overhead software could be tested, which would have costed too much time.

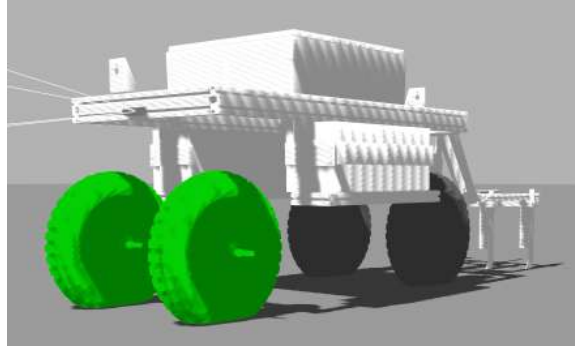


Figure 5.6: Robot in simulated environment

Implementation

The simulation was built using the model of Rosie developed in CAD as reference. The simulation model was built in URDF, using joints and links. The URDF is a tree based model which makes further extensions easily applicable. The joints were then actuated to allow to simulate an active model. Each link has its own inertia and weight and is visualized using meshes from the CAD model. This model is then spawned in a Gazebo-world - a world like condition with gravity force and friction between ground and the wheels.

The simulation only substitutes the lower control of Rosie, meaning that the control of the actuators and sensor readings, such as the encoder, are done in the simulated environment and not on the hardware.

This kind of simulation left the HLC and every software on top of that unchanged, meaning that there is no necessity to make two different HLC for the simulated and the real life application but only one unique HLC that worked in both environments. A graphical explanation can be seen in fig. 5.7.

Review

Especially due to the COVID-19 situation, the simulation has brought great advantages and made testing of the different features much more accessible.

The workflow allowed to test all new implementations in simulation first, without the need to access the hardware. Furthermore, the negative effects of hardware-related problems such as broken components or delivery issues were reduced by performing tests in simulation. Except for the wheel odometry (section 5.8.1), the software tested in simulation was working on the real robot without major problems.

5.4 High-Level Controller

While the actuator communication (section 5.1) is responsible that every motor is at its desired position, the high-level controllers translate a desired movement of the base into motor speeds and positions. As the robot moves in a plane, three inputs are needed to describe the movement of the robot: speed in x-direction, speed in

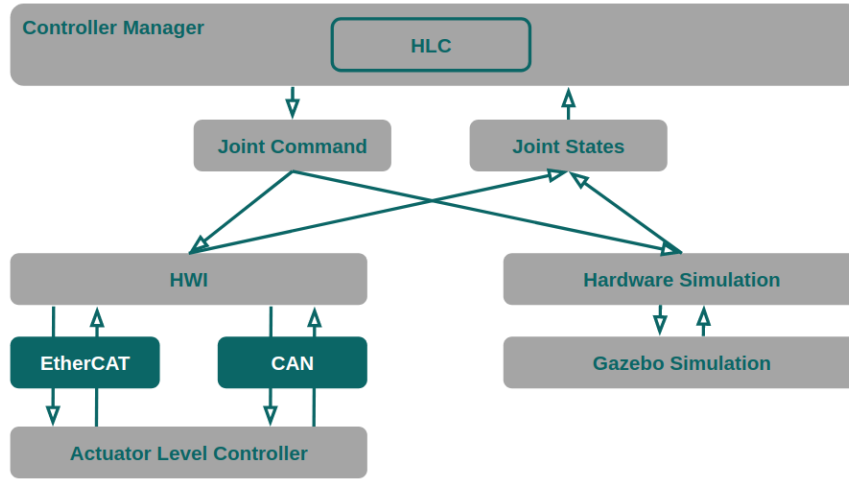


Figure 5.7: Block diagram of simulation

y-direction and angular speed around the z-axis. Using these inputs, the high-level controller calculates the corresponding positions for every steering motor and the speeds of the driving motors. Two different controllers were implemented.

5.4.1 Ackermann Steering Controller

Motivation

Especially for manual control, a simple and intuitive controller is important. Therefore, a simple ackermann steering controller was implemented, that only steers the two front wheels and keeps the others parallel to the x-axis of the robot. This mechanism is already well known as it is used in most cars and other vehicles. Therefore it is intuitive to drive the robot manually with ackermann steering. The controller is not only used for manual drive, but also for the navigation within the field where the geese feet are pulled through the ground. As the controller does not support movement in lateral direction, it is very suitable to pull the geese feet, as they can only be pulled in x-direction.

Implementation

In the following, the calculated speeds and angles of the wheels for a given velocity input of the platform are listed.

Let the desired movement of the platform be given by the speed vector $\begin{pmatrix} v_x \\ v_y \end{pmatrix}$ and the angular velocity ϕ around the z-axis. The distances between the front/back wheels is given by d_l while the distances between the left/right wheels is given by d_w . This is shown in fig. 5.8.

As ackermann steering does not support movement in y-direction, the speed v_y is not used. Also the center of rotation has to be outside of the robot, since the front wheels have to steer in the same direction. This results in the condition $|v_x/\phi| > \frac{d_w}{2}$. If this condition is not hold, the angular velocity is adapted such that the condition is tightly hold.

The calculated positions of the steering motors are the following:

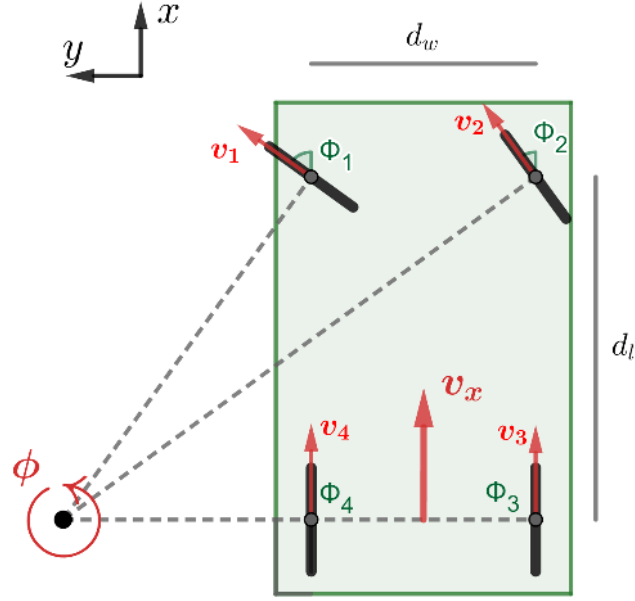


Figure 5.8: Ackermann steering

$$\phi_1 = \arctan\left(\frac{d_l}{\frac{v_x}{\phi} - \frac{v_l}{2}}\right)$$

$$\phi_2 = \arctan\left(\frac{d_l}{\frac{v_x}{\phi} + \frac{v_l}{2}}\right)$$

$$\phi_3 = 0$$

$$\phi_4 = 0$$

The speeds of the driving motors are the following:

$$v_1 = |\phi| \cdot \sqrt{\left(\frac{v_x}{\phi} - \frac{d_w}{2}\right)^2 + d_l^2} \cdot \text{sign}(v_x)$$

$$v_2 = |\phi| \cdot \sqrt{\left(\frac{v_x}{\phi} + \frac{d_w}{2}\right)^2 + d_l^2} \cdot \text{sign}(v_x)$$

$$v_3 = |\phi| \cdot \left(\frac{v_x}{\phi} + \frac{d_w}{2}\right) \cdot \text{sign}(v_x)$$

$$v_4 = |\phi| \cdot \left(\frac{v_x}{\phi} - \frac{d_w}{2}\right) \cdot \text{sign}(v_x)$$

To avoid that the motors drive while the wheels are not yet in the correct position, the controller compares the current position of the steering motors with the desired position. Only if these positions are close enough, the controller gives the speed commands to the driving motors. The allowed offset of the motor positions is set to 10° .

Review

The ackermann steering controller is one of the most frequently tested parts of the software stack, as it was always tested during manual driving. Because it was implemented when the actuator communication on the robot had not been ready yet, the controller was extensively tested in simulation first. The shift from simulation to hardware happened without major problems.

5.4.2 Swerve Steering Controller

Motivation

One of the core features of Rosie is her ability to steer all four wheels independently. This allows her to perform any planar movement. Therefore a swerve steering controller is implemented. It is the most flexible controller for the given system. The formulas for the swerve steering were derived by the team itself.

Implementation

The swerve steering controller receives three inputs, as this is the degree of freedom of the platform. Let the desired two dimensional translational speed of the robot be given as v and the angular velocity be given by ϕ . Let d_w and d_l be the distances between the front wheels and respectively the left wheels, as shown in fig. 5.9. To know the position of every wheel, there are two cases to differ.

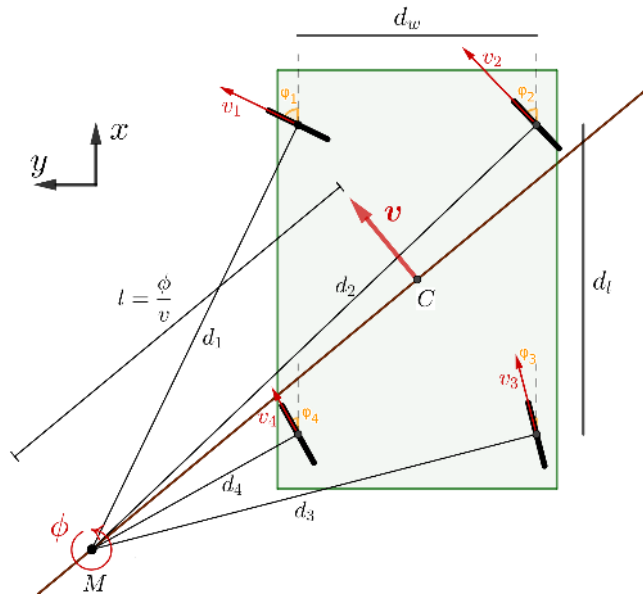


Figure 5.9: Swerve steering

Case 1: $\phi = 0$

The motion of the robot is a pure translation (fig. 5.10).

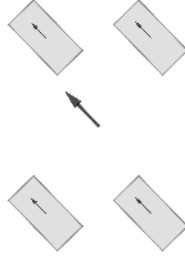


Figure 5.10: Wheel positions when driving sideways

The rotation angle of each wheel is the same. With $\mathbf{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix}$ we get

$$\phi_1 = \phi_2 = \phi_3 = \phi_4 = \begin{cases} \arctan(v_y/v_x), & \text{if } v_x \neq 0 \\ \pi/2, & \text{if } v_x = 0 \end{cases}$$

Also the speed of the wheels is the same:

$$v_1 = v_2 = v_3 = v_4 = |\mathbf{v}| = \sqrt{v_x^2 + v_y^2}$$

Case 2: $\phi \neq 0$

In this case the motion of the robot is a rotation (fig. 5.11). The resulting steering angles and driving speeds of the single wheels are given below. The derivation of this formulas are given in appendix B.5.

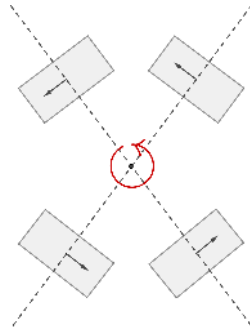


Figure 5.11: Wheel positions when turning on the spot

The steering angles are:

$$\begin{aligned}\phi_1 &= \begin{cases} \arctan\left(\frac{v_x + \phi \cdot d_l/2}{v_y - \phi \cdot d_w/2}\right), & \text{if } v_y - \phi \cdot d_w/2 \neq 0 \\ \pi/2, & \text{otherwise} \end{cases} \\ \phi_2 &= \begin{cases} \arctan\left(\frac{v_x + \phi \cdot d_l/2}{v_y + \phi \cdot d_w/2}\right), & \text{if } v_y + \phi \cdot d_w/2 \neq 0 \\ \pi/2, & \text{otherwise} \end{cases} \\ \phi_3 &= \begin{cases} \arctan\left(\frac{v_x - \phi \cdot d_l/2}{v_y + \phi \cdot d_w/2}\right), & \text{if } v_y + \phi \cdot d_w/2 \neq 0 \\ \pi/2, & \text{otherwise} \end{cases} \\ \phi_4 &= \begin{cases} \arctan\left(\frac{v_x - \phi \cdot d_l/2}{v_y - \phi \cdot d_w/2}\right), & \text{if } v_y - \phi \cdot d_w/2 \neq 0 \\ \pi/2, & \text{otherwise} \end{cases}\end{aligned}$$

The speed of the wheels are:

$$\begin{aligned}v_1 = |\mathbf{v}_1| &= \sqrt{(v_x - \phi \cdot d_l/2)^2 + (v_y + \phi \cdot d_w/2)^2} \\ v_2 = |\mathbf{v}_2| &= \sqrt{(v_x + \phi \cdot d_l/2)^2 + (v_y + \phi \cdot d_w/2)^2} \\ v_3 = |\mathbf{v}_3| &= \sqrt{(v_x + \phi \cdot d_l/2)^2 + (v_y - \phi \cdot d_w/2)^2} \\ v_4 = |\mathbf{v}_4| &= \sqrt{(v_x - \phi \cdot d_l/2)^2 + (v_y - \phi \cdot d_w/2)^2}\end{aligned}$$

To ensure the desired movement of the platform, the robot should only drive if the steering motors are in position. Therefore the calculated speeds are only given to the robot if all four wheels are close enough ($< 10^\circ$ difference) to their correct position.

Review

For this controller the simulation was very important, as the formulas are not straight forward and the correct behaviour could be tested and confirmed easily. For the change to the real robot an additional step was needed. While at the beginning in simulation the wheels could be steered by 360° without any cables blocking the movement, this is not possible on the real robot. Therefore the implementation had to be adapted, such that the steering angles ϕ_i of the wheels do not exceed $\pm 180^\circ$. After testing it in simulation, the swerve steering controller was tested extensively for several hours. This tests were successful and the implementation can be assumed to be correct.

5.4.3 Emergency Controller

Overview

In order to test the software on the robot, several safety features are needed to avoid accidents. The hardware solutions (section 4.4) are supplemented by a software

emergency mode that can easily be triggered by pushing a button on the game pad. If this emergency button is pressed, the emergency high-level controller is activated, which sets the speed of the driving motors to zero and holds the positions of the steering motors. In this way the robot can be stopped in case of an error on high-level side without restarting the whole system.

Review

Tests have shown, that this safety mechanism works reliably and as expected. It was used a lot, as the robot can be easily stopped and resumed, without turning off the motors or the whole system. This safety feature can be improved in the future by a mechanism that monitors the speeds and positions of the motors and activates the emergency controller on its own, if the high-level motor inputs are not as expected.

5.5 Manual Navigation

Motivation

For the time intervals where the system is not in autonomous mode, it has to be controlled manually. Manual control is used a lot for testing the prototype. Therefore, it is important that it is easy and intuitive to steer the robot manually, as different people need to test their code on the platform.

Implementation

To operate the system manually, a *Logitech* gamepad is used. It was chosen as there already exists a ROS library that supports this gamepad. In the following, the possibilities of the gamepad to operate the robot are listed. A complete manual about how to use the gamepad can be found in appendix A.2.

- High level controllers

In manual mode, one can switch between the ackermann and the swerve steering controller using the gamepad. In both controllers, a speed in x-direction as well as an angular velocity can be given. The swerve steering controller supports additionally a speed in y-direction.

- Emergency controller

The emergency controller can be activated to stop all motor inputs on software side. It can be deactivated by switching back into the swerve or ackermann controller.

- Autonomous mode

Manual mode, autonomous line following and autonomous turn can be activated at any time (except the emergency controller is activated).

- Geese feet

The geese feet can be lowered and lifted when the robot is in manual mode.

Review

The gamepad is as expected intuitive to use. The robot can not only be steered by the whole team, but also by externals after a short introduction. The manual completes the user friendliness.

One main problem of the used gamepad is the fact that it only sends data if the input changes. A consequence of this one way protocol is that it is not possible to distinguish between a lost connection to the gamepad and a constant speed input. The system holds the last speed until it receives a new speed. If the connection to the gamepad is lost, the robot keeps driving with the last speed input that was received correctly and can only be stopped by the remote emergency button. Right now it seems that the only way to fix this problem is to use another gamepad, which is sending its inputs continuously.

5.6 State Machine and Planners

5.6.1 State Machine

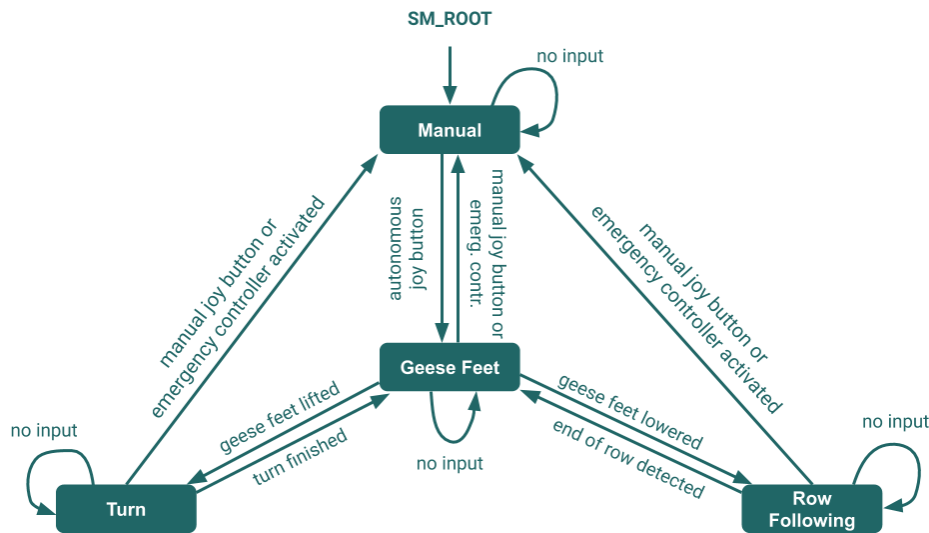


Figure 5.12: Implemented state machine

Motivation

An autonomous robotic system often contains several different states. Various events can trigger a change into a new state. Such a state handling is done by a state machine.

Implementation

The implemented system consists of three main states (fig. 5.12).

- Manual mode
The robot is completely operated using the gamepad.
- Autonomous row following
The robot detects the line and is following it autonomously.
- Autonomous turn
The robot performs a 180° turn into the next row.

The state transitions from manual to autonomous mode are triggered by gamepad inputs. As the system navigates completely autonomous through the rows, the state transitions between the two autonomous modes are triggered by an autonomous detection of the end of the row and when the turn has finished.

For autonomous weeding, an additional state was added.

- Geese feet

Depending on the next state, the geese feet are lifted or lowered.

The geese feet state is called between the transitions to the autonomous states. For autonomous navigation, the geese feet are lowered, while they are lifted for turning.

Review

The state machine is working reliable. The state transitions happen with no significant delay.

5.6.2 Row Planner

Motivation

For the autonomous line following, a planner is used. Its task is to navigate the robot over the sugar beet rows. The planner receives the position of the row to follow from the line detection algorithm. It then calculates the needed movement of the base to keep the robot driving over the detected row.

Implementation

The planner receives the position of the row with respect to the robot from the line detection. It uses the offset angle r_ϕ of the row and the distance r_d from the row to the center of the robot, as shown in fig. 5.13.

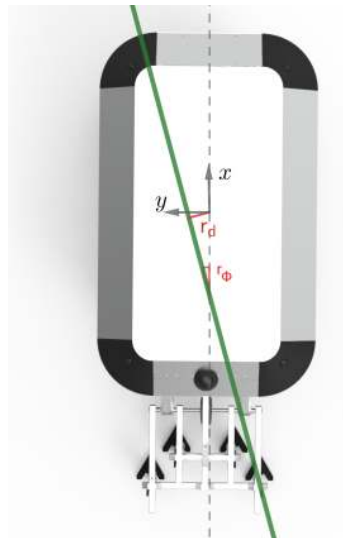


Figure 5.13: Parameters of detected line

Additionally the desired forward speed $v_{x,desired}$ of the robot is used. This speed can be set by a parameter and was set to 0.3m/s during testing.

For the ackermann steering controller (section 5.4.1 the desired speed vector and angular velocity of the platform is calculated to bring the robot back on track. Therefore the pure pursuit algorithm is used.

As the robot is in ackermann steering when following a line, this results directly in

$$v_y = 0 \quad (5.1)$$

and

$$v_x = v_{x,desired}. \quad (5.2)$$

For the angular velocity additionally the lookahead distance p_{la} of the planner is used. The lookahead distance can be set as a parameter and describes how aggressively the planner corrects its position. The bigger p_{la} is, the less aggressive are the corrections and vice versa. Field tests a value of $p_{la} = 1.1m$ worked best.

$$\dot{\phi} = v_{x,desired} \cdot 2 \cdot \sin \left(r_{\phi} + \text{sign}(r_d) \cdot \left(\arccos(|r_d|/p_{la}) - \frac{\pi}{2} \right) \right) / p_{la} \quad (5.3)$$

(5.1), (5.2) and (5.3) are the speeds given to the high-level controllers to navigate the robot over the detected line.

Testing and Review

After adjusting the parameters of the planner during field tests, it worked fine and as expected. Nevertheless it can be improved by increasing the angular velocity of the steering motors. This speed is limited for safety reasons to avoid damage of the electronic components, but this limit can be increased slowly. With faster steering wheels, the planner can correct the position faster.

For the navigation within the field, the ackermann steering controller is used. This decision was made, as the geese feet have to be pulled in x-direction. If the ackermann controller is active, the robot does not move laterally, which could lead to undesired lateral forces to the geese feet. The field test verified that the geese feet behave as desired when they are pulled in ackermann mode. Due to limited testing capacities, the behaviour of the geese feet in swerve steering mode could not be tested. This needs further investigation to observe if problems occur when the geese feet are pulled in swerve steering mode or if the navigation works even smoother when this controller is activated.

5.6.3 Turn

Motivation

Another crucial feature for a working robot pipeline is the turn at the end of a row into the next row. Therefore several turning mechanisms were evaluated, which are a combination of rotational and translational movement. This is needed to change the direction of the robot as well as move its position to the beginning of the next row. The evaluated turning mechanisms are shown in fig. 5.14. As the movement of the robot in the field is not smooth due to uneven soil, it is very difficult to rotate precisely by a given angle.

Because of this inaccuracy, it was decided to implement an algorithm that has only one rotating part to minimise this error. Also movement in lateral y-direction (as it occurs in turn nr. 3 in fig. 5.14) is on the uneven ground of a field not precise enough, to drive 50 cm lateral, as tests have shown. For this reasons, turn algorithm nr. 1 was chosen. It is only possible in the swerve steering mode, as it requires a rotation of the robot with a radius of only half of the row distance, which is 25 cm.

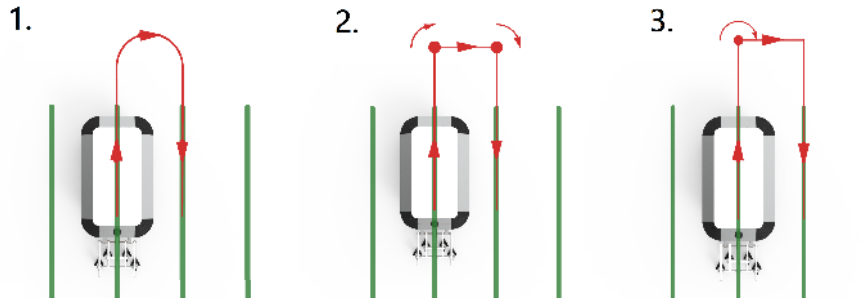


Figure 5.14: Three evaluated turn options

Implementation

The implemented turn consists of three phases. First the robot drives forward for a certain length. Afterwards, a 180° degree rotation with the diameter of the row width (50 cm) follows. Finally the robot again drives forward a certain length. The forward driving parts are needed, as the turn is activated when the robot is still in the row. Before the rotation, the robot first has to leave the row and afterwards it has to drive back into the new row, where the line detection takes over.

To drive for the given distances and for the correct rotation degree, the state estimator (section 5.8) is used as feedback. To be more stable, the straight parts are not implemented as feed forward, but as a control loop using the state estimator. In this way, angular and lateral offsets of the robot are corrected. Especially after the rotation when the robot is driving back into the row, this is important, as the rotation in the field is not very precise. The resulting errors after the rotation can be corrected.

The turn is very modular and consists of many adjustable parameters. These are the lengths of the straight parts, the speed, radius and angle of the rotation part and if the turn is to the left or to the right. The latter is used by the state machine.

Review

The turn relies highly on a precise odometry of the system as the robot needs to know when to start the rotation and when it has finished the rotation. While this was working very well in simulation, there were many problems in the field. In simulation the wheel odometry (section 5.8.1) was used, which was very accurate in simulation, but not on the real robot. The problems in the field were mainly related to the state estimator that provides the system with a fusion of odometry data. Once the state estimator problems were fixed, the turn performed well. Especially after the rotation, the system has to correct the position a lot. Thanks to the control loop, the new row is entered successfully in most of the cases. This was only tested on even ground yet and additional test in the field are needed.

5.7 System Status

5.7.1 Audio Feedback

Motivation

Having a form of feedback about the current status of robotic systems is important. The standard way for developers is system information printed in the terminal. For small programs this works fine, but the amount of information in the terminals

is increasing with larger projects. Especially for testing, it is essential to get the important information immediately and with easy access. Furthermore it is an important safety aspect to know, when the robot is in autonomous mode. One way of getting such information more easily is through audio feedback.

If the status of the system changes, the user will be directly informed and can monitor the system without following the terminal outputs. A further advantage of audio feedback is that all persons monitoring the robot know what happens and not only the person that triggers the events with the gamepad (e.g. as it is the case if the emergency controller is activated).

Not only for testing, but also for more user friendliness audio feedback is a big advantage and makes it easier to operate the robot.

Implementation

For several events, audio feedback was implemented using the ROS package `sound_play`. A custom sound package was implemented which subscribes to all necessary topics and activates the sounds via a service request to the `sound_play` package. In the following sections, the events which trigger a sound are listed.

- System is ready

When the program is started, it needs a while until the communication with the motors controllers is established and the wheels are in the correct position. As soon as the position of all motors can be read and the position of the steering motors is correct, a "system is ready" sound informs the user.

- High-level controllers

When the high-level controller is switched, the name of the new high-level controller is announced. Also the emergency controller is announced if it is activated.

- State Machine

Switches between manual mode, autonomous lane following as well as autonomous turn are announced.

- Geese feet

Whenever the geese feet are lowered or lifted, this is announced.

Review

The sounds were one of the most frequently used features during testing. Especially for the switches between the different high-level controllers, the audio feedback was very useful, as there is only one gamepad button to switch through the high-level controllers (see section 5.5). Also, the announcement of the state transitions made testing easier, as for example the end of row detection could be verified. During all field tests, several team members were evaluating the robot performance. They were directly informed if the operating team member switched from manual mode to autonomous mode, which increased the safety during testing.

5.7.2 Visual Feedback

Motivation

While audio feedback can only be used to announce certain events, visual feedback can indicate the state of the robot over a longer period of time. It is a helpful addition to audio feedback and makes the system more user friendly. Anyway, with

LEDs less information can be shared compared to sound and the user has to know the meaning of the different colors.

Realization

Four LED strips are mounted on the robot. They have a total length of 5 m and a maximal power consumption of 80 W. The brightness is sufficient for the visual feedback to be distinguishable in sunlight. One LED strip is mounted at the back of the robot while three are mounted at the front left, middle and right part. Diffuse glass ensures even light distribution. The position of these strips is shown in fig. 5.15.



Figure 5.15: Position of LED strips on Rosie

A separate visual feedback package subscribes to all necessary topics and publishes a message containing the colors of the different LED strips. The strips are controlled by a Raspberry Pi which is connected to the ROS network and subscribes to this message. In the following, the different color modes of the system are listed.

- Initialization

After starting the system, the robot is in the initialisation mode until all wheels are in the correct position and Rosie is ready to drive. During this phase, the LEDs are blinking in orange colour.

- Manual mode

Manual mode is indicated by green lights.

- Autonomous mode

Autonomous mode is indicated by blue LEDs. During the turn, the left or right LED strip is blinking, depending on the direction of the turn.

- Error

If the emergency button on the gamepad is pressed and the emergency controller is activated, the lights get red.

Review

The visual feedback makes the robot more user friendly and provides a nice feature for presenting Rosie. The status of the robot is visible at any time and the audience can see, when the robot is in autonomous mode and when it is controlled by hand. Due to time constraints and communication problems between the Intel Nuc and the Raspberry Pi, this feature has not been tested on the robot yet.

5.8 State Estimator

Motivation and Overview

Since the system is autonomous, a certain robustness in all processes is needed. One way to make some algorithms more robust is to estimate the states of the system and then check whether the estimation and the measured states are congruent. The row tracking and the turn algorithm both rely on this procedure for robustness. Different sensors can be used to estimate the odometry, and in order to gain robustness and accuracy, the fusion of multiple odometry sources is needed. The fused sensors that were implemented in this project are: visual sensors, inertial sensors and wheel sensors.

5.8.1 Wheel Odometry

Motivation

One way of getting feedback about the current motion of the robot is wheel odometry. The encoders for the steering and the driving motors are used to calculate the speed of the robot in x- and y-direction as well as the angular velocity. Because of slippage on the ground and other inaccuracies, the accuracy of this odometry is not high, but the magnitude is reliable.

Implementation

Wheel odometry was added for both implemented high-level controllers (section 5.4). Therefore the absolute position of the steering motors (given by the *Baumer* absolute encoders) and the speed of the driving motors (calculated using the encoders of the *Bosch* motors) is used.

Ackermann wheel odometry

For a rigid platform in a plane, the velocity vectors of two different points are enough to determine the motion of the whole platform. As the velocity vectors of all four wheels are given, the system is over-determined. For ackermann steering, a lateral speed of $v_y = 0$ is assumed. Furthermore it is known that the centre of rotation of the platform is on a straight line with the back wheels.

The robot speed v_x is calculated using the speeds of the back wheels.

$$v_x = \frac{v_3 + v_4}{2} \quad (5.4)$$

The angular velocity of the platform is calculated three times using the back wheels and the front left wheel as well as the front right wheel respectively. The average of these three angular velocities results in the angular velocity of the platform.

$$\phi' = \arctan \left(\frac{v_1}{d_l / \sin(\phi_1)} \right) \quad (5.5)$$

$$\phi'' = \arctan \left(\frac{v_2}{d_l / \sin(\phi_2)} \right) \quad (5.6)$$

$$\phi''' = \arctan \left(\frac{v_3 - v_4}{d_w} \right) \quad (5.7)$$

$$\phi = \frac{\phi' + \phi'' + \phi'''}{3} \quad (5.8)$$

The wheel odometry for ackermann steering is therefore given by (5.4), (5.8) as well as $v_y = 0$.

Swerve wheel odometry

The speed vectors of two points of the rigid platform are enough to calculate the odometry of this platform. Therefore first the speed vectors $\mathbf{v}_2$ and $\mathbf{v}_4$ are used to calculate the odometry for swerve steering. Afterwards $\mathbf{v}_1$ and $\mathbf{v}_3$ are used to calculate the odometry. The average of both calculations results in the wheel odometry for swerve steering.

As both calculations are analogously, only the odometry calculation using $\mathbf{v}_2 = \begin{pmatrix} v_{2,x} \\ v_{2,y} \end{pmatrix}$ and $\mathbf{v}_4 = \begin{pmatrix} v_{4,x} \\ v_{4,y} \end{pmatrix}$ is given here. There are two main cases to differ:

Case 1: No linear or no angular velocity

$$v_{4,y} \cdot v_{2,x} \approx v_{4,x} \cdot v_{2,y}$$

Case 1.1: No linear velocity (on the spot turning)

$$\begin{aligned} |v_{4,y} \cdot v_{2,y}| &> |v_{4,x} \cdot v_{2,x}|, \text{ if } v_{4,y} \cdot v_{2,y} \leq 0 \text{ or} \\ |v_{4,y} \cdot v_{2,y}| &\leq |v_{4,x} \cdot v_{2,x}|, \text{ if } v_{4,x} \cdot v_{2,x} \leq 0 \end{aligned}$$

In this case, the odometry results in

$$v_x = 0 \quad (5.9)$$

$$v_y = 0 \quad (5.10)$$

$$\phi = -\frac{v_{4,x} + v_{2,x}}{d_w} \quad (5.11)$$

Case 1.2: No angular velocity (parallel steering)

$$\begin{aligned} |v_{4,y} \cdot v_{2,y}| &> |v_{4,x} \cdot v_{2,x}|, \text{ if } v_{4,y} \cdot v_{2,y} > 0 \text{ or} \\ |v_{4,y} \cdot v_{2,y}| &\leq |v_{4,x} \cdot v_{2,x}|, \text{ if } v_{4,x} \cdot v_{2,x} > 0 \end{aligned}$$

In this case, the odometry results in

$$v_x = \frac{v_{2,x} + v_{4,x}}{2} \quad (5.12)$$

$$v_y = \frac{v_{2,y} + v_{4,y}}{2} \quad (5.13)$$

$$\phi = 0 \quad (5.14)$$

Case 2: Linear and angular velocity

$$v_{4,y} \cdot v_{2,x} \neq v_{4,x} \cdot v_{2,y}$$

First the coordinates M_x and M_y of the rotation center with respect to the middle point of the platform is calculated.

$$M_x = \frac{-v_{4,y} \cdot v_{2,y} \cdot d_w + v_{4,x} \cdot v_{2,y} \cdot d_l}{v_{4,y} \cdot v_{2,x} - v_{4,x} \cdot v_{2,y}} + \frac{d_l}{2} \quad (5.15)$$

$$M_y = \frac{-v_{4,y} \cdot v_{2,x} \cdot d_w + v_{4,x} \cdot v_{2,x} \cdot d_l}{-v_{4,y} \cdot v_{2,x} + v_{4,x} \cdot v_{2,y}} - \frac{d_w}{2} \quad (5.16)$$

This center point is used to calculate the odometry data.

$$\phi = \frac{v_{4,x}}{M_y - d_w/2} \quad (5.17)$$

$$v_x = M_y \cdot \phi \quad (5.18)$$

$$v_y = -M_x \cdot \phi \quad (5.19)$$

These cases complete the wheel odometry calculation for swerve steering. The same is done with both other wheels to get the odometry of the platform as an average of both calculations.

Review

The wheel odometry was tested in simulation first, where it is working correctly. In simulation, the environment is perfect, which is not the case in real life. On the real robot the wheel odometry, especially in swerve steering, it is not precise. While the error for the ackermann steering controller is around 10%, the wheel odometry for swerve steering is wrong by a factor of around 1.5 to 2. As the data is correct in simulation and there was no significant slippage when testing the wheel odometry on the real robot, it seems likely that the errors are related to the encoders and calculations, that are not numerically stable. Due to time constraints, this problem was not investigated further, but is an open question, that has to be solved in the future. Furthermore, the wheel odometry could be generalized by calculating the odometry of each single wheel and fuse the resulting four odometries to an overall robot odometry.

5.8.2 Visual-Inertial Odometry and Fusion of Odometries

Overview and Implementation

The tracking camera *T265* provided by *Realsense* incorporates a visual-inertial sensor system and calculates the visual-inertial odometry (VIO) on the virtual processor core of the camera. The combination of visual and inertial sensors makes the setup more robust. The IMU works well with high velocities and accelerations but has a constant bias due to subtraction of gravity which significantly affects the precision in the low speed range. Visual odometry on the other hand, can track features at slow velocities very well but is constrained by the limited frame rate. Furthermore, the *Realsense* software allows to make the odometry pipeline more robust by also fusing the wheel odometry into the workflow. The software allows to assign different covariances to wheel and visual odometry, in order to weight the two inputs such that it can be tuned for the application.

Review

The state estimator was implemented at a late stage of the project. Due to this, the testing opportunities were limited. The different standards for coordinate frames within the module made the integration of the camera cumbersome. Once the

camera was properly integrated, the pipeline only functioned properly in certain conditions and the setup never worked robustly, meaning the state estimator could only be run for a few minutes in a row and then had to be reset.

One major problem derived from the high dynamic range of the scene, since the sensor is used outdoors and is exposed to sunshine and shadows of the robot, making feature tracking hard.

Another problem could be the position of the visual sensor. First, the camera was placed in the front of the robot which led to wrong measurements because many parts of the robot were visible in the image due to the large FOV and the measured velocity resulted to be zero. At a second stage, the camera was mounted underneath the battery which might be too close to the scene. Further positions of the sensor should be iteratively tested to optimize the setup.

5.9 Row Following

5.9.1 Sensor Choice for Row Detection

Overview

As Rosie has to navigate through the rows without a map containing the exact plant positions, a suitable sensor to detect the row is needed. There are two common types of sensors to do this: A laser scanner or a camera.

A laser scanner solution would use the size of the plants for the row detection, assuming that the crops are usually bigger than the weeds. As a 2D laser scanner has to be approximately on the same height as the plants are, this can lead to difficulties if there are stones in the row and the laser scanner is very close to the ground. Mounting the 2D laser scanner further away at a specific angle to the ground is also not a good option, as this angle changes due to the suspension. Because the robot should also drive through fields where the crops are young and small, this was no way to go. Also, this method is inaccurate if there are too many weeds that have a similar height as the crops.

Hence, it was decided to place a camera on the front of the robot and do visual row detection. This method was deemed to be more stable. To work reliably, the camera has to fulfill several requirements.

Requirements

1. *Adaptivity to light changes*

To achieve a reliable row recognition, the chosen camera needs to be suitable for the variety of light conditions that come with the outdoor application.

2. *Works with Ubuntu and ROS*

As the robot is programmed using ROS, the camera must be compatible. Preferably there are already libraries to get the sensor data in ROS such that no time is lost by writing wrappers.

3. *Color image*

The color channels of a camera enable visual crop row detection.

Further an additional infrared channel could improve the possibilities to distinguish between vegetation and ground. This is not a hard constraint for the camera, but a desirable feature [34].

4. Stereo camera

Preferably, the camera should feature a stereo module to gain 3D information, as the crop height could be used to improve the row detection. Also, the depth information is more reliable than a 2D image under dynamic light conditions, as under most of the light conditions, the point cloud can still be calculated using feature tracking or an active IR projector, while the RGB data changes a lot when light varies.

Comparison of Different Cameras

Several commonly used stereo cameras were investigated and their features evaluated. An overview of the evaluated cameras can be found in appendix B.6. Every camera that did not fit the requirements mentioned above was dropped. In the end, two cameras were left: The *Intel RealSense D435i* and the *Microsoft Kinect v2*. Both use an IR projector to create a 3D point cloud using feature mapping. This is a critical point, as bright sunlight contains a lot infrared radiation, which makes the IR projector of the cameras useless. As several articles describe, this is a problem for the *Kinect* [35]. The *RealSense*, on the other hand, does not necessarily need the projector, as it also uses the RGB data to calculate the depth information. [35] confirms that the *RealSense* is suitable for agricultural outdoor applications. Considering all this, it was decided to use the *Intel RealSense D435i* for navigation. It fits the requirements and first camera test under changing light conditions were successful. The sensor range is with 10m enough for the system. Further, the near infrared channel of the *RealSense* can be extracted to use it to improve the detection of vegetation and ground.

Review

The *D435i* worked as expected. Especially the ability of the camera to adapt to changing light conditions makes the algorithms stable to these conditions. Not all of the features of the *RealSense* are used in the implemented row following algorithm. As the visual row detection algorithm using the RGB image has given adequate results as well as due to time constraints, the infrared channel and the crop height information have not been used so far. Still, they might be used to improve the accuracy of the row following algorithm in the future.

5.9.2 Row Detection

Overview

The row detection uses image analysis to determine the center row position in a single frame. This provides the basis for robust row tracking over the course of a full row. The building blocks of the implementation as well as the integration into the row tracking algorithm can be seen in fig. 5.16. The main steps of the detection are the extraction of plant pixels, enhancement of the crop rows using different image filters and finally a line detection in the resulting image.

Requirements

1. Reliability: The center row needs to be determined with sufficient reliability to enable successful row tracking. Testing determined the required reliability of the detection to be at about 80% correctly identified center rows.

2. Accuracy: Identify the center row with sufficient accuracy to prevent the robot from driving over the plants and the passive tool from pulling out sugar beet.
3. Light conditions: The robot is used in daylight throughout the entire day. This requires robustness to a high dynamic range, a variety of light conditions as well as robustness to shadows in the image.
4. Crop size: Row following should be possible during a variety of growth stages.
5. Weeds: The detection should work for low to medium weed densities.

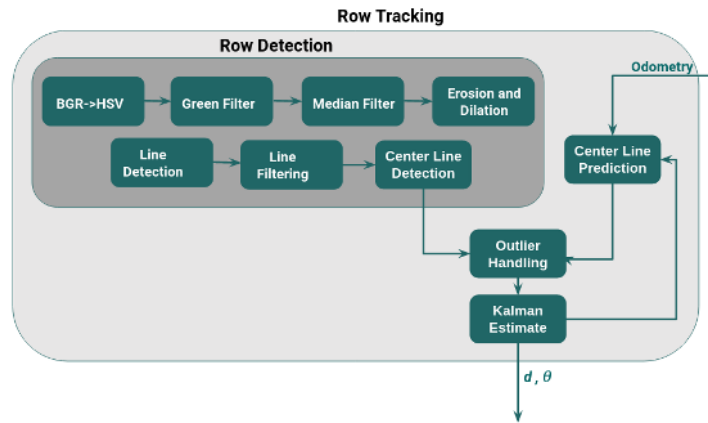


Figure 5.16: Row detection components and Row Tracking

Elaboration on the required accuracy

To specify the criteria for accuracy, the dimensions of the robot need to be considered. The inner geese feet are 15 cm apart, leaving room for the center row. Assuming that the sugar beet stem has a diameter of 5 cm, this leaves 5 cm to the side of each goosefoot share to account for inaccuracies in row following. This is visualized in fig. 5.17, where the green area represents the allowed inaccuracy in row following.

The center row can be defined by its offset and angle relative to the base frame of the robot, as shown in fig. 5.18. Combined with a distance of 110 cm of the inner tips of the geese feet from the origin of the base link, this implies that a maximum error in angle ϵ_ϕ of 2.6° is tolerated when following the row, or a maximum error in offset ϵ_d of 5 cm, assuming that the robot base is centered on the row.

As multiple components of the robot have an influence on the accuracy in row following, defining clear bounds for the desired accuracy of the row position estimation is difficult, and validation needs to be done in field tests. Thus, no specific target value was set and the fulfillment of this requirement will be only be assessed qualitatively on the recorded datasets.

Implementation

For most of the image analysis components of the implementation, the OpenCV library was used, which is an open source computer vision library. The steps to the crop row detection algorithm are outlined below and also visualized in fig. 5.23.

Transformation to HSV color space

The initial step is a transformation from RGB color space to HSV (Hue Saturation Value). This is due to the greater invariance to different light conditions that can

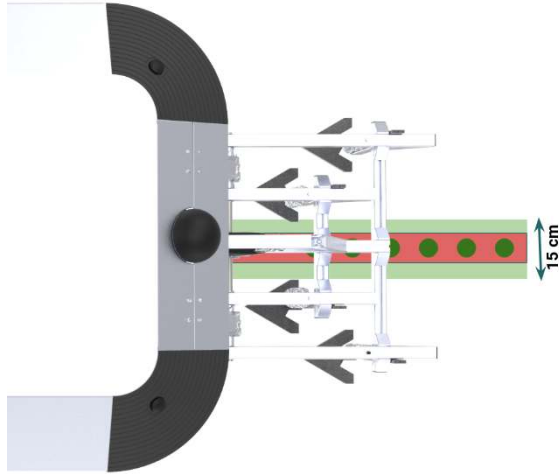


Figure 5.17: Allowed area for goosefoot sweepers (green) and damage zone (red)

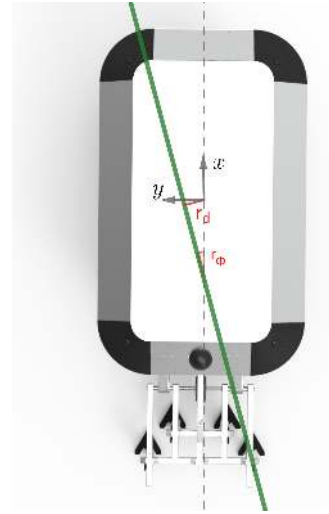


Figure 5.18: Definition of line position

be achieved when working in HSV color space. [36]

Green mask

A green mask is applied, leaving only plants in the image, which can be crops as well as weeds (see fig. 5.19). For this, the hue channel can be set to allow just values on the green spectrum, while for the saturation and value channel almost the entire spectrum is allowed.

Enhancing the crop rows: Median filter, erosion and dilation

As a first step to enhance the appearance of the crop row, a median filter is applied to reduce the appearance of weeds in the image. A median filter is typically used to reduce salt-and-pepper noise, which is what the weeds can appear like, as they are usually smaller in size than the crops themselves. The result of this process can be seen in figure 5.20. Then, a combination of image dilation and erosion is applied to further enhance the crop row (see figure 5.21).

Line Detection

A binary threshold is applied to the processed image and the Hough line transform is used to detect straight lines in the remaining image (see fig. 5.22). The Hough line parameter in the OpenCV function that was used defines how many points are needed to qualify as a line. It is dynamically calibrated such that the amount of detected lines always lies between 10 and 50. This has two desirable effects. For one, each visible crop row likely has more than one line going through it. When averaging the lines which lie close to each other, this gives a good estimate for the actual row center. Second, this can account for frames with sparse or smaller crops where the rows would not be recognized using a rigid parameter.

Rough outlier handling

Despite the filtering process for enhancing the row which was explained above, it can happen that lines are detected across the image. Due to that, the median yaw of the lines is determined and the ones which do not lie within a threshold of this value are discarded, leaving only lines which point to the dominant direction. This is possible because as the rows are parallel, their yaws in the image plane are similar.

Center line estimation

After this rough outlier filter, the remaining lines which lie close to each other are averaged. Finally, the row which passes closest to the robot center is chosen as the center line and two 3D points in the camera frame are computed using the depth stream of the camera. These row points are then published by the detection node.

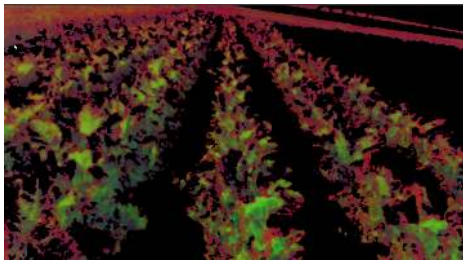


Figure 5.19: Green Filter

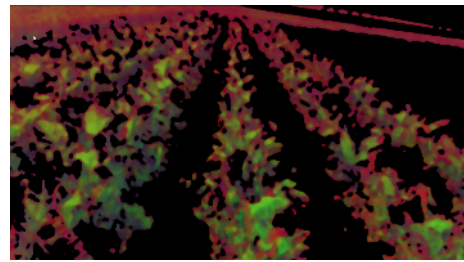


Figure 5.20: Median Filter

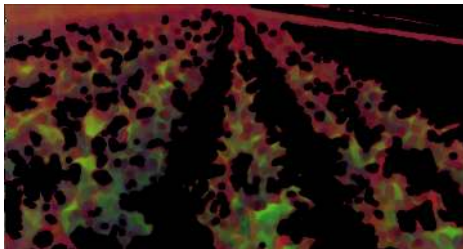


Figure 5.21: Eroded Crop Map



Figure 5.22: Detected center line

Figure 5.23: Steps in row detection for big crops

Review

With this implementation, the rates of correctly identified center rows vary from 86.7% to 98% in the analyzed datasets (appendix B.7). The tested conditions will be further examined in the following sections.

Spring Stage

The most challenging growth stage for row detection is the spring stage where the plants are tiny. Nevertheless, it was possible to correctly identify about 86.7% of the center rows in the analyzed dataset. Under these conditions, it is even difficult for humans to recognize the crop rows (shown in fig. 5.24). To the left side of the image the green filter, the enhanced crop rows and the result of the Hough line detection can be seen.

The factor which impedes the detection most at this growth stage are the frames where only very few crops are visible in the center row. As it is the earliest growth stage, some sugar beets have not sprouted yet, creating these gaps.

The algorithm was only tested with low weed density as no other data was available, but it can be assumed that the performance would suffer significantly in the presence of weed (see fig. 5.25). In the spring stage, the plants themselves have similar size as the weeds, which makes an reliable row detection difficult. An approach to counter these effects could be a pattern detection, where not one line at a time is extracted (as was done in this implementation), but several lines at once [37].



Figure 5.24: Barely visible crop rows at the spring stage

2-4 leaf stadium

With increasing plant size, the algorithm showed a more reliable performance, correctly detecting the center row in 91.4% of the cases (medium weed density). As in this stage, while not all plants are equally sized, almost all have sprouted. Although there can still be some visible gaps in the rows, the amount decreases compared to the spring stage, thus improving the detection. It was observed that when the correct center line was identified, the accuracy was rather high.

4-6 leaf stadium

This growth stage has shown the best performance, with about 98% correctly identified center rows in rows with medium weed density. Performance was impaired by the appearance of bigger weeds, which were typically the cause of incorrectly or inaccurately detected rows. Gaps were no longer an influential factor at this growth stage. The accuracy was found to be high due to the clearly visible crop rows at this stage.

8 leaf stadium

For plants that have reached their last growth stage of 8 leaves, new challenges arise. As can be seen in fig. 5.23, the crop rows are not clearly visible anymore in the analyzed image, as from the robot's perspective, almost everything except for the center row looks green and lines can be detected all across the image. To counter this, a region of interest was chosen for the detection. Narrowing the image about 35% enabled correct detection in 98.1% of the analyzed frames with moderate weed density. The accuracy decreased compared to the earlier growth stages due to the broad crop rows.

Conclusion

The implemented algorithm shows good results in terms of reliability with only slight necessary adaptations for different growth stages.

The accuracy of the detection depends on the growth stage.

The goal of light-independent performance can be reached by working in HSV color space.

A key factor which impairs the reliability and accuracy of the algorithm is the weed density, which can cause the center row to be detected incorrectly (fig. 5.25).

Moreover, it can happen that if the center row is clearly less dense compared to the surrounding rows, it will not be detected because only the 50 most dominant line combinations are allowed. But setting this parameter higher typically raised the inaccuracy.



Figure 5.25: Failure due to weed patches

5.9.3 Row Tracking

Overview

As shown in the previous section, re-detecting the row in each frame is not sufficient to follow the center row. The detection can have outliers in multiple consecutive frames, which would cause the robot to drive over the crop rows. This can be avoided by row tracking: once the first detection is made, the row position can be predicted using odometry data. Having two informations about the row position; the detection and this prediction, allows to detect outliers and to get a better overall estimate.

Requirements

1. **Outlier Handling:** The row tracking should include outlier handling for incorrectly detected center rows.
2. **Robustness:** The correct center row needs to be tracked over the course of full rows to fulfill the task of autonomous navigation.
3. **Accuracy:** The row tracking should improve the accuracy compared to the detection to be sufficiently accurate to prevent the goosefoot sweepers from pulling out the sugar beets.

Implementation

The row tracking uses the velocity data from the state estimation (section 5.8). This information can be used to predict the new row position by applying a coordinate transformation on the row points from the previous estimate and determining the resulting offset and angle with respect to the robot. The transformation matrices are found by integrating the velocities over time.

By comparing the predicted values to the detection, outliers are identified and removed. A line qualifies as an outlier if for either row state r_d, r_ϕ it holds:

$$|r_{det} - r_{pred}| < 2 \cdot (\sigma_{det} + \sigma_{pred} + \sigma_q) \quad (5.20)$$

with σ being the standard deviation and q the associated process noise of the state. If an outlier is detected, the prediction is kept as the estimate because it is assumed that the state estimation, although it will drift over time, gives continuous odometry outputs. An example of such an outlier is shown in fig. 5.26(b). Therefore it can not cause huge discrepancies from one frame to the next, unlike the detection. In return, the uncertainties of the row states are increased in the next prediction. An

example of successful outlier handling can be seen in fig. 5.26(a).

After the outlier handling, the two informations on the crop row position can be fused using a Kalman filter (fig. 5.16). As this is a recursive filter the knowledge from previous estimates is not lost but instead improves the row tracking.

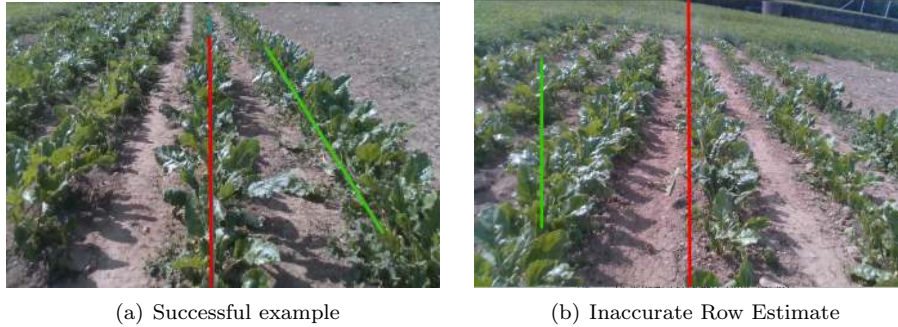


Figure 5.26: Outlier Handling: Detection (green), Prediction (blue) and Estimate (red)

Review

Performance on datasets

At the earlier growth stages, only the visual inertial odometry was recorded. But it was found not to be sufficiently accurate for row tracking as it contained a lot of drift, which is why as a consequence the wheel odometry was added. But due to this change, the final row tracking could only be tested on datasets at the 8 leaf stadium (appendix B.7).

Because at this growth stage the detection rate was already around 98%, outlier removal was successful over the course of all three recorded rows at our test field (30 m row length) and the right center row was always determined.

To examine the reliability for lower detection rates, some parameter values in the detection were changed to lower the success rate to 83.0%, which was still sufficient for successful row tracking. But the accuracy suffered compared to the more reliable detection.

There were five instances over the course of these three rows, compared to two for the properly tuned setting, where the center row was identified correctly, but rather imprecisely, such as in fig. 5.26(b). It can be assumed that such an estimate would have caused plants to be damaged. The shown example was caused by five consecutive faulty detections and a coincidentally inaccurate prediction.

The maximum amount of successfully removed consecutive outliers in the examined data was 9.

Field tests

The first successful field tests, where a row could be followed in full length, were performed using only the wheel odometry as an input. This was sufficient to track the center row robustly.

The accuracy on the other hand was not sufficient. In each autonomous drive there were instances where the robot drove close enough to the plants to cause damage with the tools. In a majority of the cases the prediction seemed to be at fault, clearly drifting to one side. In some instances, the detection repeatedly placed the center row closer to one edge of the crop rows.

The row tracking with the improved state estimation, fusing wheel odometry and

VIO, could not be validated on the field anymore as the weather did not allow it.

Conclusion

The addition of the described outlier handling and Kalman filter has improved the center row estimation significantly. The algorithm is able to track the correct center row over the course of full crop rows. The accuracy of the estimated center row has been improved as well.

But as shown in instances such as fig. 5.26(b), even with these features, the accuracy is likely still too low to autonomously follow the rows without causing significant damage to the crops.

These problems might be overcome by improving the state estimation as this is the least tested component and was often the cause for inaccurate estimates.

The task of successful row following also depends on several other software components like the planner and the controllers, as well as external factors such as the soil conditions. For this reason, more field tests need to be conducted in order to draw a conclusion on whether the desired accuracy can even be achieved in the current setup. There is also the possibility that some components need to be reevaluated completely, e.g. the approach of vision-based navigation or the arrangement of the passive tool.

5.9.4 End of Row Detection

Motivation

To complete the navigation within the field, the end of a row needs to be detected. This detection triggers the turn mode, allowing Rosie to change rows autonomously. The reliability of this detection is very important. If it is wrongly triggered, the robot will start its turn in the middle of the field or it will keep driving in line following mode, if the end is not detected.

Realization

For the detection, the image data of the D435i camera is used. To detect the end of a row, a simple algorithm was chosen due to time constraints. Supposing that the field is surrounded by grass, the algorithm is looking for green areas of grass. Therefore, the green image pixels are extracted using the HSV color space. This makes the algorithm more stable to changing light conditions. Erosion and dilation is used to remove single outlier pixels. In the next step, the image is divided into horizontal strips of same size and the green pixels in every strip are counted. If the amount of green points in a certain strip is above a certain threshold, this strip is detected as an end of row strip. This threshold was set 95% which led to good testing results. If the end of row strip that is closest to the robot is closer than a given threshold, the end of row has been detected. To combat wrong detections in single frames, this has to hold for several frames in a row, until the end of row trigger is activated. An analyzed image frame where the end of row is detected (visualized by a green horizontal bar) is shown in fig. 5.27.

Review

Although this algorithm is very simple, it worked reliable. After adjusting the different threshold parameters and the parameters for the green color range, the algorithm works under bright as well as very shady light conditions without new adjustments to the parameters. This is not only because of the hsv color space, but also because of the D435i camera, that adapts automatically to changing light conditions. Next to this parameters that define the green color range, also the



Figure 5.27: Camera image analyzed by end of row detection algorithm

threshold parameter for the minimal amount of green pixel needed for an end of row strip, is very important. It is set to a rather high value so that the end of row is not triggered if the situation is not clear. The reason therefore is that a wrong detection in the field leads to more sugar beet damage as when the real end of row is not detected and the robot keeps driving in line following mode.

While this simple approach works very reliable for the given test field, it has some limitations where the algorithm is not working as expected. One problem occurs in the evening and morning, when the sun shines very flat and can glare the camera. As this happens only in a small time slot, it is not a big problem. The biggest limitation is that the field needs to be surrounded by green grass for the end of row detection. In other scenarios where at the end of the row there is only soil or new plants, the algorithm is not working.

One approach to face this different conditions could be an extended line detection, that can not only detect the lines, but also analyses the start and end point of the detected line. The detection can be further improved by measuring the length of the first row. This length can be used for the following rows to prevent a detection in the middle of the field.

5.10 Crop Detection

In order for Rosie to act as a platform for weeding in the rows (see fig. 3.30), it was decided to provide a feature which distinguishes crops from weed and locates the stem of the crop. This allows for system improvements such as actively weeding within close proximity to the crops without damage, resulting in more yield (see section 6.2 and section 6.2). Another application is phenotyping, which is an important indicator which farmers rely on to know the state of their crops and prevent diseases early on (see section 6.2). Hence, the team concluded crop detection to be a necessary feature for an all-round weeding system which shall be integrated using visual sensors.

The stated feature was accomplished by using a camera module which identifies crops, weeds and soil (see section 5.10.1) as well as an algorithm for extracting the crops centroid locations (see section 5.10.2). The resulting predictions made by crop detection is visualized in fig. 5.28.

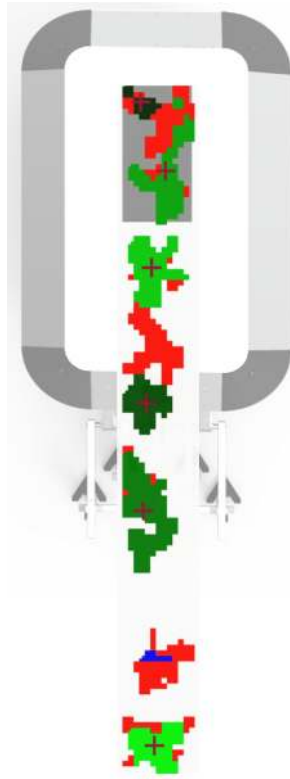


Figure 5.28: Crop Segmentation and Centroid Extraction

5.10.1 Segmentation Software

The segmentation software which is required for crop detection must distinguish crops from weeds and soil. This is a challenging and time consuming task to achieve, as each crop is different in size, health and profile. The team considered implementing a segmentation software from scratch by implementing a neural network and training it on data collected on sugar beet fields. However, by taking into account that collecting and labeling the necessary data is a time consuming task and would unlikely lead to reliable results with only one possible crop cycle, the team decided against this option. Another reason being that there are existing segmentation software which have perfected this task and could be implemented into the system.

Existing Solutions

After comparing the scarce amount of existing solutions, the company *Deepfield Robotics* was contacted which has worked around this problem for the last 5 years and has trained their algorithm on over 2 million labelled images.¹ This is more than any other trained segmentation software that was encountered during research. *Deepfield Robotics* provides a product named *Aquila*, which is a camera module with an integrated software for sugar beet detection. This module contains a 4 channel camera, an on-board computer as well as LEDs for constant illumination. Fortunately, the *Aquila* camera module was able to be borrowed throughout the duration of the project and be integrated in the system.

¹Another option the team looked into is *Bonnet*, an open-source based CNN which has been trained on various data, including sugar beet. This was not implemented because it was discovered towards the end of the project and there was no time left to try out this subsidiary approach. More information on this can be found here: [PRBonn - bonnet](https://github.com/PRBonn/bonnet)

Camera Module *Aquila*

The *Aquila* module takes RGB as well as NIR images in order to create a segmentation matrix. This matrix contains three possible segmentation values: crop, weed and soil (shown as green, red and transparent respectively in fig. 5.29). In order to speed up the process, the segmentation algorithm is applied on a region of interest which covers $20\text{ cm} \times 40\text{ cm} = 800\text{ cm}^2$ of the ground.

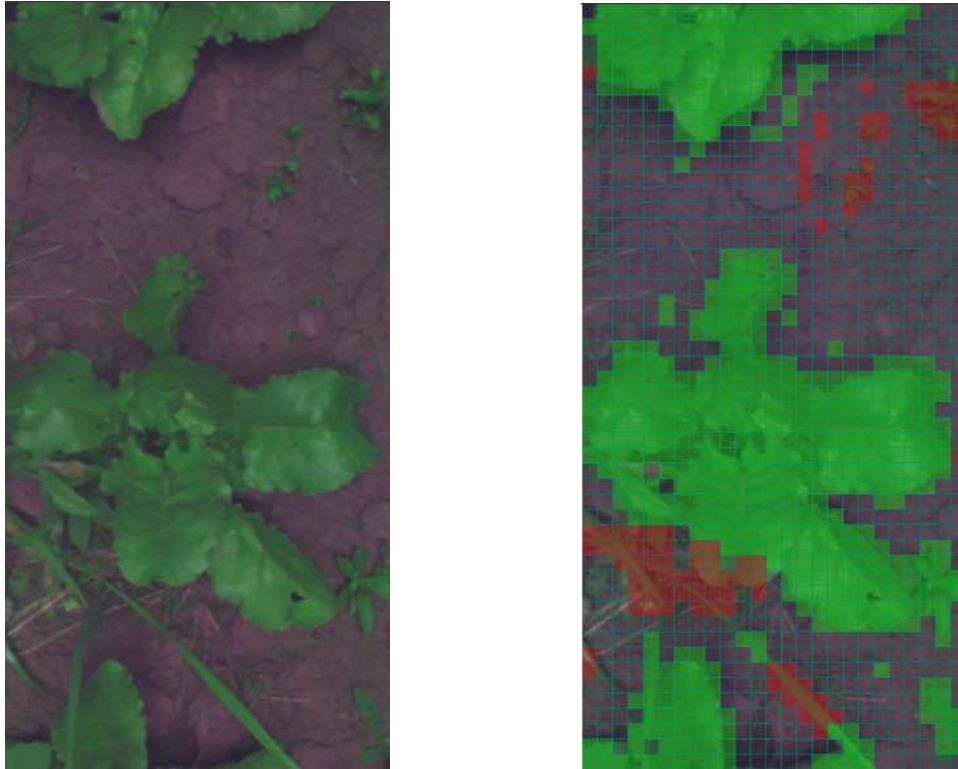


Figure 5.29: Aquila Segmentation Software, ROI $20\text{ cm} \times 40\text{ cm}$

Aside from other attributes, the *Aquila* provides, one notable feature is the constant lighting condition it creates using the LEDs mounted aside the camera. This light source is turned on and off alongside the frame-rate of the camera to reduce power consumption (see fig. 5.30).



Figure 5.30: Aquila Module from *Deepfield Robotics*

Implementation

Mounting the camera module

To implement the *Aquila* into the system, two item rods were placed for mounting the module at the required height of 50 cm above the ground with respect to the camera lens (see fig. 5.31).



Figure 5.31: *Aquila* mounting position (left) and bottom view (right)

Rosmaster running on camera module

The module launches a master in *ROS*, which is why the on-board computer of the system and the raspberry pi were connected to the module's rosmaster. This is a suboptimal set up, since it is preferred to run the rosmaster on the on-board computer because most of the nodes are launched there. Despite that, it did not affect the systems capabilities to run all of the nodes simultaneously. Implementing *multimaster*² would be a possible solution to launch a second rosmaster on the on-board computer, however, adding this feature is a time-consuming task and was not deemed necessary.

5.10.2 Centroid Extraction

In order for weed manipulation to be added for future implementation of for instance the active weeding, a centroid extraction algorithm was built in addition to the segmentation software. The goal of this algorithm is to distinguish crops from another and specify their stem / centroid. Since the segmentation software provides reliable crop segmentation, the pixel locations of said crops are used for the centroid extraction.

Overview

By comparing different clustering algorithms on labeled data, it was concluded that DBSCAN leads to the most consistent results with a relatively fast processing time of 0.01 ms. Hence, DBSCAN (*Density-based spatial clustering of applications with noise*) was chosen and implemented for the centroid extraction process.

Requirements

For precise removal of weeds in close proximity to crops, it is necessary to rely on accurate predictions of the crops centroid positions.

Since the mechanism to be chosen is optional, the requirement was oriented towards the bachelor thesis *active weeding*. For this mechanism, a requirement of 2 cm

²find more information on multimaster <http://wiki.ros.org/multimasterhere>.

accuracy is necessary. This means, that the algorithm shall locate centroids within a radius of $r_{pred} \leq r_{max} = 2 \text{ cm}$ from its ground truth (see fig. 5.32).

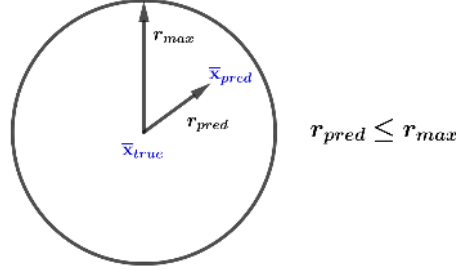


Figure 5.32: Centroid Accuracy Requirement

Approach

The segmentation software provides a segmentation matrix with detected crops. Hence, segments of the image are identified as sugar beet. A straight forward approach would be to calculate the center of all the detected segments, assuming they all belong to the same sugar beet. However, with a ground coverage of 40 cm in length, the region of interest of the Aquila module can detect up to three crops per frame. Therefore, the segments detected as sugar beet do not all belong to the same crop and have to be distinguished from each other. A common approach for such problems are clustering algorithms. Hence, research and evaluation was done on different clustering algorithms to determine the one most suited for centroid extraction.

Research and Evaluation

Suited clustering algorithms

There are a lot of different clustering algorithms with various applications. To narrow it down, focus was set to those which were suited for the stated problem mentioned previously:

- Affinity Propagation
- Mean-Shift
- DBSCAN *Density-based spatial clustering of applications with noise*

What differentiates these clustering algorithms from others is that they do not rely on the previous knowledge of the absolute number of clusters. This is necessary, since the amount of sugar beet within one segmentation matrix may vary.

Qualitative Evaluation

Of the mentioned clustering algorithms, DBSCAN proved to be the best because it allows to bridge elements of the segmentation matrix which are located close to each other, however not adjacent. This distance can be chosen by tweaking the *epsilon* parameter. The speed of this algorithm occurs because it does not calculate the centroid iteratively. It checks and labels each element one at a time and therefore does not require a lot of computing power.

Quantitative Evaluation

In order to compare the various clustering algorithms on the specific problem of

sugar beet distinction, a quantitative evaluation was made on labeled data. This data was extracted from the Aquila module and comes in the form of a segmentation matrix including the ground truth of the crops centroid locations.

The three clustering algorithms were then applied on the labeled data. The accuracy was measured by counting the number of correct centroid locations of varying radius and number of detected clusters found. The results are shown in fig. 5.33.

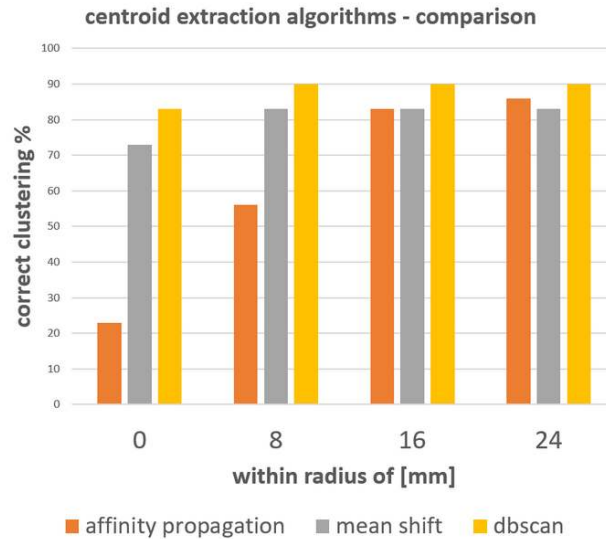


Figure 5.33: Quantitative comparison of clustering algorithms

As can be seen in the results, DBSCAN is the most accurate. Hence, DBSCAN was chosen to be implemented as it suits the input data the best.

DBSCAN

DBSCAN, *Density-Based Spatial Clustering of Applications with Noise*, is an algorithm which takes a list of the positions of each element as input and outputs a list of clustered elements. This implicates, that the algorithm finds elements which belongs together and gives each cluster an individual *cluster-ID*.

The algorithm goes through each individual element of the list and checks for neighbouring elements. If the neighbouring elements are within a certain proximity they get added to the cluster. This proximity is determined by the *epsilon* parameter.

A second parameter, *minpoints*, defines the minimum amount of elements to make up a cluster. If an identified cluster contains less elements than required by said parameter, the cluster is identified as noise. This is visualized by the blue pixels within the image shown in section 5.10.2 (more information on DBSCAN can be found here).

Implementation

As other software packages within the system, the centroid extraction algorithm is a node within the ROS-environment of the robot. The key steps for the implementation of the centroid extraction are the following:

1. Crop Segmentation
2. Clustering crops with DBSCAN
3. Centroid calculation of individual clusters

In a first step (1), the segmentation software identifies crop positions within the ROI. This information is converted into a list of each identified crop element including information about its position.

The second step (2) uses this list in order to distinguish various crops from one another. This is achieved by implementing the DBSCAN. This clustering algorithm then adds a cluster-ID to each element of the list which groups different elements together to one cluster.

In a last step (3), the centroids then get calculated for each crop cluster. The centroid is simply the weighted average of each location of the elements which make up a cluster. Therefore, this is achieved by the following eq. (5.21):

$$\mathbf{c}(i) = \frac{1}{n(i)} \sum_{j=1}^{n(i)} \mathbf{x}_j \quad (5.21)$$

i : cluster-ID

$n(i)$: number of elements of the cluster-ID

$\mathbf{x}_j$: element position

As an additional step, the node outputs the crops centroids with respect to the robots coordinate system. In order to verify the correct crop distinction, each detected crop is displayed with a different shade of green.

Parameters

One challenge when working on clustering crops comes in the form of different growth stages. In order to make up for these growth stages, the *minpoints* and *epsilon* parameters are used to tweak the different stages.

Conclusion

Functionality

The centroid extraction works reliably on leaf stadiums 0, 2, 4 and 6. The last growth stage (8 leaf stage) leads to flaws, since the leaves of neighbouring sugar beets overlap and the clustering algorithm does not distinguish them from each other. This leads to faulty centroids and wrong number of detected crops. However, since weeding is not done for the last growth stage, it does not hinder the systems functionality.

Improvements

One improvement would be to add a *kalman filter*, for an even more accurate evaluation of the centroid position. This could be achieved by using the state estimation and taking the distance between each crop into account (on average these distances vary between 15 cm and 20 cm).

Another improvement could come in the form of autonomously detecting the growth stage, which would save the user from changing the DBSCAN parameters. A rough implementation of this could be achieved by counting the amount of pixels detected as sugar beet and changing the parameters for each range from the number of detected crop pixels.

Chapter 6

Outcome and Future Work

6.1 Outcome

The following section contains a discussion of the goals set in section 2.6.2 compared to the ones eventually achieved. Table 6.1 shows an overview of how well each goal was fulfilled. A more detailed explanation is found further below.

Goals set	Percentage achieved
Sustainable	80%
Protect the Crops	80%
No extra Labour for Farmers	70%
Safety	95%

Table 6.1: Comparison of the goals set and the goals achieved

The goal of the project was to build a prototype which shows that sustainable weeding can be achieved with an autonomous solution and that overcoming common obstacles such as getting stuck or relying on a preprogrammed field map is feasible. Moreover, it wanted to prove that it is possible to build a robot which does no harm to the crops, puts no extra labour on the farmers and can be operated safely.

Sustainable

The energy for operating the system can be taken from Li-Ion Batteries, which - if electricity from renewable sources is used - provide an ecological power source as outlined in section 4.2. Furthermore, the implemented passive tool introduced in section 3.5.2 removes 70% of the weeds in the field without the use of herbicide. This value, however, will be enhanced by the implementation of an active tool, which is part of a bachelor thesis (see section 6.2). Therefore, the 80% in the sustainability goal resulted.

Protect the Crops

The goal to protect the crops was mostly achieved. There is no occurrence of severe soil compaction despite the weight of the robot. The strict adherence to the clearance zone around the crops and the correspondingly designed hardware provides that parts of the robot rarely get in way with the plants. As mentioned in section 5.9.3, following full rows could be achieved, although the precision needs to be improved in order to protect the crops. The initial goal of only harming 1 out

of 100 plants is clearly missed, which is why a score of 80% results in this category.

No extra Labour for Farmers

Rowesys' vision to provide an autonomous solution for organic weeding implies that no extra labour must be added for the farmers. The robust locomotion provides that the robot rarely gets stuck on rough agricultural terrain and therefore there is not often a need for human interventions to ensure mobility. Moreover, no additional infrastructure such as GPS RTK or seeding data is needed and the requirement of a surface power of one hectare per day can be adhered to. However, the end of row detection as described in section 5.9.4 only works under certain conditions and needs to be improved in order to allow for everyday use of the robot. Furthermore, additional features such as a field changing algorithm need to be implemented to ensure full autonomy. Hence, a score of 70% was allocated.

Safety

All of the safety measures described in section 5.4.3 provide a suitable protection for the people working with the robot and the robot itself in an academic context. Additional safety features, which allow for a commercial use, were not considered and therefore not implemented. Furthermore, almost the whole system is protected from light rain and dust (IP 54) as well as sunlight, which is tackled by the casing, as required by the SRS. Only the incremental encoders of the *maxon* motors do not have an IP rating. They are, however, additionally sheltered by the casing in the edges of the platform. This results in a score of 95% in this category.

The requirements met and the goals achieved make Rowesys a unique solution for autonomous weeding. The foundation for a reliable navigation pipeline has been laid and a well-working platform has been developed. It is the hope of team Rowesys that its system helps to kick off the next revolution in agriculture, which will be robotic.

6.2 Future Work

Even though project Rowesys achieved most of its goals, there is still a lot of room for improvement. The team paid a lot of attention on making the robot versatile and possible to adapt as well on hardware as on software side. A selection of future development possibilities will be discussed in the following paragraphs.

More reliable navigation

One subsystem, which still leaves room for further development is the autonomous navigation. It could be made more reliable, adapted for work during the night and several additional features could be implemented. Those include for example an integration of a LiDAR sensor for a sophisticated obstacle avoidance, which does not rely on satellite data, or the development of a field changing algorithm, as it is done in a bachelor thesis described further below. Furthermore, an autonomous deployment function from the robot's garage to the field could be added to enhance the autonomy of the robot.

Phenotyping

Another possibility is to implement a phenotyping mechanism not only for crop mapping as it is done in a bachelor thesis but also for weeds such that they can be analysed, mapped and further measures taken by the farmer if needed. Moreover, the weeds, which are beneficial for the crops because they support their growth, could be avoided by an active weeding tool.

Sophisticated active weeding tool

For the project, the team decided on using a mechanical solution for the active weeding tool. However, there are many other possibilities to fight weed such as using lasers or microwave radiation. According to the results from tests of the *Technical College Bern* [11], none of these technologies has reached a level such that it might get interesting for the everyday use in agriculture for the time being. However, as there is a lot of research in these areas [38], this might change.

Repurposing

Another possible option is to repurpose the robot for other applications. A lot of the implemented features such as the suspended platform or powerful propulsion are not only valuable in agriculture but also in other areas. Rosie could be used as a robotic platform for other engineering challenges.

Bachelor Theses

Some of the aforementioned additional features inspired the team to implement them in the form of a bachelor thesis. These are described below.

Autonomous Navigation between Fields

One thesis approaches the challenge of changing fields. The goal is to enable the robot to weed multiple fields and thus increase its autonomy. To achieve this, obstacle avoidance and path planning are implemented.

Phenotype Crop Mapping of Sugar Beet Fields

In order to prevent diseases early on it is necessary for farmers to know about the health of their crops. This bachelor thesis focuses on recreating a field map and highlighting the state of health of the sugar beet by using the data gathered through on-board cameras.

Development of Active Weeding Tool

Another bachelor thesis concentrates on the development of an active weeding tool as outlined in section 3.5.3. The thesis focuses on the primary weeding mechanism, independent of the moving platform of the robot. The mechanism allows the destruction of weeds and dodging the crops even if the platform is not centered around the lane.

Integration and Control of Active Weeding Tool

The last thesis builds upon the aforementioned one and tackles the design of a mechanical interface between the platform and the active tool. This includes the synchronization of the propulsion motors of the active tool with the output of the endeffector's trajectory planning algorithm such that the tool dodges crops and destroys weed. Furthermore, the implementation of a lifting mechanism for the tool including its low- and high-level controller is part of this thesis.

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Appendix A

Robot Manual

A.1 General Manual

A.1.1 Start System

There are four steps to start the whole system. The positions of the buttons to press are shown in fig. A.1. The switches (C) and the red button (D) will be mounted at the outside of the electrical box soon. Also the electrical box will be labeled in detail soon.

1. Initialisation

Make sure that the three switches (C) are switched to the black side (off) and that the red button (D) is pressed (off). This avoids an over-current error of the batteries.

2. Turn on battery

To turn on the batteries, press the battery button (A) for 2 seconds until it shines green.

3. Turn on system

Switch the three switches (C) to the not black side (on) and unpress the red button (D) (on). The switches connect the fans, the *Elmo* drives and the Nuc, Router, Switch and USB hub with the power supply. The red button connects the *Linak* motor with the power supply.

4. Turn on PC

Press the NUC button (E) such that it shines green. This turns on the Intel NUC. After approximately one minute, the screen should show the Ubuntu desktop.

To shut down the system, do these steps in reverse order: Power off the NUC, turn off the switches and finally press the battery button until it does not glow anymore.

To turn off the system immediately in case of an emergency, one of the four emergency buttons in the top corners of the robot can be pressed. This turns off the batteries.

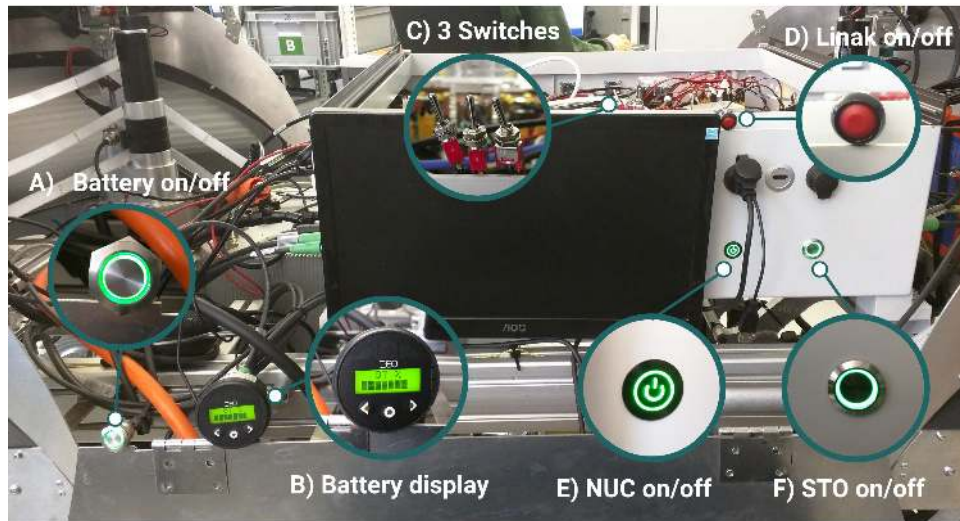


Figure A.1: Positions of buttons and switches

A.1.2 Start Software

The system has to be turned on. If this is not the case, follow the steps in appendix A.1.1.

To start the software, four steps are necessary.

1. Enable STO

Press the STO button (F) in the electrical box such that it shines green. Take the sender of the remote emergency button and pull the red button out. Important: Every 10 minutes, a button on the remote emergency sender has to be pressed. If for a time interval of 15 minutes, no button is pressed, the sender turns off as a safety feature. If this happens, press a button on the remote emergency sender and afterwards press the "Ethercat reset" button on the gamepad.

In case of an emergency regarding the motors, press the red button of the remote emergency sender to disconnect the motors from the power.

2. Open terminal

3. Type `sudo -s`

The shell needs to run with root privileges for the motor communication.

4. Start desired launch file

After launching, the person that is responsible for operating the robot needs to be able to press the remote emergency button at any time. As long as the STO is activated, this person has to monitor the system without exception.

Be careful! The absolute encoders often lose their position after the system is turned off. In this case, the steering motors turn more than 180° and can damage the cables. If one wheel turns more than 90° away from the straight position after starting the software, press the remote emergency button and refer to appendix A.1.3.

There are several possible launch files that can be launched.

- `roslaunch rowesys.launch rowesys.all.launch`
This starts the whole autonomous and manual pipeline. The robot can be operated manually and it can be switched to autonomous mode.
- `roslaunch rowesys.launch manual_drive.launch`
This starts the manual mode.
- `roslaunch rowesys.launch manual_state_estimator.launch`
This starts the manual mode as well as the state estimation (for testing).
- `roslaunch rowesys.launch manual_line_detection.launch`
This starts the manual mode as well as the line detection with the state estimation (for testing).

After a launch file is launched, it takes 10 to 30 seconds until the communication with the motor drivers is established and the motors are in position. A "system is ready" sound informs the user as soon as this is the case. From now on, the system can be operated using the gamepad (appendix A.2).

To stop the system, go back to the terminal and press Ctrl+C. If this does not work for some reason, the on/off button of the NUC (E) can be used to power off the NUC and stop the software.

A.1.3 Dealing with Errors

This section will never be complete. It contains the most common errors that can happen and how to deal with them.

Motors do not find their position when initializing system

This is the case, if an absolute encoder has lost its position.

1. Press the remote emergency switch to stop the motors.
2. Steer the wheel manually in the direction it has steered wrongly, until it is slightly more than 180° away from the straight position.
3. Switch the encoder reset switch for the corresponding wheel on and off (fig. A.2).
4. Steer the wheel back by 90°.
5. Switch the encoder reset switch on and off.
6. Steer the wheel into the correct straight position.
7. Switch the encoder reset switch on and off one last time. Now the encoder should have the correct position.
8. Re-activate the remote emergency button and press the "Ethercat reset" button on the gamepad to activate the motors again. If this is not working, restart the software.

Motors suddenly start steer although no driving input is given

This seems to be a problem with the Ethercat. Press the "Ethercat reset" button immediately once or twice. If this is not working, restart the software.



Figure A.2: Position of encoder reset switches

Turn is activated but robot keeps driving forward at the end of the row in autonomous mode

This is most likely a problem with the state estimation. Restart the software.

Robot keeps driving in manual mode and does not react to gamepad inputs

The connection to the gamepad is lost. Make sure to stay in close distance to the robot, but maximal 10m away.

Robot suddenly stops and motors are turned off

Probably the STO is triggered. Most likely no button on the remote emergency sender was pressed for more than 15 minutes. Press a button on the remote emergency sender and press afterwards the "Ethercat reset" button on the gamepad.

When trying to turn on the batteries, a "PRECHAR." error occurs on the battery display

Turn batteries off, ensure that the three switches are turned off and the red button is pressed. Than try again.

When trying to turn on the batteries, a "RELAIS +" error occurs on the battery display

Probably one of the four emergency buttons is pressed. Unpress it and try again.

A.1.4 Charge Batteries

If the battery display shows under-voltage, the batteries have to be charged.

1. Turn off the system
2. Connect the charging station with the charging outputs of the electrical box
3. Connect the white CAN cable of the battery with the CAN cable on the robot
4. Plug in the charging station in a socket

After plugging in the charging station, the batteries are turned on. If the batteries are charged completely, the charging station turns off and can be disconnected from the electrical box.

A.1.5 Passwords

The password for the Intel NUC is *Rowesys1719*.

The password for the router is *RowesysBot*.

The password for the Raspberry Pi is *rowesys*.

A.1.6 Contact

For further questions or problems please contact

- the team for general questions
rowesys@ethz.ch
- Timo Schöneegg in urgent cases
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A.2 Manual for Gamepad Control

This manual refers to the Logitech F710 gamepad.

Manual - Autonomous

Switch between the manual and autonomous modes by using the buttons A, B and Y.

To switch to manual mode, press B.

To switch to the autonomous mode line_following, press A.

To switch to the autonomous mode turning, press Y.

Steering in manual mode

Driving

You drive by giving twist commands using the gamepad. Please be aware, that some inputs can not be fulfilled by the currently active high-level controller. For example in the ackermann steering mode, can not have an angular velocity, but no linear velocity in x-direction. In this case, the robot prefers the linear inputs.

Linear twist in x-direction There are two possibilities to give a forward speed.

- 1) Press the RT trigger to drive forward and the LT trigger to drive backwards.
- 2) Move the right mini joystick up to drive forward and down to drive backward.

If the trigger and the mini joystick give inputs, the trigger input is preferred.

Linear twist in y-direction Move the right mini joystick to the left, to drive lateral to the left and move it to the right to drive to the right.

Angular twist in z-direction To turn the robot, you can give an angular velocity to it. Move the left mini joystick to the left for a positive angular velocity and to the right, for a negative angular velocity.

Switching between high-level controllers

Press X to switch between the different high-level controllers in the order:

ackermann - swerve

Additional features

Emergency stop

To stop the robot immediately, no matter in which mode it is, press the RB button. The robot switches to the emergency controller and to manual mode.

To resume with driving, press X again. This switches back to the last high-level controller before the emergency controller was activated.

Ethercat

To reset the ethercat, press the LB button.

Geese feet

Press key UP to lift the geese feet.

Press key DOWN to lower the geese feet.

Overview

B	manual mode
A	autonomous line detection mode
Y	autonomous turn mode
X	switching between high-level controllers (only in manual mode possible)
RB	activate emergency controller to stop robot and switch to manual mode
LB	reset the ethercat

Left and right trigger/vertical axis right minijoy

drive forward and backward

Horizontal axis right minijoy

drive lateral (only in swerve steering mode possible)

Horizontal axis left minijoy

angular velocity for turning

Key UP/DOWN

lift/lower geese feet

Appendix B

Additional Information

B.1 Stable Platform Calculation for Suspension

This calculation defines the maximum angle the platform of the robot is allowed to tilt. The maximum angle is defined as the angle such that each goosefoot sweeper is still in touch with the soil. This is the case if $\alpha \leq \tan^{-1}(\frac{3}{55}) = 3.12^\circ$ (see fig. B.1). If $\alpha = 3.12^\circ$ then the outer goosefoot sweeper is just in touch with the soil which ensures the weed is still cut off. If $\alpha > 3.12^\circ$ then the outer goosefoot sweeper is not in touch with the soil anymore and thus the weed is not cut off. Therefore, the maximum tilt angle of the platform is $\alpha_{max} = 3.12^\circ$.

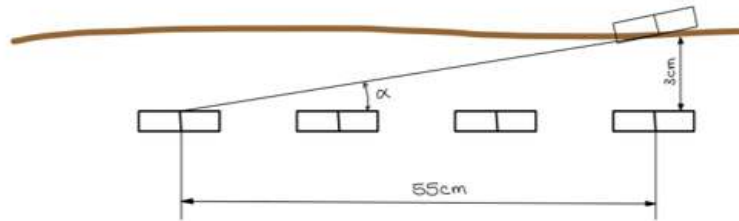


Figure B.1: Sketch for the maximum allowed angle of the platform based on the requirement that each goosefoot sweeper must be in touch with the soil.

B.2 Tipping Over Calculation

The situation of the robot which was used was an inclined plane with an angle θ . The robot is moving at maximal speed and turning with the maximal angular velocity defined in a field. As environmental condition a wind of 40 km/h blowing from the side has been superposed.

The first thing which had to be done was calculating the center of gravity. This was accomplished by taking the weights of the single parts into account and superposing them with respect to the soil in order to get the height of the cog.

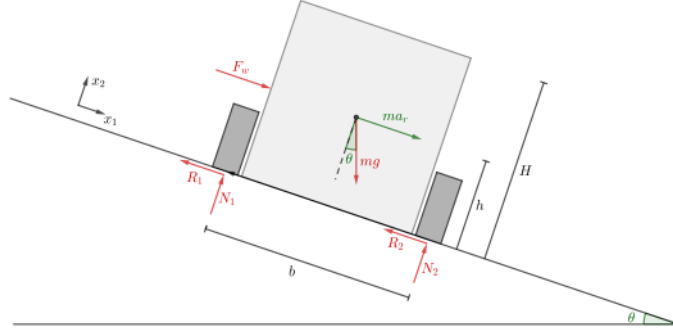


Figure B.2: The model of the robots worst case situation for tipping over

$$x_{cog} = \sum_i^N \frac{x_i m_i}{M} \text{ where } M = \sum_i^N m_i$$

Then linear momentum balance and the angular momentum balance was applied, with all the inputs previously mentioned and also shown in the graphic.

AMB:

$$-N_1 \cdot b - F_w \cdot \frac{H}{2} - m \cdot a_r \cdot h - h \cdot \sin(\theta) \cdot m \cdot g + \cos(\theta) \cdot m \cdot g \cdot \frac{b}{2} = 0$$

$$N_1 = \cos(\theta) \cdot m \cdot g \cdot \frac{1}{2} - \frac{F_w \cdot H}{2 \cdot b} - \frac{m \cdot a_r \cdot h}{b} - \frac{h \cdot \sin(\theta) \cdot m \cdot g}{b} > 0!$$

LMB:

$$x_1 : \sin(\theta) \cdot m \cdot g + F_w - R_1 - R_2 = -m \cdot a_r$$

$$x_2 : N_1 + N_2 - \cos(\theta) \cdot m \cdot g = 0$$

$$R = R_1 + R_2$$

$$\Rightarrow R = m \cdot a_r + F_w + \sin(\theta) \cdot m \cdot g$$

$$N_2 = \cos(\theta) \cdot m \cdot g - N_1$$

But as the reader may have noticed, the radial acceleration is still unknown.

$$a_r = \underbrace{\ddot{r}}_{\text{Circular trajectory}} + r \cdot \dot{\varphi}^2 = r \cdot \dot{\varphi}^2$$

$$\sin(\theta) = \frac{l}{2 \cdot r}$$

$$\tan(\varphi) = \frac{l}{t+b/2}$$

$$\dot{\varphi} = \frac{v}{r}$$

$$r = \sqrt{(t+b/2)^2 + (l/2)^2}$$

$$\rightarrow t = \frac{l}{\tan(\varphi)} - b/2$$

$$r = \sqrt{\left(\frac{l}{\tan(\varphi)} - b/2 + b/2\right)^2 + (l/2)^2} = \sqrt{\left(\frac{l}{\tan(\varphi)}\right)^2 + \left(\frac{l}{2}\right)^2}$$

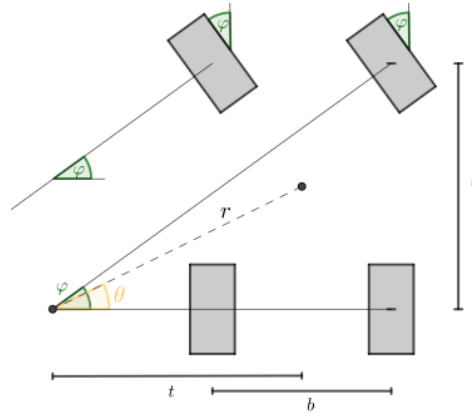


Figure B.3: Car model that was used to calculate radial acceleration

$$a_r = r \cdot \dot{\varphi}^2 = r \cdot \left(\frac{v}{r} \right)^2 = \frac{v^2}{\sqrt{\left(\frac{l}{\tan(\varphi)} \right)^2 + \left(\frac{l}{2} \right)^2}}$$

All the forces were implemented inside an excel file in order to be able to tweak and tune the different parameters.

maximal Slope Acceleration for half car model	
parameter	value
length [m]	1.0
weight [kg]	300.0
width [m]	0.5
CoG_x [m]	0.5
CoG_y [m]	0.25
CoG_z [m]	0.4
slope θ [degree]	10.0
F_{wind} [N]	132.0
total height [m]	0.9
speed [m/s]	3.0
turning angle [deg]	20.0
radial acceleration [m/s^2]	3.222799525
minimal normal force [N] (for 2 wheels)	148.035453
resistance [N] (for 2 wheels)	1609.886444
maximal normal force(for 2 wheels)	2750.253764
Fs2(N)	368.4210526
Wheel torque(Nm)	70.0
diameter(m)	0.38
Ms1=Ms2	257.8947368
h:height of susp attachment(m)	0.7
Ms3=steering torque(Nm)	75.0
FS1(lateral)	804.9432222
Fs3(vertical)	1375.126882

B.3 Steering Speed Calculation

Given parameters

width track $d = 0.30\text{m}$
width wheel $w = 0.13\text{m}$
lateral offset $h = \frac{d-w}{2} = 8.5\text{cm}$
 $x(t_0) = 0, x(t_1) = 0.15\text{m}, x(t_f) = 0.3\text{m}$
 $y(t_0) = 0, y(t_1) = h, y(t_f) = h$
 $\psi(t_0) = 0, \psi(t_f) = 0$
 $v = 0.2\text{m/s}$
Assumption: $\dot{\psi} = \text{const}$

Derived equations

$$\begin{aligned}\dot{x}(t) &= v \cdot \cos(\dot{\psi}t) \\ \dot{y}(t) &= v \cdot \sin(\dot{\psi}t) \\ \Rightarrow y(t) &= \int_0^t \dot{y}(t) dt = v * \int_0^t \sin(\dot{\psi}t) dt = \frac{v}{\dot{\psi}} * (1 - \cos(\dot{\psi}t))\end{aligned}$$

Condition 1: $y(t_1) = \frac{h}{2}$

$$\begin{aligned}t_1 &= \frac{1}{\dot{\psi}} \cdot \arccos(1 - \frac{h \cdot \dot{\psi}}{2 \cdot v}) \\ x(t_1) &= \int_0^{t_1} \dot{x}(t) dt = v \cdot \int_0^{t_1} \cos(\dot{\psi}t) dt = \frac{v}{\dot{\psi}} \cdot \sin(\dot{\psi}t_1) \\ \Rightarrow \dot{\psi} &= \frac{v}{x(t_1)} \cdot \sin(\dot{\psi}t_1) = \frac{v}{x(t_1)} \cdot \sin(\arccos(1 - \frac{h \cdot \dot{\psi}}{2 \cdot v}))\end{aligned}$$

Condition 2: $x(t_1) = \frac{x(t_f)}{2} = 0.15\text{m}$

$$\Rightarrow \dot{\psi} \geq 0.699409\text{rad/s} = 40.1^\circ/\text{s}$$

Condition 3: $t_f \leq 1.5\text{s}$

$$\begin{aligned}\dot{\psi} &= \frac{1}{t_1} \cdot \arccos(1 - \frac{h \cdot \dot{\psi}}{2 \cdot v}) \\ \Rightarrow \dot{\psi} &\geq 0.777324\text{rad/s} = 44.5^\circ/\text{s}\end{aligned}$$

B.4 Swerve Drive Decision Process

The decision of swerve drive was evaluated for different aspects which are listed in fig. B.4.

	Relevance [1-3]	Skid	Ackermann	Swerve Drive	Articulated
Steering					
Wheel alignment ensured	3	yes	yes	no	yes
No-slip condition	3	no	yes	yes	yes
High agility within row	3	no	no	yes	no
High agility for turning	2	yes	no	yes	no
Low complexity for turning	2	yes	no	yes	no
Reduced ground damage	2	no	yes	yes	no
Implementation					
Suspension harmony	3	yes	yes	yes	yes
Easy to handle torque for turning wheel	2	yes	yes	yes	no
Other					
Space-efficient	3	yes	no	yes	no
Light	2	yes	yes	no	yes
Cheap	1	yes	yes	no	yes
Personal interest					
Fun	3	no	no	yes	no

Figure B.4: Decision steering mechanism

B.5 Swerve steering calculation: Case 2

Case 2: $\phi \neq 0$

In this case the motion of the robot is a rotation. The center of motion is on the line through the middle point of the robot perpendicular to the given speed vector $\mathbf{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix}$. The distance l of the center of motion M to the center of the robot C calculates to $l = \phi/|\mathbf{v}|$ (fig. 5.9). This results in the vector

$$\mathbf{MC} = \begin{pmatrix} v_y \\ -v_x \end{pmatrix} \cdot \frac{1}{\phi}.$$

Now the vectors from M to the centers of the wheels can be calculated.

$$\mathbf{d}_1 = \mathbf{MC} + \begin{pmatrix} d_w/2 \\ d_l/2 \end{pmatrix} = \begin{pmatrix} v_y/\phi + d_w/2 \\ -v_x/\phi + d_l/2 \end{pmatrix}$$

$$\mathbf{d}_2 = \mathbf{MC} + \begin{pmatrix} d_w/2 \\ -d_l/2 \end{pmatrix} = \begin{pmatrix} v_y/\phi + d_w/2 \\ -v_x/\phi - d_l/2 \end{pmatrix}$$

$$\mathbf{d}_3 = \mathbf{MC} + \begin{pmatrix} -d_w/2 \\ -d_l/2 \end{pmatrix} = \begin{pmatrix} v_y/\phi - d_w/2 \\ -v_x/\phi - d_l/2 \end{pmatrix}$$

$$\mathbf{d}_4 = \mathbf{MC} + \begin{pmatrix} -d_w/2 \\ d_l/2 \end{pmatrix} = \begin{pmatrix} v_y/\phi - d_w/2 \\ -v_x/\phi + d_l/2 \end{pmatrix}$$

This results in the following speed vectors for the single wheels, as $\mathbf{v}_i$ has to be perpendicular to $\mathbf{d}_i$.

$$\mathbf{v}_1 = \phi \cdot \begin{pmatrix} v_x/\phi - d_l/2 \\ v_y/\phi + d_w/2 \end{pmatrix} = \begin{pmatrix} v_x - \phi \cdot d_l/2 \\ v_y + \phi \cdot d_w/2 \end{pmatrix}$$

$$\mathbf{v}_2 = \phi \cdot \begin{pmatrix} v_x/\phi + d_l/2 \\ v_y/\phi + d_w/2 \end{pmatrix} = \begin{pmatrix} v_x + \phi \cdot d_l/2 \\ v_y + \phi \cdot d_w/2 \end{pmatrix}$$

$$\mathbf{v}_3 = \phi \cdot \begin{pmatrix} v_x/\phi + d_l/2 \\ v_y/\phi - d_l/2 \end{pmatrix} = \begin{pmatrix} v_x + \phi \cdot d_l/2 \\ v_y - \phi \cdot d_w/2 \end{pmatrix}$$

$$\mathbf{v}_4 = \phi \cdot \begin{pmatrix} v_x/\phi - d_l/2 \\ v_y/\phi - d_w/2 \end{pmatrix} = \begin{pmatrix} v_x - \phi \cdot d_l/2 \\ v_y - \phi \cdot d_w/2 \end{pmatrix}$$

Analogously to case 1, the wheel positions and speeds can be calculated.

For the wheel positions this results in:

$$\begin{aligned}\phi_1 &= \begin{cases} \arctan\left(\frac{v_x + \phi \cdot d_l/2}{y - \phi \cdot d_w/2}\right), & \text{if } v_y - \phi \cdot d_w/2 \neq 0 \\ \pi/2, & \text{otherwise} \end{cases} \\ \phi_2 &= \begin{cases} \arctan\left(\frac{v_x + \phi \cdot d_l/2}{v_y + \phi \cdot d_w/2}\right), & \text{if } v_y + \phi \cdot d_w/2 \neq 0 \\ \pi/2, & \text{otherwise} \end{cases} \\ \phi_3 &= \begin{cases} \arctan\left(\frac{v_x - \phi \cdot d_l/2}{v_y + \phi \cdot d_w/2}\right), & \text{if } v_y + \phi \cdot d_w/2 \neq 0 \\ \pi/2, & \text{otherwise} \end{cases} \\ \phi_4 &= \begin{cases} \arctan\left(\frac{v_x - \phi \cdot d_l/2}{v_y - \phi \cdot d_w/2}\right), & \text{if } v_y - \phi \cdot d_w/2 \neq 0 \\ \pi/2, & \text{otherwise} \end{cases}\end{aligned}$$

For the speed of the wheels this results in:

$$\begin{aligned}v_1 &= |\mathbf{v}_1| = \sqrt{(v_x - \phi \cdot d_l/2)^2 + (v_y + \phi \cdot d_w/2)^2} \\ v_2 &= |\mathbf{v}_2| = \sqrt{(v_x + \phi \cdot d_l/2)^2 + (v_y + \phi \cdot d_w/2)^2} \\ v_3 &= |\mathbf{v}_3| = \sqrt{(v_x + \phi \cdot d_l/2)^2 + (v_y - \phi \cdot d_w/2)^2} \\ v_4 &= |\mathbf{v}_4| = \sqrt{(v_x - \phi \cdot d_l/2)^2 + (v_y - \phi \cdot d_w/2)^2}\end{aligned}$$

B.6 Camera choice

The evaluated cameras for the navigation are listed in fig. B.5.

B.7 Row Detection and Tracking Reliability

B.7.1 Row Detection at different growth stages

Growth stage	Date	Location	Weed Density	Correct	Incorrect	Success rate
spring stage	4.24.20	Eschikon	low	110	15	86.7%
2-4 leaf	5.4.20	Sins	moderate	141	11	92.2%
4-6 leaf	5.12.20	Eschikon	medium	198	4	98.0%
8 leaf	5.26.20	Eschikon	medium	270	5	98.1%

The outliers were identified visually. A detection was classified to be an outlier if either the start or end point were not placed on the row.

Presets

Differences in the preset parameters used for the above analysis concerned the median kernel size. For the spring stage and the 2-4 leaf stage this was set to 2, while

for the bigger plants it was set to 9. This difference is due to the far lower plant size at the earlier growth stages. A big median kernel size would simply remove the crops as well. For bigger plants it can remove weeds between the rows.

Also, the erosion and dilation parameters were changed. Dilating the image can fuse plant parts which were previously not connected, this can be leaves or neighboring plants. The following erosion process will then remove from the outer edges, concentrating the plants towards their center. This process is applied 3 times. If the erosion size is bigger than the dilation size, the row width will shrink in total,

Camera	E-con Systems Tara	Occipital Structure	Orbbec Persee	Intel RealSense R200	Orbbec Astra Pro	Intel RealSense F200	Stereolabs ZED Stereo Camera	DUO mini lx for Xbox 360	Microsoft Kinect Kinect v2	RealSense D435i
Picture										
Released	2016	2014	2016	2015	2015	2015	2015	2013	2014	2017
Price (CHF)	250	499	240	99	150	99	449	695	100	200
Tracking sensing range	Stereo mono cameras 0.5-3m	IR 0.4-3.5m	IR 0.4-8m	IR 0.5-3.5m	IR 0.4-8m	IR 0.2-1.2m	Stereo rgb cameras 1.5-20m	Passive IR 0.3-2.4m	IR 0.5-4.5m	IR and rgb 0.28-10m
rgb		IOS camera resolution	1280x 720, 30 fps	1920x 1080, 30 fps	1280x 960, 10 fps	1920x 1080, 30 fps	1280x 480, 120 fps	640x 480, 30 fps	1920x 1080, 30 fps	1920x 1080, 30 fps
resolution	752x480	640x480	640x480	640x480	640x480	512x424	640x480	320x240	512x424	1280x720
fps	30	30	30	60	30	30	120	20-200	30	90
interface	USB 3.0	USB-C	Ethernet	USB 3.0	USB 2.0	USB 3.0	USB 3.0	USB 2.0	USB 3.0	USB-C
outdoor	☒	☐	☐	☐	☐	☒	☒	☐	☒	☒
tools	C++, ROS	iOS, Unity3D, OpenNI	C++, ROS	Unity3D	OpenNI	Unity3D	ROS Kinetic	Unity 3D	Unity, ROS Melodic	ROS Melodic
sensor	two OnSemi MT9V024 sensors	Structure sensor	Astra Pro		Astra Pro					Global Shutter
IMU		☒								☒
Linux/C++	C++ SDK for Linux	some support for Linux	SOC supports Ubuntu 14.04, C++, ROS		Linux					Ubuntu 18.04, C++, Ros Melodic
features	height estimation, face detection, pointcloud generation	ultra-wide angle	ARM based SOC (System on chip)		Hand tracking 2 Microphones					IMU
SDK	lightweight set camera para- meters (e.g. auto exposure)	OpenNI2	OpenNI2 or Astra SDK		OpenNI2 or Astra SDK					SDK 2.0
environment	ambient lighting, for buildings (no IR projector)	Infrared cameras works outside			Infrared depth sensor, Indoor		no infrared, but nevertheless works outdoor	indoor	problems with bright sunlight, 3D just de- pends on IR	indoor and outdoor
mounting	plastic									M3 thread mounting points

Figure B.5: Considered cameras and evaluation

which is desirable for bigger plants as it increases the accuracy of the detection. Thus, for the growth stage from 4-8 leaves, the erosion size was set to 4 and the dilation size to 2.

For the smaller plants, the opposite applies. The aim is to enhance the crops, therefore the dilation size is set to 4 and the erosion size to 2.

B.7.2 Row Tracking

Growth stage	Date	Weed Density	Correct	Incorrect	Correct d.	Correct t.
8 leaf	5.26.20	medium	198	4	98%	100%
8 leaf	5.26.20	medium	200	34	83.0%	100%

For both levels of tuning, rosbags containing data over a full row on our test field (30 m length per row) were examined. The outlier percentage of the detection was calculated over the course of approximately half a row. For the final estimate from the row tracking containing the same presets for the detection the outliers were counted in three different rows, but none were found.

B.8 Suspension

B.8.1 Free Body Diagram Suspension

the linear and angular momentum balance for the free body diagram 3.14 leads to the following set of equations:

$$\sum F_x \stackrel{!}{=} 0 : B_x - A_x = 0 \quad (\text{B.1})$$

$$\sum F_y \stackrel{!}{=} 0 : B_y + A_y + F_{ext} - F_{SD} = 0 \quad (\text{B.2})$$

$$\sum F_z \stackrel{!}{=} 0 : B_z + A_z = 0 \quad (\text{B.3})$$

$$\sum M_x \stackrel{!}{=} 0 : M_T + l \cdot B_y + l \cdot A_y - \frac{2}{3} \cdot l \cdot F_{SD} = 0 \quad (\text{B.4})$$

$$\sum M_y \stackrel{!}{=} 0 : M_S + h \cdot B_z - h \cdot A_z + l \cdot A_x - l \cdot B_x = 0 \quad (\text{B.5})$$

$$\sum M_z \stackrel{!}{=} 0 : h \cdot F_{SD} - h \cdot A_y - r \cdot A_x + r \cdot B_x + h \cdot B_y = 0 \quad (\text{B.6})$$

with $l = \cos(14^\circ) \cdot L_3 = 0.179m$ and $h = 0.09m$.

solving this equations one ends up with the following bearing forces:

$$F_{SD} = 8kN$$

$$A_z = -B_z = 416N$$

$$B_y = -1.5kN$$

$$A_y = 6.5kN$$

$$B_z = A_z = 0$$

Thus the resulting bearing force is

$$F_{res} = \sqrt{A_y^2 + A_z^2} = 6.5kN \quad (\text{B.7})$$

B.9 Locomotion and Propulsion

B.9.1 Soil Compaction

The locomotion concept must cause as little soil compaction as possible even on humid soil such that the the growth of the crops is not restricted and it is possible

to pull the weeding tool through the soil.

Soil compaction occurs when the pressure applied on the ground by the robot exceeds the soil strength and has a negative impact on plant growth. Therefore, the aim is to minimize it.

Soil compaction occurs as soon as the pressure applied to the soil is $p_{soilcompaction} > 49050Pa$. According to the ETH Pflanzenforschungsstation Eschikon this is a common used value in agriculture. Thus, the contact area per wheel must be bigger than

$$A_{min} = \frac{m_R \cdot g}{p_{soilcompaction}} = \frac{300kg \cdot 9.81m/s^2}{49050Pa} = 0.015m^2 \quad (B.8)$$

B.9.2 Tracks and Wheels

A table of all the arguments for and against tires and/or tracks is shown below. The scores in table 3.2 originated from the following train of thoughts:

- Scoring: 1-2 worst imaginable solution, 3-5 much worse specs. than other solution, 6-7 not satisfying, 8-9 satisfying, 10 best imaginable solution.
- Surface Area: 0.015cm is the minimum area required, which is the absolute minimum because with less, the crops would be severely hindered in growth. The tracks, on the other hand, have a 4 times larger contact area, which is close to the largest area possible (which would be if the whole space between the rows would be filled). Therefore, the tracks scored 9/10 and the tires 3/10.
- Traction: There is little data available for traction of tires and tracks. However, an interview with a salesman of the largest importer of tires and tracks for snowblowers in Europe has shown that this scoring is justified.
- Little Complexity: Since tires consist of only a wheel made out of rubber, they score 10/10. Tracks, on the other hand, consist out of at least 6 parts [9], which include pulleys and the band around the tracks, which make them much more complex in maintenance and implementation. Therefore, they scored only 3/10.
- Personal Interest: This is an arbitrary scoring depending on the team's interest, which has been evaluated by poll.
- Continuous Force Distribution: Considering the aforementioned scoring scheme and fig. 3.20, wheels are the best possible solution and tracks are very bad.
- Availability: There is an abundant amount of wheel resellers, which makes the supply situation perfect. On the other hand, it is very hard to receive tracks in the sizes Rosie would need, and they would need to be customized.

Tracks	
PRO	CONTRA
great traction (especially in wet soil) [9]	supplier from China
easy to implement (suspension; hub higher)	need to be customized in required sizes
smoother drive	issues with obstacles -- > small pulleys
more transfer torque to ground	not produced in this size
narrower	peak pressures under boggy wheels
more fun	more expensive
smoother ride on rough fields	deliver less power to the ground (7/10) [9]
more maneuverable	higher rolling resistance
cannot go flat	Maintenance and Wear
includes suspension (damping from rubber)	no change in tread spacing
	more parts which can break if dirt get inside

Tires	
PRO	CONTRA
easier to overcome obstacles	less traction than Tracks on wet soil
more information available	large wheels get broad
examples around	no suspension included
very good availability	hub lower
reputable suppliers	less fun
more uniform pressure distribution	rough ride on rough fields
cheaper	complex to make them maneuverable
deliver more power to the ground (9/10) [9]	can go flat
lower rolling resistance	
less wear; no maintenance	
tread spacing adjustable	
flexing property -- > self cleaning	
Hub Motors possible	

B.9.3 Motor Power and Torque Calculation

The required torque and power of the propulsion motors was calculated as follows:

Mass of robot: $m = 300 \text{ kg}$

Diameter of wheel: d , whereas 1" equals 2.54 cm

Slope (10°) that has to be overcome: $\alpha = \frac{\pi}{180} * 10$

Velocity while on the field: $v = 0.2 \text{ m/s}$

Velocity while moving between fields: $v_2 = 3 \text{ m/s}$

Coefficient of friction on the field (Worstcase): $\mu = 0.35$ [39] Coefficient of friction off the field: $\mu_2 = 0.2$

Gravitational acceleration: $g = 9.81 \text{ m/s}^2$

The rotational speed is therefore calculated as:

$$n = \frac{v}{2 * \pi * d / 2}$$

respectively,

$$n_2 = \frac{v_2}{2 * \pi * d / 2}$$

Therefore, the frictional force is for the two cases,

$$F_r = m * g * \cos(\alpha) * \mu$$

$$F_{r2} = m * g * \cos(\alpha) * \mu_2$$

and the gravitational force is calculated as

$$F_g = m * g * \sin(\alpha)$$

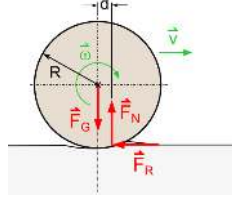


Figure B.6: Friction and other forces

The maximal force that is applied from the goose feet onto the robot is $F_{gf} = 1000N$. This value has been evaluated through testing at ETH Crop Science in Eschikon.

Therefore, the max. torque applied onto all of the wheels on the field with the geese feet down is

$$M_{gf} = (F_r + F_g + F_{gf}) * \frac{d}{2}$$

and the max. torque without the goose feet on and off (2) the field is calculated as

$$M_{nogf} = (F_r + F_g) * \frac{d}{2}$$

$$M_{nogf2} = (F_{r2} + F_g) * \frac{d}{2}$$

This leads to a required torque per wheel of

$$M_{req} = \frac{M_{gf}}{4}$$

Furthermore, the required power per motor is

$$P_{gf} = \frac{2 * \pi * M_{gf} * n}{4 * \eta}$$

with goose feet and

$$P_{nogf} = \frac{2 * \pi * M_{nogf} * n_2}{4 * \eta}$$

without geese feet. $\eta = 0.8$ is the efficiency of the motor and comes from the manufacturer's information.

The results are summarized in table 3.3 in section 3.3.2. For the sake of brevity, only the common standardized wheel sizes are listed.

Matlab Script for calculation of propulsion motor requirements

```
m = 200;
d = 14*0.0254; // 12"
alpha = pi/180*10; // here 10 degrees
```

```

v = 0.2;
v2 = 3;
mue = 0.35; // worstcase
source: https://www.schweizer-fn.de/stoff/reibwerte/reibwerte\_sonstige.php
- rolling resistance
mue2 = 0.2; // for calculation of max power - 3m/s only on compact soil
n = v/(2*pi*d/2); // n = v/(2*pi*r)
n2 = v2/(2*pi*d/2);

F_r = m*9.81*cos(alpha)*mue;
F_r2 = m*9.81*cos(alpha)*mue2;
F_g = m*9.81*sin(alpha);
F_gf = 1000; // Goosefeet (gf) - Worstcase 100kg, otherwise 60kg
M_gf = (F_r+F_g+F_gf)*d/2;
M_ogf = (F_r+F_g)*d/2; // ogf == without Goosefeet
M_ogf2 = (F_r2+F_g)*d/2;
req-torque-each = M_gf/4
P_with_gf = 2*pi*M_gf*n/(0.8); // 0.8 is efficiency of motor (manufacturer information)
P_with_gf_each = P_with_gf/4

P_w/o_gf = 2*pi*M_ogf2*n2/(0.8); // Calculation of required power at max. speed
P_w/o_gf_each = P_w/o_gf/4

```

B.9.4 Hub Motor Evaluation

	Bosch Motors	unbranded	unbranded	Bafang H800
Diameter Wheel	18"	19" x 1.85"	12"	9"
	Moped Tire	120/70-12 Tire		
Required Torque	115 Nm	121 Nm	76.9 Nm	57.6 Nm
required Power für 3 m/s	700 W	700 W	700 W	700 W
Max. Torque Motor	110 Nm	150 Nm	Probably too little	55 Nm
Power Motor	550-5000 W	3000 W	1000 W	500 W
Weight Motor	?	8.9 kg	13.5 kg (w/ wheel)	?
Voltage Motor	48 V	72 V	48 V	48 V
Problem	good option	Heat production	Torque	Heat Production
Max speed 1 m / s				

Maxon BikeDrive	Swincar motor
8"	16"
51.2 Nm	102.5 Nm
700 W	700 W
50 Nm	85 Nm
500 W	1000 W (max 3000 W)
3.5 kg	
48 V	
Max speed 1 m/s	good option
narrow wheels -> high soil compression	SRS need to be adjusted

B.10 Weeding Tools

B.10.1 Spring Selection

According to fig. 3.35 the relevant dimensions are as follows:

lever spring $l_s = 0.26\text{m}$ (frontal sweeper) resp.

lever spring $l_s = 0.41\text{m}$ (rear sweeper)

lever sweeper $h_{GF} = 0.35\text{m}$

traction force $F_{GFmax} = 300\text{N}$ (frontal sweeper) resp.

traction force $F_{GFmax} = 200\text{N}$ (rear sweeper)

spring force F_S

A deflection of 5 cm should occur at $\frac{2}{3}$ of the maximum traction force. The equivalent force on the spring is then given by

$$\Rightarrow F_{S2/3} = F_{GFmax} \cdot \frac{2}{3} \cdot \frac{h_{GF}}{l_s}$$

This gives the maximum spring constant for the spring:

$$\Rightarrow R_{max} = \frac{F_{S2/3}}{5\text{cm}}$$

Which gives the following results for the two different springs:

$$R_{max} = 5.39 \frac{\text{N}}{\text{mm}} \text{ (frontal sweeper)}$$

$$R_{max} = 2.28 \frac{\text{N}}{\text{mm}} \text{ (rear sweeper)}$$

The smaller the actual spring constant, the greater the preload on the spring needs to be.

The maximum force of the spring (at 7 cm deflection) is calculated to:

$$F_{Smax} = R_{max} \cdot 7\text{cm} = 377.3\text{N} \text{ (frontal sweeper)}$$

$$F_{Smax} = R_{max} \cdot 7\text{cm} = 159.6\text{N} \text{ (rear sweeper)}$$

Under consideration that the springs must allow a maximum deflection of 7 cm, a maximum force of F_{GFmax} (safety-factor included) and they need to fit into the design, the right springs could be selected.

B.10.2 Linear Motor Selection

The total possible vertical force (all four springs at its maximum deflection) is $2 \cdot 377.3 + 2 \cdot 159.6 = 1'073.8\text{N}$. The vertical bar of the four-bar linkage acts just in the center of the horizontal position of the two springs, so to simplify the calculations, we suppose the spring force acts directly on the bottom end of the vertical bar.

According to fig. B.7, the vertical forces in the elbow of the linkage system have to be counteracted between the spring force and the vertical projection of the force on the linear motor:

$$F_S = F_{lm} \cdot \cos(\beta)$$

With F_S as the sum of all the spring forces, the maximum force on the linear motor is then given by:

$$F_{lm} = \frac{F_S}{\cos(\beta)} = 1'216\text{N}$$

with $\beta = 28$.

Therefore a linear motor with a maximum force of 1'500 N was chosen.

Since the assumed forces are the maximum estimated forces, and the calculated situation represents all four goosfoot sweepers getting stuck at the same time, a safety factor of only 1.23 was tolerated.

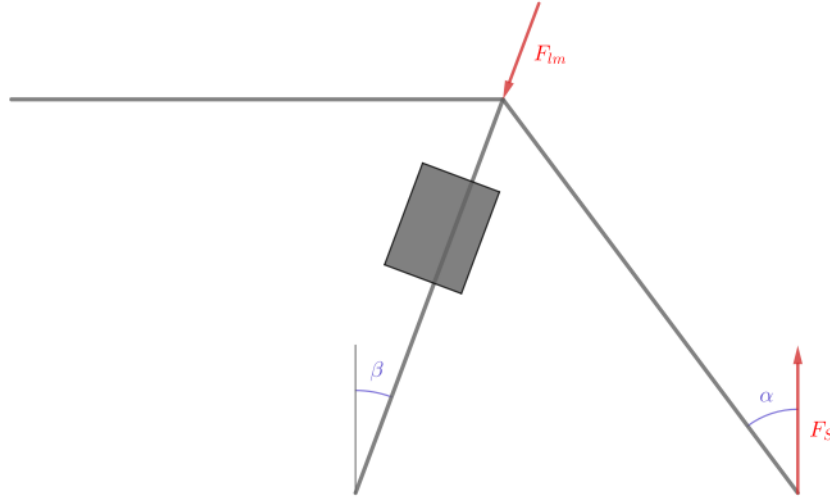


Figure B.7: Forces on the linear motor.

Appendix C

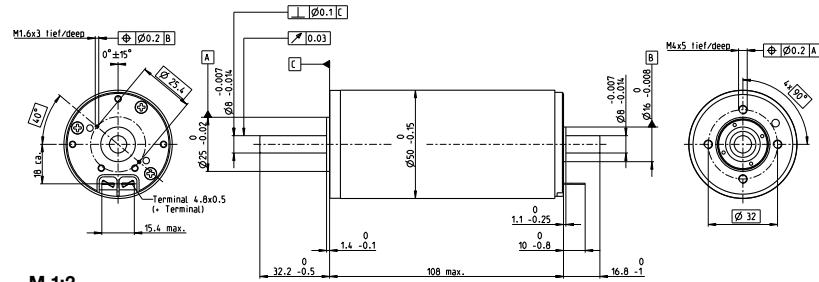
Datasheets

The referenced Data Sheets are presented below.

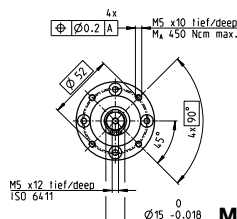
C.1 Steering Data Sheets

C.1.1 Steering Motor Data Sheet

RE 50 Ø50 mm, Graphitbürsten, 200 Watt



Planetengetriebe GP 62 A Ø62 mm, 8.0–50.0 Nm



M 1:4

Planetengetriebe	geradeverzahnt			
Abtriebswelle	Stahl			
Abtriebswellenlagerung	Kugellagerlagerung			
Radialspiel, 7 mm ab Flansch	max. 0,08 mm			
Axialspiel	max. 1 mm			
Max. axiale Belastung (dynamisch)	120 N			
Max. axiale Aufpresskraft	1000 N			
Drehsein, Antrieb zu Abtrieb	=			
Max. Eingangs-drehzahl dauernd	3000 min ⁻¹			
Empfohlener Temperaturbereich	-30...+140 °C			
Stufenzahl	1	2	3	4
Max. radiale Belastung, 24 mm ab Flansch	240 N	360 N	570 N	740 N

Artikelnummern

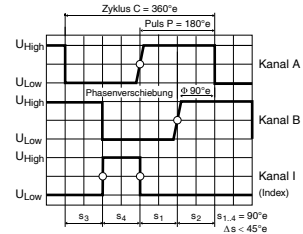
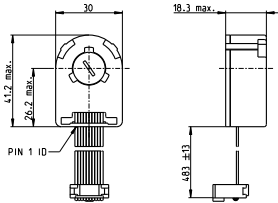
☐ Standardprogramm ☐ Sonderprogramm (auf Anfrage)		110499	110501	110502	110503	110504	110505	110506	110507	110508
Getriebedaten										
1	Untersetzung	5,2:1	19:1	27:1	35:1	71:1	100:1	139:1	181:1	236:1
2	Untersetzung absolut	$\frac{57}{11}$	$\frac{3591}{187}$	$\frac{3249}{121}$	$\frac{1039}{44}$	$\frac{226223}{3179}$	$\frac{204687}{2057}$	$\frac{186189}{1331}$	$\frac{87723}{644}$	$\frac{41059}{176}$
3	Max. Motorwelledurchmesser	mm	8	8	8	8	8	8	8	8
4	Stufenzahl		1	2	2	3	3	3	3	3
5	Max. Dauerdrehmoment	Nm	8	25	25	25	50	50	50	50
6	Kurzzeitig zulässiges Drehmoment	Nm	12	37	37	37	75	75	75	75
7	Max. Wirkungsgrad	%	80	75	75	75	70	70	70	70
8	Gewicht	g	950	1250	1250	1250	1540	1540	1540	1540
9	Mittleres Getriebeispiel unbelastet	°	1.0	1.5	1.5	1.5	2.0	2.0	2.0	2.0
10	Massenträgheitsmoment	gcm ²	109	100	105	89	104	105	102	88
11	Getriebelänge L1	mm	72.5	88.3	88.3	88.3	104.2	104.2	104.2	104.2

[illegible]

C.1.3 Incremental Encoder Steering

Encoder HEDL 5540 500 Impulse, 3 Kanal, mit Line Driver RS 422

maxon sensor



Drehrichtung cw (Definition cw S. 64)

- Lagerprogramm
- Standardprogramm
- Sonderprogramm (auf Anfrage)

Artikelnummern

	110512	110514	110516	110518	X drives
Impulszahl pro Umdrehung	500	500	500	500	500
Anzahl Kanäle	3	3	3	3	3
Max. Impulsfrequenz (kHz)	100	100	100	100	100
Max. Drehzahl (min ⁻¹)	12 000	12 000	12 000	12 000	12 000
Wellendurchmesser (mm)	3	4	6	8	2-4

maxon Baukastensystem

+ Motor	Seite	+ Getriebe	Seite	+ Bremse	Seite	Gesamtlänge [mm] / • siehe Getriebe
RE 25	129/131					75.3
RE 25	129/131	GP 26/GP 32	346/348			•
RE 25	129/131	KD 32, 1.0 - 4.5 Nm	359			•
RE 25	129/131	GP 32, 0.75 - 6.0 Nm	349/352			•
RE 25	129/131	GP 32 S	382-387			•
RE 25, 20 W	130					63.8
RE 25, 20 W	130	GP 22, 0.5 Nm	340			•
RE 25, 20 W	130	GP 26/GP 32	346/348			•
RE 25, 20 W	130	KD 32, 1.0 - 4.5 Nm	359			•
RE 25, 20 W	130	GP 32, 0.75 - 6.0 Nm	349/352			•
RE 25, 20 W	130	GP 32 S	382-387			•
RE 25, 20 W	130		AB 28	491		94.3
RE 25, 20 W	130	GP 26/GP 32	346/348	AB 28	491	•
RE 25, 20 W	130	KD 32, 1.0 - 4.5 Nm	359	AB 28	491	•
RE 25, 20 W	130	GP 32, 0.75 - 6.0 Nm	349/352	AB 28	491	•
RE 25, 20 W	130	GP 32 S	382-387	AB 28	491	•
RE 25, 20 W	131		AB 28	491		105.8
RE 25, 20 W	131	GP 26/GP 32	346/348	AB 28	491	•
RE 25, 20 W	131	KD 32, 1.0 - 4.5 Nm	359	AB 28	491	•
RE 25, 20 W	131	GP 32, 0.75 - 6.0 Nm	349/352	AB 28	491	•
RE 25, 20 W	131	GP 32 S	382-387	AB 28	491	•
RE 30, 15 W	132					88.8
RE 30, 15 W	132	GP 32, 0.75 - 4.5 Nm	350			•
RE 30, 60 W	133					88.8
RE 30, 60 W	133	GP 32, 0.75 - 6.0 Nm	348-355			•
RE 30, 60 W	133	KD 32, 1.0 - 4.5 Nm	359			•
RE 30, 60 W	133	GP 32 S	382-387			•
RE 35, 90 W	134					91.7
RE 35, 90 W	134	GP 32, 0.75 - 8.0 Nm	348-356			•
RE 35, 90 W	134	GP 42, 3.0 - 15.0 Nm	361			•
RE 35, 90 W	134	GP 32 S	382-387			•
RE 35, 90 W	134		AB 28	491		124.3
RE 35, 90 W	134	GP 32, 0.75 - 8.0 Nm	348-356	AB 28	491	•
RE 35, 90 W	134	GP 42, 3.0 - 15.0 Nm	361	AB 28	491	•
RE 35, 90 W	134	GP 32 S	383-387	AB 28	491	•

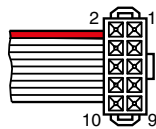
Technische Daten

Versorgungsspannung V_{CC}	5 V $\pm$ 10%
Typische Stromaufnahme	55 mA
Ausgangssignal	EIA Standard RS 422
verwendeter Treiber:	DS26LS31
Phasenverschiebung Φ	90° $\pm$ 45°
Signalanstiegszeit (typisch, bei $C_L = 25$ pF, $R_L = 2.7$ k Ω , 25 °C)	180 ns
Signalabfallzeit	40 ns
(typisch, bei $C_L = 25$ pF, $R_L = 2.7$ k Ω , 25 °C)	90°
Indexpulsbreite	90°
Betriebstemperaturbereich	-40...+100 °C
Trägheitsmoment der Impulsscheibe	≤ 0.6 gcm ²
Max. Winkelbeschleunigung	250 000 rad s ⁻²
Strom pro Kanal	± 20 mA

Das Indexsignal I ist synchronisiert mit Kanal A und B.

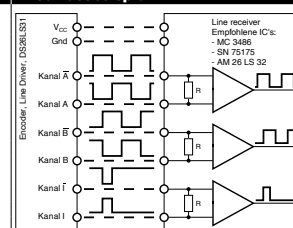
442 maxon sensor

Pinbelegung



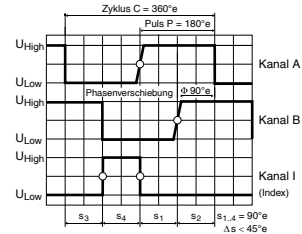
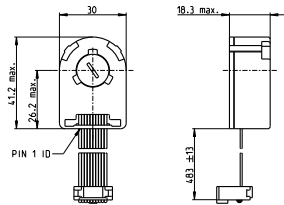
- 1 N.C.
 - 2 V_{CC}
 - 3 GND
 - 4 N.C.
 - 5 Kanal A
 - 6 Kanal A
 - 7 Kanal B
 - 8 Kanal B
 - 9 Kanal I (Index)
 - 10 Kanal I (Index)
- Steckertyp DIN 41651/
EN 60603-13
Flachbandkabel AWG 28

Anschlussbeispiel

Abschlusswiderstand R = typisch 120 Ω

Ausgabe April 2019 / Änderungen vorbehalten

Encoder HEDL 5540 500 Impulse, 3 Kanal, mit Line Driver RS 422



Drehrichtung cw (Definition cw S. 64)

- Lagerprogramm
- Standardprogramm
- Sonderprogramm (auf Anfrage)

Artikelnummern

110512	110514	110516	110518	X drives
500	500	500	500	500
3	3	3	3	3
100	100	100	100	100
12000	12000	12000	12000	12000
3	4	6	8	2-4

Typ

Impulszahl pro Umdrehung	500	500	500	500	500
Anzahl Kanäle	3	3	3	3	3
Max. Impulsfrequenz (kHz)	100	100	100	100	100
Max. Drehzahl (min ⁻¹)	12000	12000	12000	12000	12000
Wellendurchmesser (mm)	3	4	6	8	2-4



maxon Baukastensystem

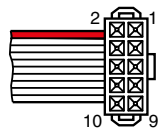
+ Motor	Seite	+ Getriebe	Seite	+ Bremse	Seite	Gesamtlänge [mm] / • siehe Getriebe
RE 40, 25 W	135					91.7
RE 40, 150 W	136					91.7
RE 40, 150 W	136	GP 42, 3.0 - 15.0 Nm	361			•
RE 40, 150 W	136	GP 52, 4.0 - 30.0 Nm	366			•
RE 40, 150 W	136			AB 28	491	124.3
RE 40, 150 W	136	GP 42, 3.0 - 15.0 Nm	361	AB 28	491	•
RE 40, 150 W	136	GP 52, 4.0 - 30.0 Nm	366	AB 28	491	•
RE 50, 200 W	137					128.7
RE 50, 200 W	137	GP 52, 4.0 - 30.0 Nm	367			•
RE 50, 200 W	137	GP 62, 8.0 - 50.0 Nm	368			•
RE 65, 250 W	138					157.3
RE 65, 250 W	138	GP 81, 20.0 - 120.0 Nm	369			•
A-max 26	155-158					63.1
A-max 26	155-158	GP 26, 0.75 - 4.5 Nm	346			•
A-max 26	155-158	GS 30/GP 32	347/350			•
A-max 26	155-158	GP 32, 0.75 - 6.0 Nm	349/352			•
A-max 26	155-158	GS 38, 0.1 - 0.6 Nm	360			•
A-max 26	155-158	GP 32 S	382-387			•
A-max 32	160					82.3
A-max 32	160	GP 32, 0.75 - 6.0 Nm	348-353			•
A-max 32	160	GS 38, 0.1 - 0.6 Nm	360			•
A-max 32	160	GP 32 S	382-387			•
EC 32, 80 W	209					78.4
EC 32, 80 W	209	GP 32, 0.75 - 6.0 Nm	348-355			•
EC 32, 80 W	209	GP 32 S	382-387			•
EC 40, 170 W	210					103.4
EC 40, 170 W	210	GP 42, 3.0 - 15.0 Nm	361			•
EC 40, 170 W	210	GP 52, 4.0 - 30.0 Nm	366			•

Technische Daten

Versorgungsspannung V_{CC}	5 V $\pm$ 10%
Typische Stromaufnahme	55 mA
Ausgangssignal	EIA Standard RS 422
verwendeter Treiber	DS26LS31
Phasenverschiebung Φ	90°e $\pm$ 45°e
Signalanstiegszeit	(typisch, bei $C_L = 25$ pF, $R_L = 2.7$ k Ω , 25°C) 180 ns
Signalabfallzeit	(typisch, bei $C_L = 25$ pF, $R_L = 2.7$ k Ω , 25°C) 40 ns
Indexpulsbreite	90°e
Betriebstemperaturbereich	-40...+100°C
Trägheitsmoment der Impulsscheibe	≤ 0.6 gcm ²
Max. Winkelbeschleunigung	250 000 rad s ⁻²
Strom pro Kanal	± 20 mA

Das Indexsignal I ist synchronisiert mit Kanal A und B.

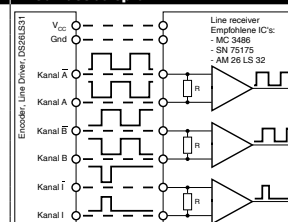
Pinbelegung



- 1 N.C.
- 2 V_{CC}
- 3 GND
- 4 N.C.
- 5 Kanal A
- 6 Kanal A
- 7 Kanal B
- 8 Kanal B
- 9 Kanal I (Index)
- 10 Kanal I (Index)

Steckertyp DIN 41651/
EN 60603-13
Flachbandkabel AWG 28

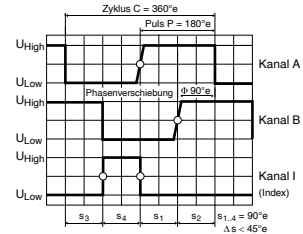
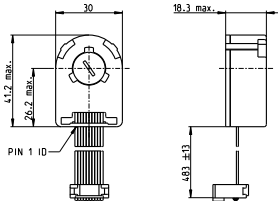
Anschlussbeispiel

Abschlusswiderstand R = typisch 120 Ω

maxon sensor

Encoder HEDL 5540 500 Impulse, 3 Kanal, mit Line Driver RS 422

maxon sensor



Drehrichtung cw (Definition cw S. 64)

- Lagerprogramm
- Standardprogramm
- Sonderprogramm (auf Anfrage)

Artikelnummern

	110512	110514	110516	110518	X drives
Typ					
Impulszahl pro Umdrehung	500	500	500	500	500
Anzahl Kanäle	3	3	3	3	3
Max. Impulsfrequenz (kHz)	100	100	100	100	100
Max. Drehzahl (min ⁻¹)	12 000	12 000	12 000	12 000	12 000
Wellendurchmesser (mm)	3	4	6	8	2-4



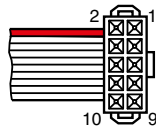
maxon Baukastensystem

+ Motor	Seite	+ Getriebe	Seite	+ Bremse	Seite	Gesamtlänge [mm] / • siehe Getriebe
EC-max 30, 40 W	222					62,6
EC-max 30, 40 W	222	GP 32, 1.0 - 8.0 Nm	353/356			•
EC-max 30, 40 W	222	KD 32, 1.0 - 4.5 Nm	359			•
EC-max 30, 40 W	222	GP 32 S	382-387			•
EC-max 30, 40 W	222		AB 20	488		98,4
EC-max 30, 40 W	222	GP 32, 1.0 - 8.0 Nm	353/356	AB 20	488	•
EC-max 30, 40 W	222	KD 32, 1.0 - 4.5 Nm	359	AB 20	488	•
EC-max 30, 40 W	222	GP 32 S	382-387	AB 20	488	•
EC-max 30, 60 W	223					84,6
EC-max 30, 60 W	223	GP 32, 1.0 - 8.0 Nm	353/356			•
EC-max 30, 60 W	223	KD 32, 1.0 - 4.5 Nm	359			•
EC-max 30, 60 W	223	GP 42, 3.0 - 15.0 Nm	362			•
EC-max 30, 60 W	223		AB 20	488		120,4
EC-max 30, 60 W	223	GP 32, 1.0 - 8.0 Nm	353/356	AB 20	488	•
EC-max 30, 60 W	223	KD 32, 1.0 - 4.5 Nm	359	AB 20	488	•
EC-max 30, 60 W	223	GP 42, 3.0 - 15.0 Nm	362	AB 20	488	•
EC-max 40, 70 W	224					81,4
EC-max 40, 70 W	224	GP 42, 3.0 - 15.0 Nm	362			•
EC-max 40, 70 W	224		AB 28	490		110,7
EC-max 40, 70 W	224	GP 42, 3.0 - 15.0 Nm	362	AB 28	490	•
EC-max 40, 120 W	225					111,4
EC-max 40, 120 W	225	GP 52, 4.0 - 30.0 Nm	367			•
EC-max 40, 120 W	225		AB 28	490		140,7
EC-max 40, 120 W	225	GP 52, 4.0 - 30.0 Nm	367	AB 28	490	•
EC-4pole 22, 90 W	229					70,1
EC-4pole 22, 90 W	229	GP 22/GP 32	343/353			•
EC-4pole 22, 90 W	229	GP 32 S	382-387			•
EC-4pole 22, 120 W 230						87,5
EC-4pole 22, 120 W 230		GP 22/GP 32	343/353			•
EC-4pole 22, 120 W 230		GP 32 S	382-387			•

Technische Daten

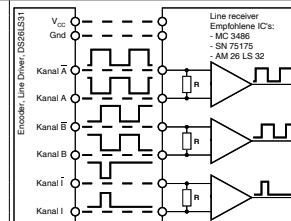
Versorgungsspannung V_{CC}	5 V $\pm$ 10%
Typische Stromaufnahme	55 mA
Ausgangssignal	EIA Standard RS 422
verwendeter Treiber:	DS26LS31
Phasenverschiebung Φ	$90^\circ \pm 45^\circ$
Signalanstiegszeit	
(typisch, bei $C_L = 25$ pF, $R_L = 2.7$ k Ω , 25°C)	180 ns
Signalabfallzeit	
(typisch, bei $C_L = 25$ pF, $R_L = 2.7$ k Ω , 25°C)	40 ns
Indexpulsbreite	90°
Betriebstemperaturbereich	-40...+100°C
Trägheitsmoment der Impulsscheibe	≤ 0.6 gcm ²
Max. Winkelbeschleunigung	250 000 rad s ⁻²
Strom pro Kanal	± 20 mA

Pinbelegung



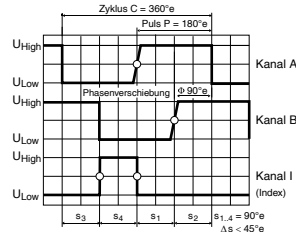
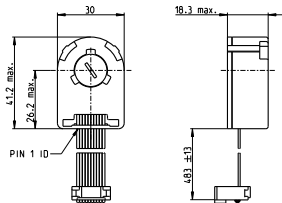
- N.C.
 - V_{CC}
 - GND
 - N.C.
 - Kanal A
 - Kanal A
 - Kanal B
 - Kanal B
 - Kanal I (Index)
 - Kanal I (Index)
- Steckertyp DIN 41651/
EN 60603-13
Flachbandkabel AWG 28

Anschlussbeispiel

Abschlusswiderstand R = typisch 120 Ω

Das Indexsignal I ist synchronisiert mit Kanal A und B.

Encoder HEDL 5540 500 Impulse, 3 Kanal, mit Line Driver RS 422



Drehrichtung cw (Definition cw S. 64)

- Lagerprogramm
- Standardprogramm
- Sonderprogramm (auf Anfrage)

Artikelnummern				
110512	110514	110516	110518	X drives
500	500	500	500	500
3	3	3	3	3
100	100	100	100	100
12000	12000	12000	12000	12000
3	4	6	8	2-4

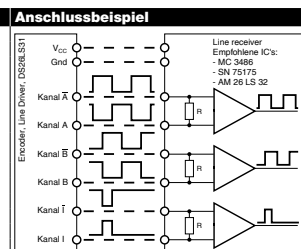
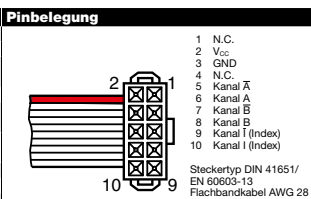
Typ	Impulszahl pro Umdrehung	Anzahl Kanäle	Max. Impulsfrequenz (kHz)	Max. Drehzahl (min ⁻¹)	Wellendurchmesser (mm)
	500	3	100	12000	3
					4
					6
					8
					2-4

maxon Baukastensystem					
+ Motor	Seite	+ Getriebe	Seite	+ Bremse	Seite
EC-4pole 30, 100 W 231		GP 32, 4.0 - 8.0 Nm	356		
EC-4pole 30, 100 W 231		GP 42, 3 - 15 Nm	362		
EC-4pole 30, 100 W 231				AB 20	488
EC-4pole 30, 100 W 231		GP 32, 4.0 - 8.0 Nm	356	AB 20	488
EC-4pole 30, 100 W 231		GP 42, 3 - 15 Nm	362	AB 20	488
EC-4pole 30, 200 W 233					
EC-4pole 30, 200 W 233		GP 32, 4.0 - 8.0 Nm	356		
EC-4pole 30, 200 W 233		GP 42, 3 - 15 Nm	362		
EC-4pole 30, 200 W 233				AB 20	488
EC-4pole 30, 200 W 233		GP 32, 4.0 - 8.0 Nm	356	AB 20	488
EC-4pole 30, 200 W 233		GP 42, 3 - 15 Nm	362	AB 20	488
EC-i 30, 30 W	240				
EC-i 30, 30 W	240	GP 32, 1.0 - 6.0 Nm	353		
EC-i 30, 30 W	240	GP 32 S	382-387		
EC-i 30, 45 W	241				
EC-i 30, 45 W	241	GP 32, 1.0 - 6.0 Nm	354		
EC-i 30, 45 W	241	GP 32 S	382-387		
EC-i 30, 50 W	242				
EC-i 30, 50 W	242	GP 32, 1.0 - 6.0 Nm	354		
EC-i 30, 50 W	242	GP 32 S	382-387		
EC-i 30, 75 W	243				
EC-i 30, 75 W	243	GP 32, 1.0 - 6.0 Nm	354		
EC-i 30, 75 W	243	GP 32 S	382-387		

Technische Daten	
Versorgungsspannung V _{CC}	5 V ± 10%
Typische Stromaufnahme	55 mA
Ausgangssignal	EIA Standard RS 422
verwendeter Treiber	DS26LS31
Phasenverschiebung Φ	90°e ± 45°e
Signalanstiegszeit	(typisch, bei C _L = 25 pF, R _L = 2.7 kΩ, 25 °C) 180 ns
Signalabfallzeit	(typisch, bei C _L = 25 pF, R _L = 2.7 kΩ, 25 °C) 40 ns
Indexpulsbreite	90°e
Betriebstemperaturbereich	-40...+100 °C
Trägheitsmoment der Impulsscheibe	≤ 0.6 gcm ²
Max. Winkelbeschleunigung	250 000 rad s ⁻²
Strom pro Kanal	± 20 mA

Das Indexsignal I ist synchronisiert mit Kanal A und B.

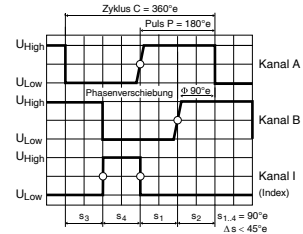
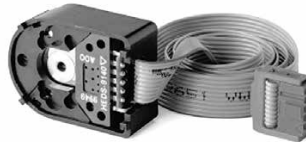
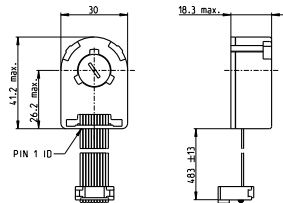
Ausgabe April 2019 / Änderungen vorbehalten



Abschlusswiderstand R = typisch 120 Ω

maxon sensor 445

Encoder HEDL 5540 500 Impulse, 3 Kanal, mit Line Driver RS 422



Drehrichtung cw (Definition cw S. 64)

- Lagerprogramm
- Standardprogramm
- Sonderprogramm (auf Anfrage)

Artikelnummern

110512	110514	110516	110518	X drives
500	500	500	500	500
3	3	3	3	3
100	100	100	100	100
12000	12000	12000	12000	12000
3	4	6	8	2-4

Typ

Impulszahl pro Umdrehung	500	500	500	500	500
Anzahl Kanäle	3	3	3	3	3
Max. Impulsfrequenz (kHz)	100	100	100	100	100
Max. Drehzahl (min ⁻¹)	12000	12000	12000	12000	12000
Wellendurchmesser (mm)	3	4	6	8	2-4



maxon Baukastensystem

+ Motor	Seite	+ Getriebe	Seite	+ Bremse	Seite	Gesamtlänge [mm] / • siehe Getriebe
EC-i 40, 50 W	244/245					49.0
EC-i 40, 50 W	244	GP 32, 1.0 - 6.0 Nm	353			•
EC-i 40, 50 W	244/245	GP 42, 3.0 - 15.0 Nm	362			•
EC-i 40, 50 W	244	GP 32 S	382-387			•
EC-i 40, 70 W	246/247					59.0
EC-i 40, 70 W	246	GP 32, 1.0 - 6.0 Nm	353			•
EC-i 40, 70 W	246/247	GP 42, 3.0 - 15.0 Nm	362			•
EC-i 40, 70 W	246	GP 32 S	382-387			•
EC-i 40, 100 W	248					79.0
EC-i 40, 100 W	248	GP 42, 3.0 - 15.0 Nm	362			•
EC-i 52, 180 W	249					100.7
EC-i 52, 180 W	249	GP 52, 4.0 - 30.0 Nm	366			•
EC-i 52, 200 W	250					
EC-i 52, 200 W	250	GP 52, 4.0 - 30.0 Nm	366			
DCX 22 S	84-85					online
DCX 22 L	86-87					online
DCX 26 L	88-89					online
DCX 32 L	90					online
DCX 35 L	91					online

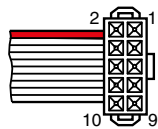
Technische Daten

Versorgungsspannung V_{CC}	5 V $\pm$ 10%
Typische Stromaufnahme	55 mA
Ausgangssignal	EIA Standard RS 422
verwendeter Treiber:	DS26LS31
Phasenverschiebung Φ	90° $\pm$ 45°
Signalanstiegszeit (typisch, bei $C_L = 25$ pF, $R_L = 2.7$ k Ω , 25°C)	180 ns
Signalabfallzeit (typisch, bei $C_L = 25$ pF, $R_L = 2.7$ k Ω , 25°C)	40 ns
Impulsbreite	90°
Betriebstemperaturbereich	-40...+100°C
Trägheitsmoment der Impulsscheibe	≤ 0.6 gcm ²
Max. Winkelbeschleunigung	250000 rad s ⁻²
Strom pro Kanal	± 20 mA

Das Indexsignal I ist synchronisiert mit Kanal A und B.

446 maxon sensor

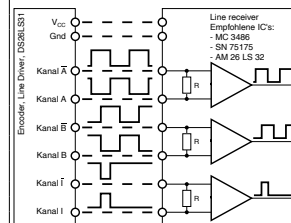
Pinbelegung



- N.C.
- V_{CC}
- GND
- N.C.
- Kanal A
- Kanal A
- Kanal B
- Kanal B
- Kanal I (Index)
- Kanal I (Index)

Steckertyp DIN 41651/
EN 60603-13
Flachbandkabel AWG 28

Anschlussbeispiel

Abschlusswiderstand R = typisch 120 Ω

Ausgabe April 2019 / Änderungen vorbehalten

C.1.4 Absolute Encoder Steering

Absolute Drehgeber - SSI

Vollwelle mit Servoflansch

Magnetische Single- oder Multiturn-Drehgeber 14 Bit ST / 18 Bit MT

EAM360-S - SSI - MAGRES



EAM360 SSI mit Vollwelle

Merkmale

- Drehgeber Single- oder Multiturn / SSI
- Präzise magnetische Abtastung
- Winkelgenauigkeit bis $\pm 0,15^\circ$
- Auflösung max. 32 Bit (14 Bit ST, 18 Bit MT)
- Zusätzliche Inkrementalsignale
- Taktfrequenz bis 2 MHz
- Hohe Schutzart bis IP 67
- Hohe Schock- und Vibrationsfestigkeit

Optional

- Korrosionsschutz C5-M

Technische Daten - elektrisch

Betriebsspannung	4,5...30 VDC (SSI, SSI + TTL/ RS422)
	5,5...30 VDC (SSI + HTL/ Gegentakt)
Betriebsstrom typ.	60 mA (5 VDC, ohne Last) 20 mA (24 VDC, ohne Last)
Initialisierungszeit	≤ 170 ms nach Einschalten
Datenaktualität	Typ. 2 μ s (zyklische Abfrage)
Schnittstellen	SSI, SSI + inkremental
Funktion	Multiturn, Singleturn
Betriebsart	Ringregisterbetrieb (auf Anfrage)
Schrittzahl pro Umdrehung	≤ 16384 / 14 Bit
Anzahl der Umdrehungen	≤ 262144 / 18 Bit
Absolute Genauigkeit	$\pm 0,15^\circ$ (+20 $\pm 15^\circ$ C) $\pm 0,25^\circ$ (-40...+85 $^\circ$ C)
Abtastprinzip	Magnetisch
Code	Gray oder binär
Codeverlauf	CW: aufsteigende Werte bei Drehung im Uhrzeigersinn; Blick auf den Flansch
Eingänge	SSI-Takt: Linereceiver RS422 Nullsetzeingang Zählrichtung
Ausgangsstufen	SSI-Daten: Linedriver RS422 Inkremental: Linedriver RS422 oder Gegentakt (optional)
Inkremental-Ausgang	1024, 2048, 4096 Imp./Umdr. (weitere auf Anfrage)
Ausgangssignale	A+, A-, B+, B-
Ausgabefrequenz	≤ 350 kHz
Störfestigkeit	DIN EN 61000-6-2
Störaussendung	DIN EN 61000-6-4
Diagnosefunktion	DATAVALID (auf Anfrage)

Technische Daten - mechanisch

Baugröße (Flansch)	$\varnothing 36$ mm
Wellenart	$\varnothing 10 \times 16$ mm, Vollwelle mit Fläche
Flansch	Servoflansch
Schutzart DIN EN 60529	IP 65 (ohne Wellendichtung), IP 67 (mit Wellendichtung)
Betriebsdrehzahl	≤ 6000 U/min
Anlaufdrehmoment	≤ 2 Ncm (+20 $^\circ$ C, IP 65) $\leq 2,5$ Ncm (+20 $^\circ$ C, IP 67)
Trägheitsmoment	15,38 gcm ²
Zulässige Wellenbelastung	≤ 40 N axial ≤ 80 N radial
Werkstoffe	Gehäuse: Stahl verzinkt Flansch: Aluminium Welle: Edelstahl
Betriebstemperatur	-40...+85 $^\circ$ C (siehe allgemeine Hinweise)
Relative Luftfeuchte	95 %
Widerstandsfähigkeit	DIN EN 60068-2-6 Vibration 30 g, 10-2000 Hz DIN EN 60068-2-27 Schock 500 g, 1 ms
Masse ca.	170 g
Anschluss	Flanschdose M12, 8-polig Flanschdose M12, 12-polig Kabel 2 m

Magnetische Single- oder Multiturn-Drehgeber 14 Bit ST / 18 Bit MT

Bestellbezeichnung

EAM360-S	W	A	.			.				.A
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W	Servoflansch, Servonut $\varnothing 33$ mm, M3
---	--

* Nur erhältlich bei SSI mit Inkrementalsignale

Absolute Drehgeber - SSI

Vollwelle mit Servoflansch

Magnetische Single- oder Multiturn-Drehgeber 14 Bit ST / 18 Bit MT

EAM360-S - SSI - MAGRES

Zubehör

Stecker und Kabel

10146775	Kabeldose M12, 8-polig, gerade, ohne Kabel
11170528	Kabeldose M12, 8-polig gerade, geschirmt, 5 m Kabel (ESG 34FH0500GVS)
11177375	Kabeldose M12, 8-polig gerade, geschirmt, 10 m Kabel (ESG 34FH1000GVS)
11091511	Kabeldose M12, 8-polig, gerade, geschirmt, 20 m Kabel
11078614	Kabeldose M12, 12-polig gerade, ohne Kabel
11048452	Kabeldose M12, 12-polig gerade, geschirmt, 2 m Kabel (ESG 34JP0200G)
11043780	Kabeldose M12, 12-polig gerade, geschirmt, 5 m Kabel (ESG 34JP0500G)
11048455	Kabeldose M12, 12-polig gerade, geschirmt, 10 m Kabel (ESG 34JP1000G)

Montagezubehör

10106004	Spannriden-Set ø10 mm
----------	-----------------------

Allgemeine Hinweise

Für eine präzise thermische Auslegung ist die Eigenerwärmung abhängig von Drehzahl, Schutzart, Anbau und Umgebungsbedingungen sowie der Elektronik und Versorgungsspannung zu berücksichtigen. Näherungsweise gilt für die Eigenerwärmung 3 K (Variante IP 65) bzw. 8 K (Variante IP 67) pro 1000 U/min. Wird der Drehgeber nahe der maximalen Kennwerte betrieben, sollte die tatsächliche Temperatur am Flansch des Drehgebers gemessen werden.

Anschlussbelegung

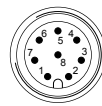
Kabel / Flanschdose M12, 8-polig

für Anschlusskennziffern -L und -B

Pin	Aderfarbe	Signale	Beschreibung
1	weiss	0 V	Betriebsspannung
2	braun	+Vs	Betriebsspannung
3	grün	Clock+	Taktleitung
4	gelb	Clock-	Taktleitung
5	grau	Data+	Datenleitung
6	rosa	Data-	Datenleitung
7	blau	SET	Nullsetzeingang
8	rot	DIR	Zählrichtungseingang*

Kabelschirm: Schirm mit Gehäuse verbunden

Kabeldaten: 4 x 2 x 0,14 mm², paarweise verdreht



Stift, A-codiert

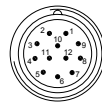
Kabel / Flanschdose M12, 12-polig

für Anschlusskennziffern -L und -K

Pin	Aderfarbe	Signale	Beschreibung
1	braun	+Vs	Betriebsspannung
2	blau	SET	Nullsetzeingang
3	weiss	0 V	Betriebsspannung
4	grün	Clock+	Taktleitung
5	rosa	Data-	Datenleitung
6	gelb	Clock-	Taktleitung
7	schwarz	A+	Inkremental-Signal
8	grau	Data+	Datenleitung
9	rot	DIR	Zählrichtungseingang*
10	violett	A-	Inkremental-Signal
11	grau/rosa	B+	Inkremental-Signal
12	rot/blau	B-	Inkremental-Signal

Kabelschirm: Schirm mit Gehäuse verbunden

Kabeldaten: 6 x 2 x 0,14 mm², paarweise verdreht



Stift, A-codiert

* Entfällt bei Option: DATAVALID

Absolute Drehgeber - SSI

Vollwelle mit Servoflansch

Magnetische Single- oder Multiturn-Drehgeber 14 Bit ST / 18 Bit MT

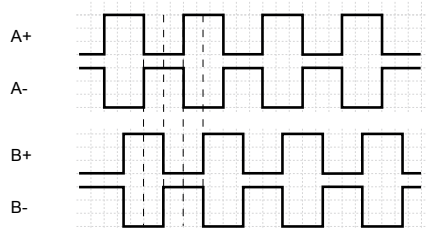
EAM360-S - SSI - MAGRES

Beschreibung der Anschlüsse

SET	Nullsetzeingang. Zum Setzen eines Nullpunktes an jeder beliebigen Stelle. Der Nullsetzvorgang wird durch einen High-Impuls ausgelöst und muss nach der Zählrichtungsauswahl (DIR) erfolgen. Impulsdauer >100 ms. Für max. Störfestigkeit nach dem Nullsetzen an 0 V legen.
DIR	Zählrichtungseingang. Unbeschaltet liegt der Eingang auf High. Für max. Störfestigkeit je nach Drehrichtung an +Vs bzw. 0 V legen. CW HIGH - CCW LOW (Bei Ausführung mit DATAVALID entfällt der Zählrichtungseingang).

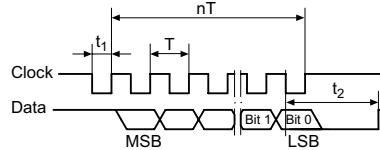
Ausgangssignale

Inkrementalsignale: Drehrichtung im Uhrzeigersinn bei Blick auf den Flansch.



Datenübertragung

Ausgangssignal

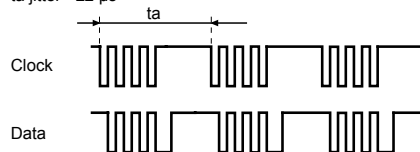


$T = 0,5 \dots 10 \mu s$	$t_1 = 0,25 \dots 5 \mu s$
$t_2 = 20 \pm 2 \mu s$	$f_{max.} = 2 \text{ MHz}$

Datenerfassungszeit t_a

Voraussetzung für eine Datenaktualität von typ. 2 μs ist folgendes Timing des SSI Masters. Bei Nichteinhaltung beträgt die Datenaktualität <50 μs .

$t_a < 5000 \mu s$
 $t_a \text{ jitter } < \pm 2 \mu s$



Schaltpegel

Steuereingänge	Eingangsschaltung
Maximal	$0 \dots +V_s$
Eingangspegel Low	<1 V
Eingangspegel High	>2,1 V

RS422

Ausgangspegel High	>2,3 V
Ausgangspegel Low	<0,5 V
Belastung	<20 mA

Gegentakt

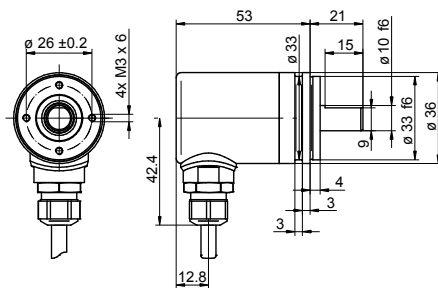
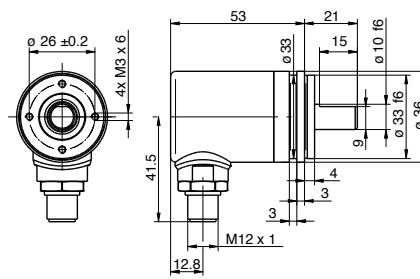
Ausgangspegel High	$\geq +V_s - 2,2 \text{ V}$
Ausgangspegel Low	<0,7 V
Belastung	<20 mA

Gilt für Standardleitungslänge bis 2 m, bei längeren Leitungen ist der Spannungsabfall zu berücksichtigen.

Absolute Drehgeber - SSI

Vollwelle mit Servoflansch

Magnetische Single- oder Multiturn-Drehgeber 14 Bit ST / 18 Bit MT

EAM360-S - SSI - MAGRES**Abmessungen****EAM360, Kabel****EAM360, M12 radial**

C.2 Suspension Bearing Data Sheet

SCHAEFFLER

Needle roller/angular contact ball bearings

NKIA5901-XL (Series NKIA)

single direction axial component, to DIN 5 429-2

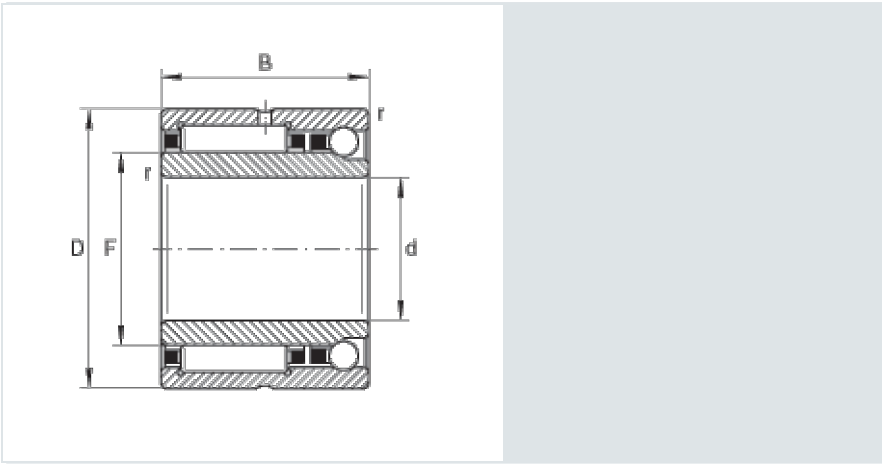


The datasheet is only an overview of dimensions and basic load ratings of the selected product. Please always observe all the guidelines in these overview pages. Further information is given on many products under the menu item "Description". You can also order comprehensive information via the Catalogue ordering system (https://www.schaeffler.de/content.schaeffler.de/en/news_media/index.jsp) or by telephone on +49 (91 32) 82 - 28 97.

d	12 mm
D	24 mm
B	16 mm
F	16 mm
r _{min}	0,3 mm
m	40 g Mass
C _r	8600 N Basic dynamic load rating, radial
C _{0r}	8300 N Basic static load rating, radial
C _a	2700 N Basic dynamic load rating, axial
C _{0a}	3450 N Basic static load rating, axial

SCHAEFFLER

Cur	1660 N	Fatigue limit load, radial
Cua	157 N	Fatigue limit load, axial
nG	23600 1/min	Limiting speed
nGr	21000 1/min	Reference speed



Leistungs-Kostenrechnung Pflanzenbau

Zuckerrüben, wendend, gezogene Saatbettbereitung, Saat, ökologisch, Schlaggröße 2 ha, Ertragsniveau mittel, mittlerer Boden, 67-kW-Mechanisierung, Hof-Feld-Entfernung 2 km

Arbeitsvorgänge		Arbeitsvorgang	Menge	Arbeits- zeit- bedarf	Diesel- bedarf	Ab- schrei- bung	Maschinenkosten			Betriebs- stoffe
Häufig- keit	Zeit- raum		..l/ha	h/ha	l/ha		Zins- kosten	Sonsti- ges	Repara- turen	
0,2	SEP2	Bodenprobe						€/ha		
1	OKT2	Entnahme von Hand, Fahren mit Pick-up Pflügen mit Drehpflug		0,04	0,03	0,09	0,01	0,04	0,03	0,02
1	FA	4 Schare, 1,4 m, angebaut, 67 kW		1,89	23,50	15,14	4,34	1,95	23,59	17,63
1	FEB2	Eggen mit Federzinkenegge								
1	FA	3 m; 67 kW		0,74	7,19	6,60	1,91	1,73	8,47	5,39
1	FEB2	Nmin-Probenahme								
1	BP	Entnahme von Hand, Fahrt mit Pick-up Gülle ausbringen, ab Hof		0,50	0,18	0,63	0,09	0,25	0,18	0,14
1	MRZ1	Pumpe/kwagen, 5 m³, Schleppschlauchverteiler, 7,5 m; 45 kW Gülle, Rind	15,00 m³	1,59	5,69	14,40	3,48	1,93	14,29	4,27
1	MRZ1	Eggen mit Saattbettkombination								
1	FA	4 m; 67 kW		0,58	5,73	7,32	2,17	1,31	7,85	4,30
1	MRZ2	Einzelkornsaat								
1	FA	6 Reihen; 45 kW Zuckerrüben, Z-Saatgut, ökologisch	1,23 U	0,98	4,27	25,74	4,93	1,50	14,60	3,20
1	APR1	Bestandesbonitur								
1	FA	Visuelle Bonitur; Fahren mit Pick-up		0,13	0,12	0,43	0,06	0,18	0,10	0,09
1	APR2	Hacken, 1. und 2. Hacke								
1	FA	6-reihig; 45 kW /		1,15	5,40	9,23	2,49	1,25	9,53	4,05
1	MAI1	Hacken, 1. und 2. Hacke								
1	FA	6-reihig; 45 kW		1,15	5,40	9,23	2,49	1,25	9,53	4,05
1	MAI1	Bestandesbonitur								
1	FA	Visuelle Bonitur; Fahren mit Pick-up		0,13	0,12	0,43	0,06	0,18	0,10	0,09
1	MAI1	Handhacke (1. Hacke)								
1	FA	Kleinbus, 54 kW; 9 AK		84,40	0,30	0,74	0,13	0,86	0,88	0,23
1	MAI2	Hacken, 3. und 4. Hacke								

